Note) Slides and example codes are available:
https://github.com/mint-lab/dl_tutorial

Deep Learning Tutorial with PyTorch

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Why Deep Learning?

- **AlphaGo** beat Sedol Lee in a five-game Go match. (2016)

Image: BBC

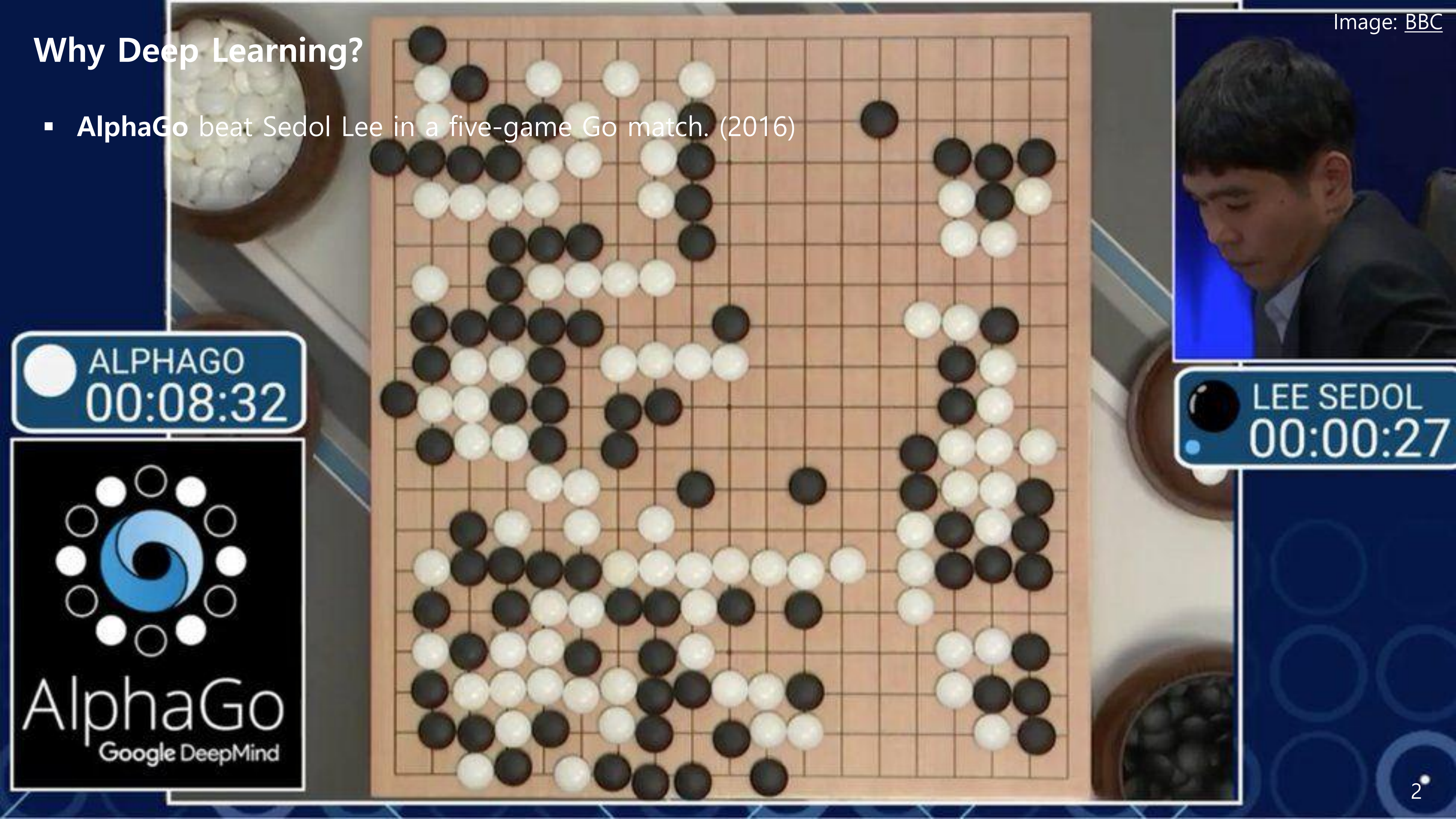
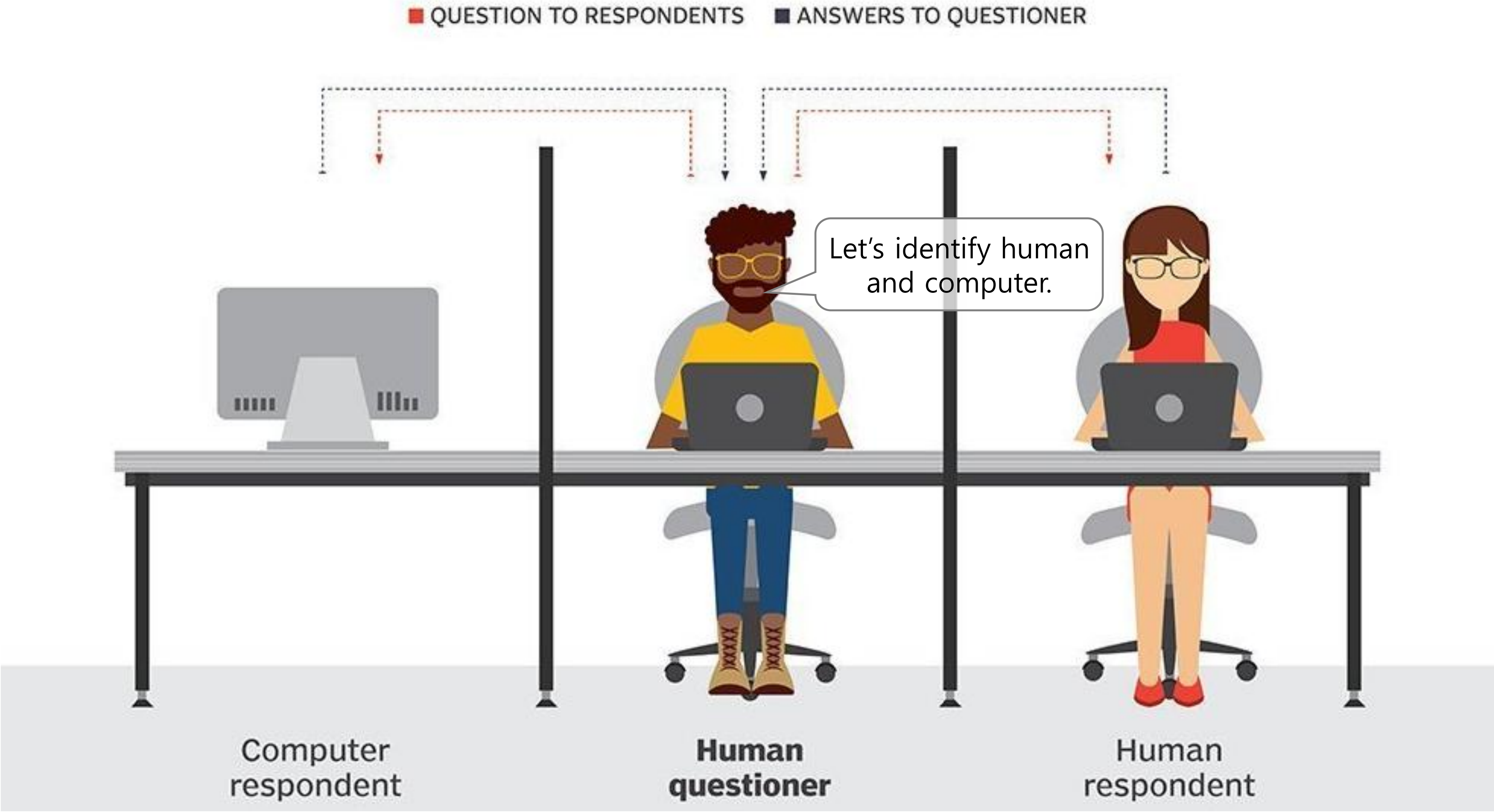


Image: TechTarget

Why Deep Learning?

- **ChatGPT** broke the Turing test. (2022)
  - Reference) Nature, Vol. 619, 2023 DOI



Why Deep Learning?

- **ChatGPT** gives ***exact*** *answers* to my questions, but **Google** retrieves the ***nearest*** *webpages* to my queries.

Image: pexels (Brett Sayles)

Why Deep Learning?

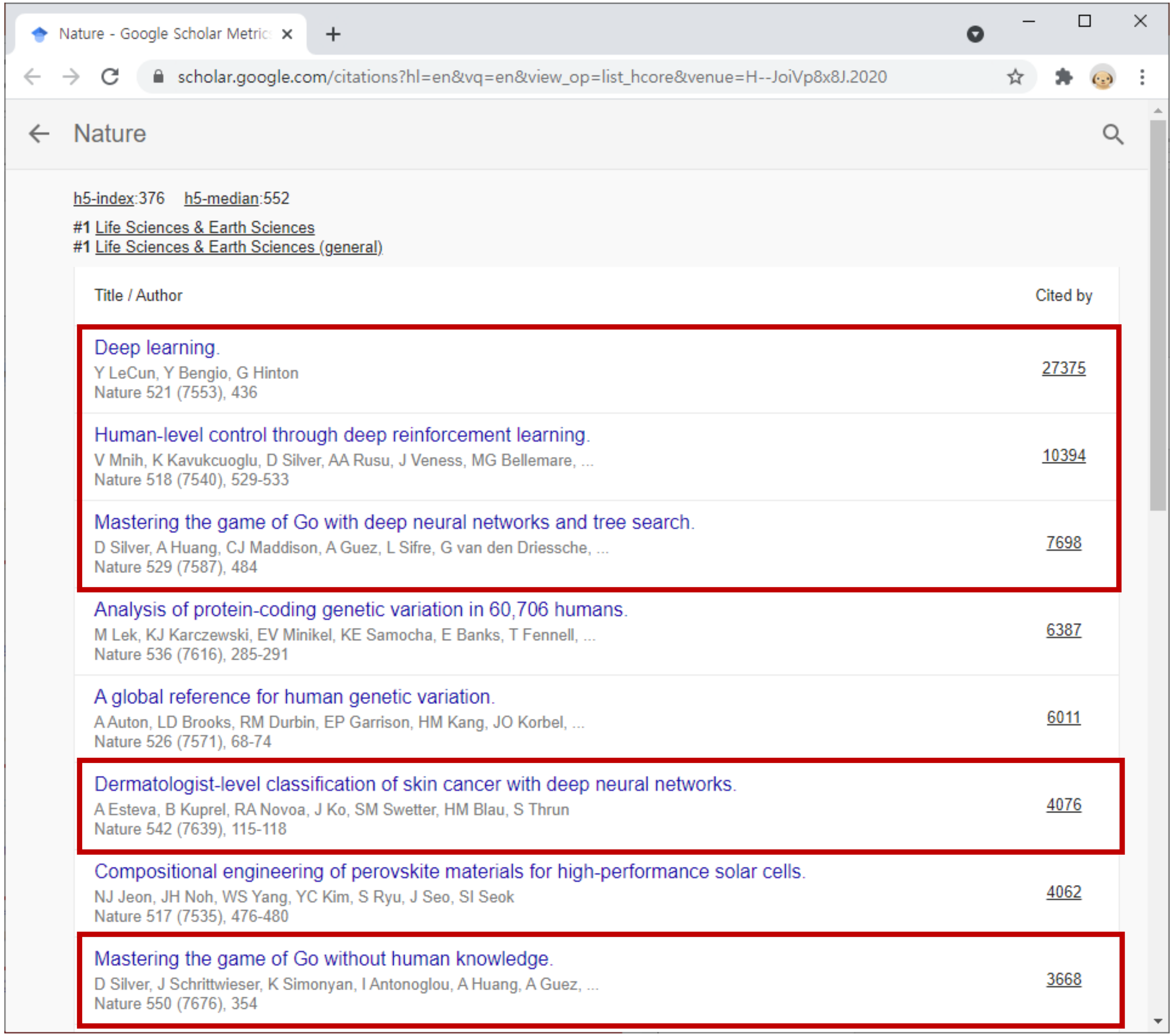


Image: [Google Scholar] (retrieved on April 7th, 2021)

Why Deep Learning?

- Yoshua Bengio, Geoffrey Hinton, and Yann LeCun won the **Turing Award (2018)**.
  - Note) Chronological listing of A.M. Turing Award Winners

Yoshua Bengio

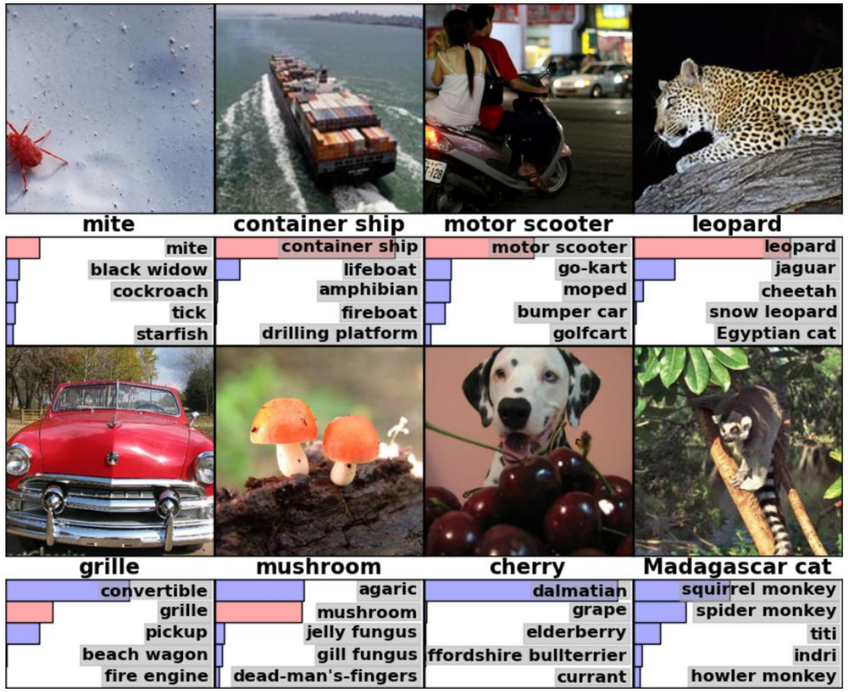
Geoffrey Hinton

Yann LeCun

Image: ACM

Why Deep Learning?

- **AlexNet** won the ILSVRC (ImageNet Large Scale Visual Recognition Competition) 2012 (top-5 error: **16.4%**).
  - An **8-layer** neural network (approx. 61 million parameters) with 2 x GPUs
  - ReLU (rectified linear unit) is used for the vanishing gradient problem and speed-up.
  - Data augmentation (more data) and dropout (regularization) was used to solve the overfitting problem.
  - Local response normalization (~ batch normalization) is used for stable training and generalization.



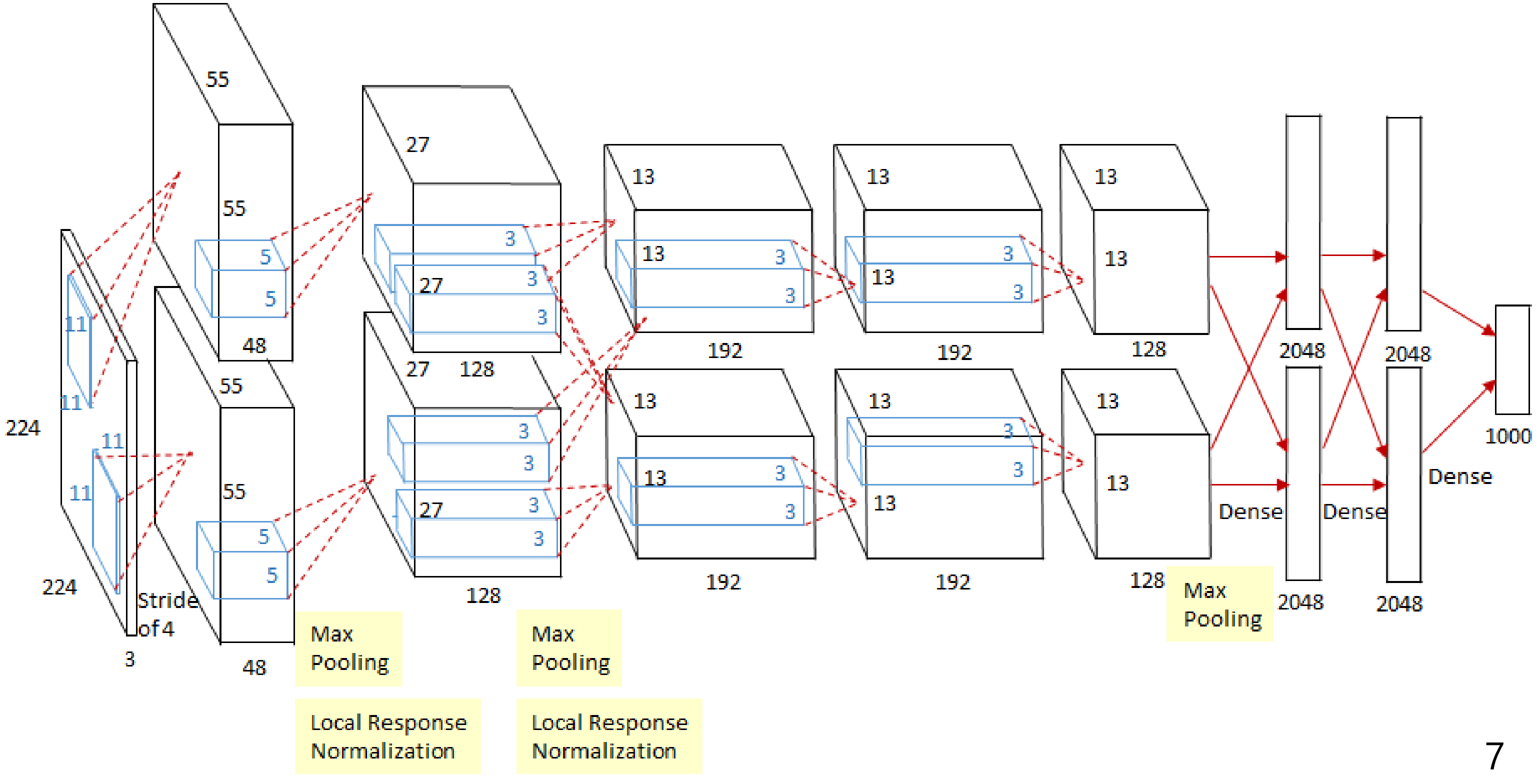
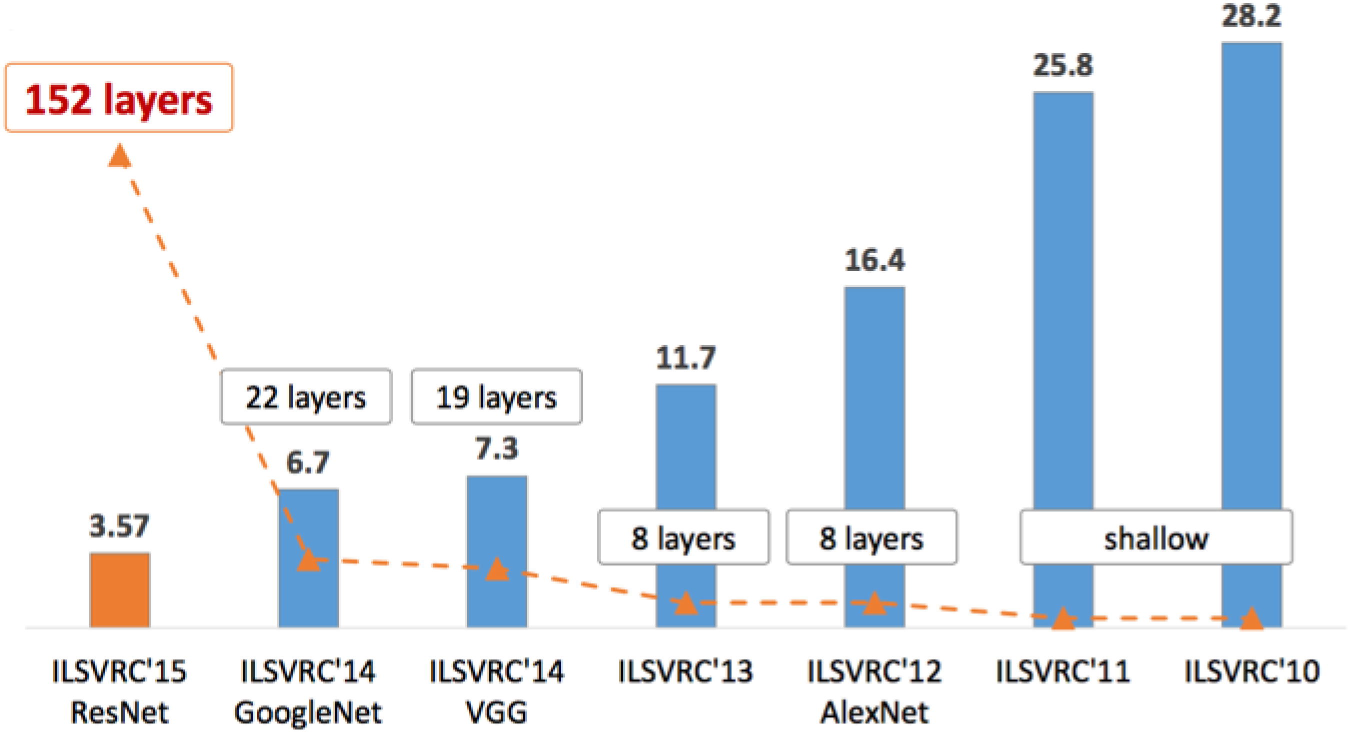


Image: Medium article (by Sik-Ho Tsang)

What is Deep Learning?

- *Deep learning* is machine learning with **deep neural networks** (shortly DNN).
  - e.g. **ResNet** (residual neural network)
    - A **152-layer** network with skip connections
    - The winner of ImageNet Challenge 2015 (top-5 error: **3.57%**)



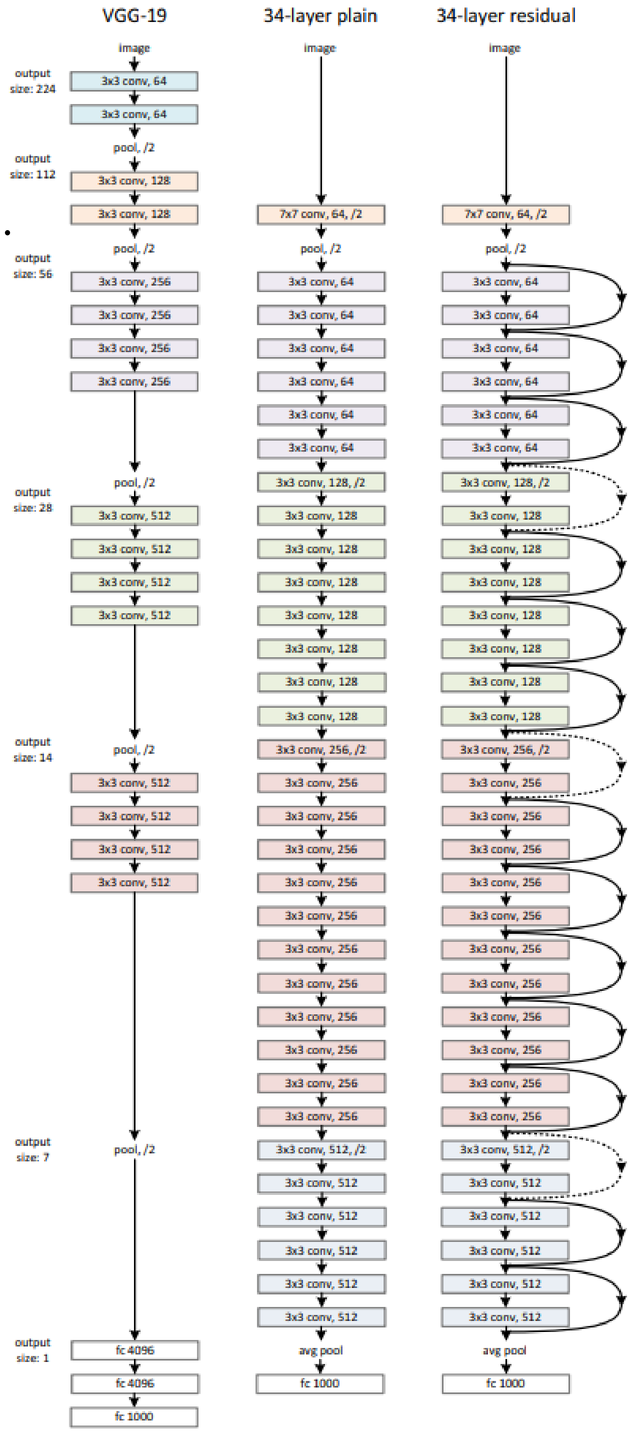
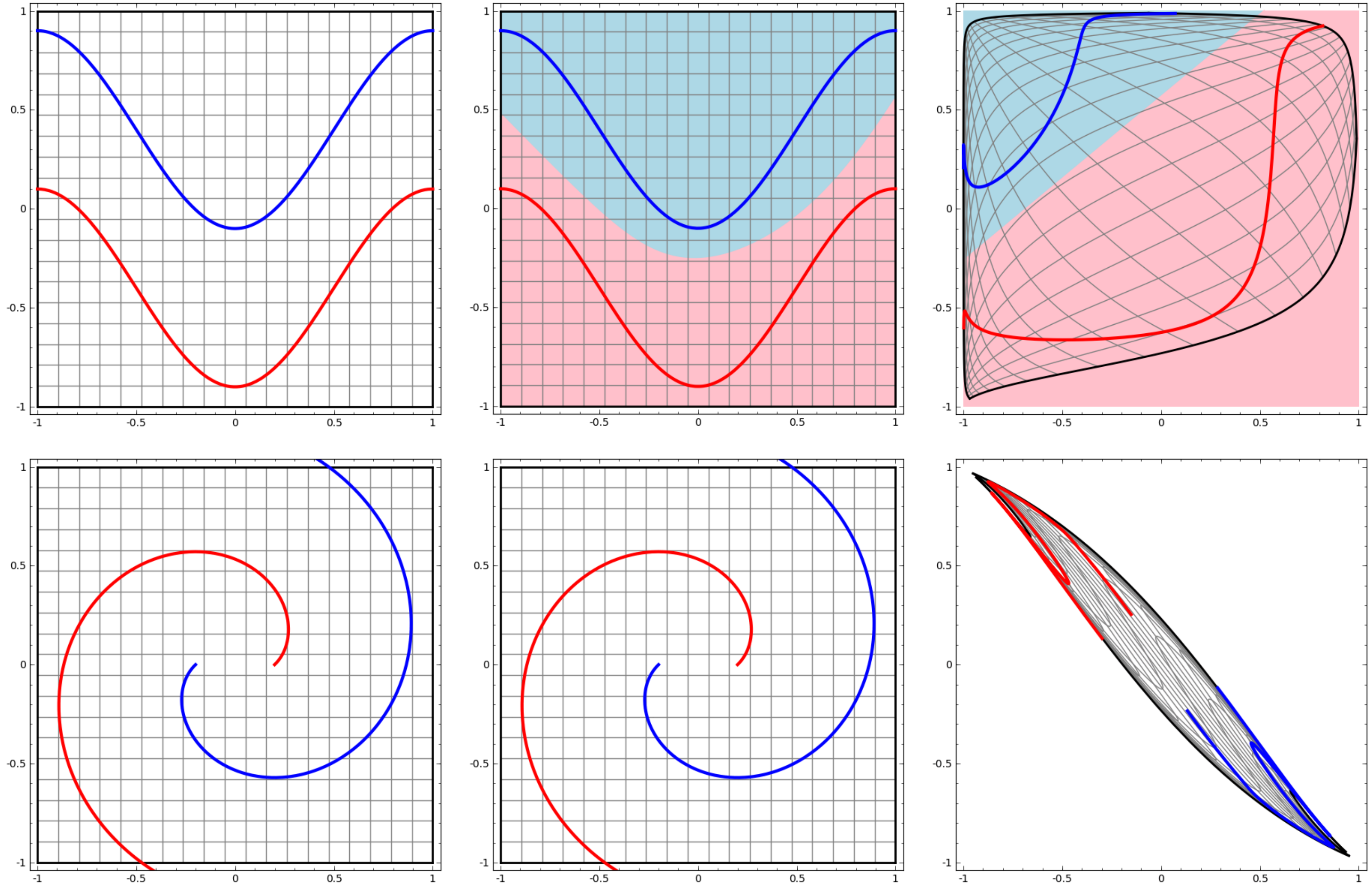


Image: arXiv:1512.03385 and OpenGenus IQ

What is Deep Learning?

- *Deep learning* is **representation learning** (a.k.a. feature learning).
  - e.g. Task: Draw a line to separate the red line and blue line.

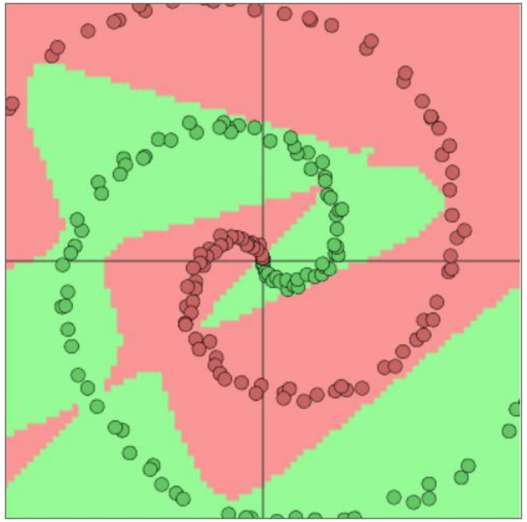


Image/Animation: colah's blog

What is Deep Learning?

- *Deep learning* is **representation learning** (a.k.a. feature learning).
  - e.g. Task: Draw a line to separate the red circles and green circles.
    - Note) Try it at ConvnetJS or TensorFlow Playground.

```
layer_defs = [];
layer_defs.push({type:'input', out_sx:1, out_sy:1, out_depth:2});
layer_defs.push({type:'fc', num_neurons:6, activation: 'tanh'});
layer_defs.push({type:'fc', num_neurons:2, activation: 'tanh'});
layer_defs.push({type:'softmax', num_classes:2});

net = new convnetjs.Net();
net.makeLayers(layer_defs);

trainer = new convnetjs.SGDTrainer(net, {learning_rate:0.01, momentum:0.1, batch_size:10, l2_decay:0.001});
```



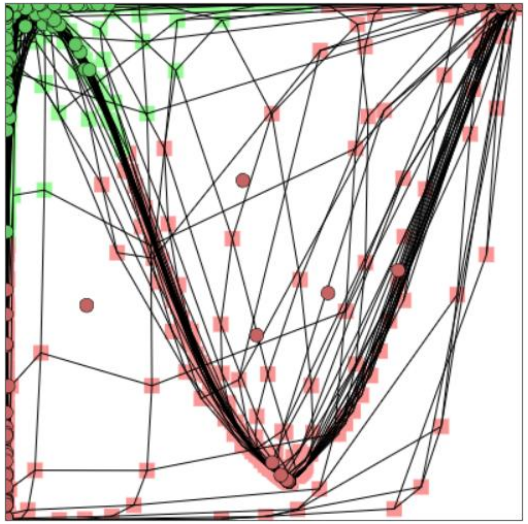


Image: ConvNetJS (retrieved on April 15th, 2021)

What is Deep Learning?

- *Deep learning* is **representation learning** (a.k.a. feature learning).

Machine learning:

Deep learning:

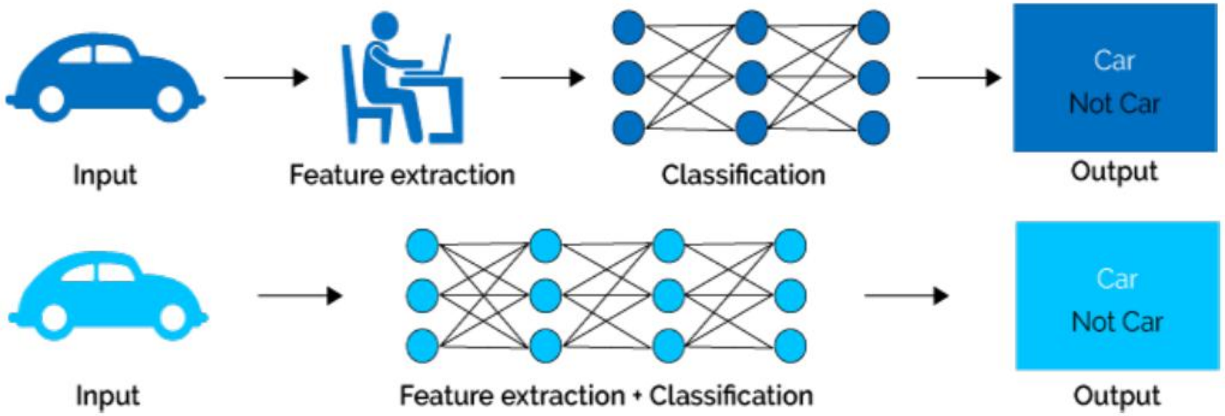


Image: MIT DL Lectures (by Lex Fridman)

What is Deep Learning?

- *Deep learning* is **scalable**.

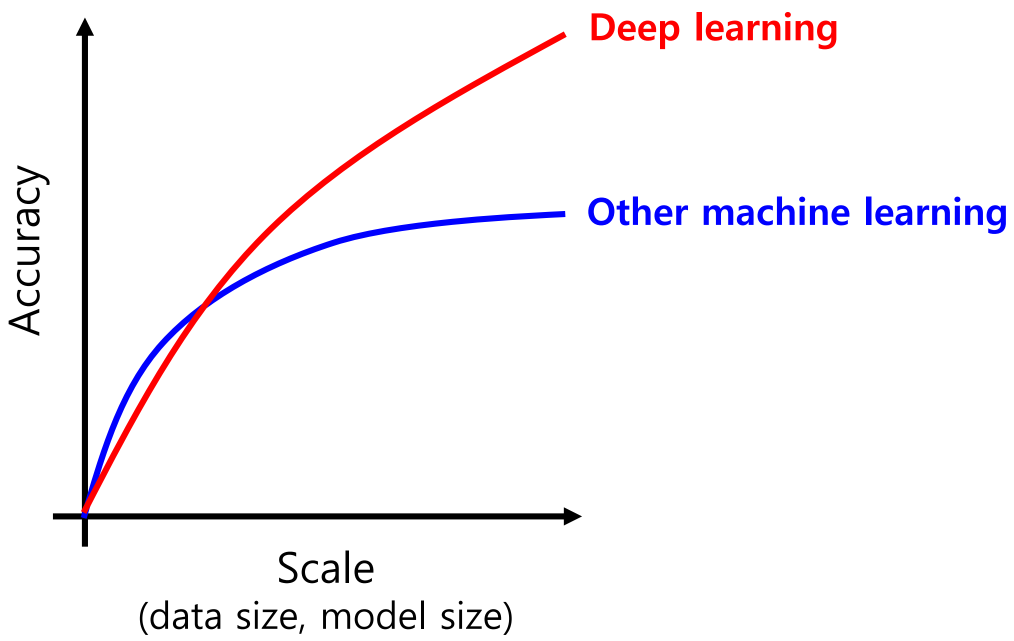


Table of Contents

Why PyTorch?

- **10 reasons why PyTorch is the DL framework of the future**

(by Dhiraj K, September 18th, 2019)

1. PyTorch is Pythonic
2. Easy to learn
3. Higher developer productivity
4. Easy debugging
5. Data parallelism
6. Dynamic computational graph support
7. Hybrid front-end
8. Useful libraries
9. ONNX (Open Neural Network Exchange) support
10. Cloud support

Why PyTorch?

- Popularity (vs. TensorFlow) in academic communities

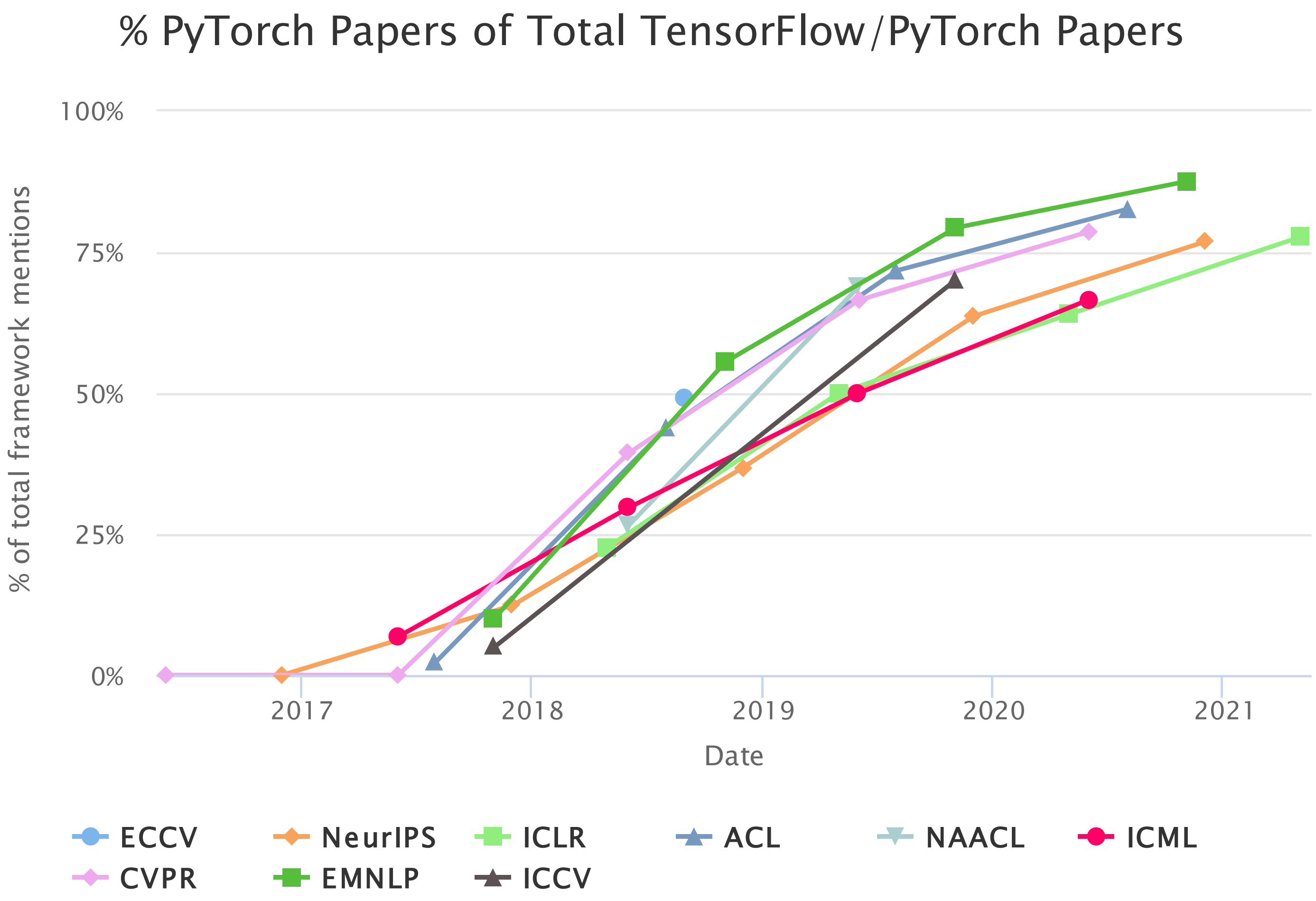


Image: [PyTorch vs TensorFlow (by Horace He)]()

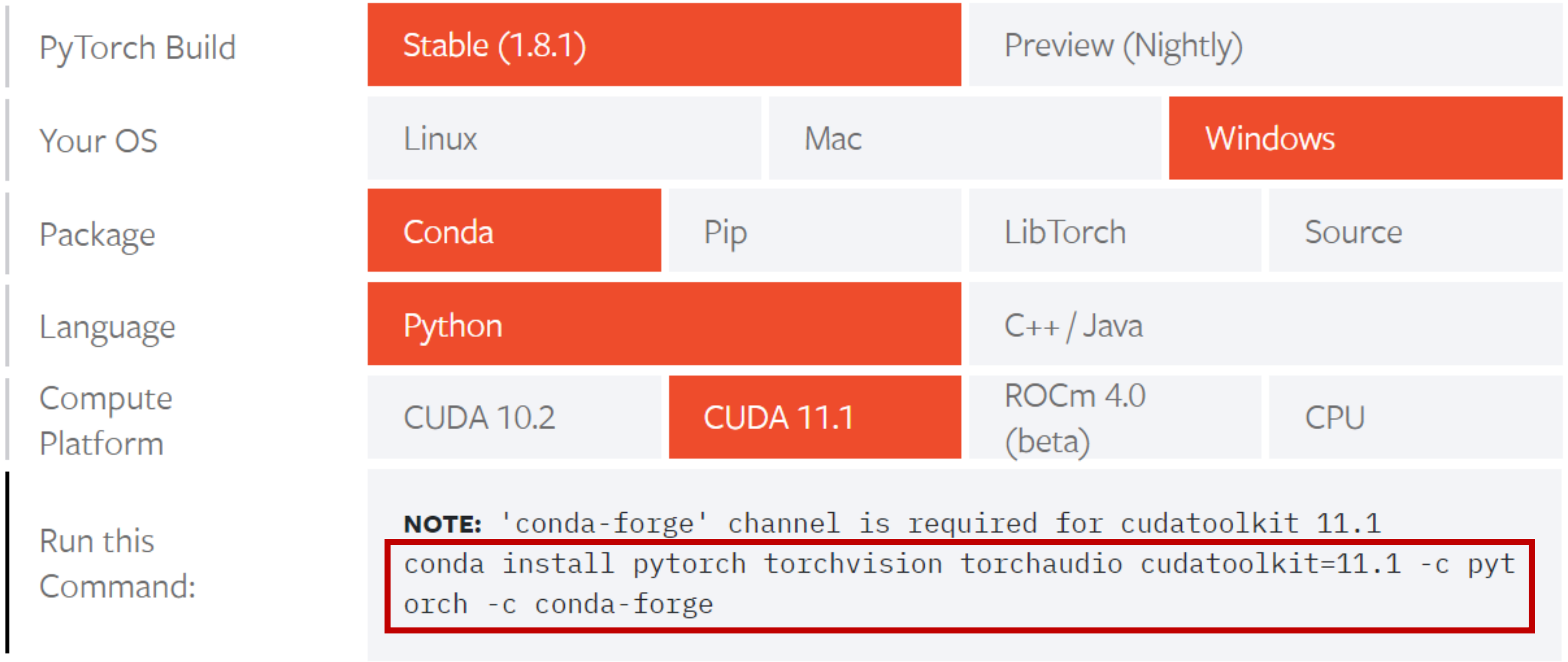
What is PyTorch?

PyTorch

- **PyTorch** (2016) is an open source deep learning library based on Python.
  - It originated from *Torch* (2002) based on the Lua script language.
  - It is primarily developed by Facebook's AI Research lab (FAIR).
- (From my point of view) PyTorch is a **NumPy extension** with supports of
  - \+ An n-dimensional array with GPU acceleration and automatic differentiation
  - \+ Useful modules for neural networks
- My useful lists on deep learning and PyTorch

Practice) PyTorch Installation

- Please follow PyTorch's installation instruction for your system.
  - Note) If you want GPU acceleration, please install the matched version of CUDA in advance. Please visit CUDA Toolkit Archive to download a specific version of CUDA.



- Note) You can use Google Colab without local installation of PyTorch.

Image: PyTorch (retrieved on April 7th, 2021)

Note) Python Virtual Environment

- A **Python virtual environment** is an isolated Python runtime environment that contains a particular version of Python and its additional packages.
  - Note) Virtual machine: A computer entirely implemented by software
- Why virtual environments?
  - We can run Python applications with different versions of packages (e.g. PyTorch 0.9 and PyTorch 1.8) together.
- Tool #1) venv in the Python Standard Library
- Tool #2) conda in Anaconda
  - *Conda* is an open source package/dependency/environment management system for programming languages such as Python, R, Java, JavaScript, C/C++, and more.
  - Usage

`conda create --name` *`venv_name`* `python=3.6`	Create a virtual environment
`conda create --name venv_dst --clone venv_src`	Copy a virtual environment
`conda activate` *`venv_name`*	Activate the virtual environment
`conda deactivate`	Deactivate the virtual environment
`conda env remove --name` *`venv_name`*	Remove the virtual environment
`conda env list`	List all virtual environments
`conda list`	List all installed packages

Learning PyTorch with Practice

(From my point of view) PyTorch is a **NumPy extension** with supports of

+ An n-dimensional array with GPU acceleration and automatic differentiation

  so-called a ***tensor*** (Note: A matrix is a second-order tensor.)

+ Useful modules for neural networks

- Practice with tensors
  - Creating a tensor
  - Reshaping a tensor
  - Line fitting from two points
  - CPU vs. GPU-acceleration
  - Automatic differentiation (so called *Autograd*)
- Practice with useful modules for neural networks
  - Gradient descent by hands and `torch.optim`
  - More examples with DNNs, CNNs, and RNNs.

Practice) Creating a Tensor (1/2)

```
import numpy as np
import torch

# 1. Create a tensor from a composite data
x = np.array([[3, 29, 82], [10, 18, 84]])
y = torch.tensor(x)
print(y.ndim, y.dim())      # 2                       Note) x.ndim
print(y.nelement())         # 6                       Note) x.size
print(y.shape, y.size())    # torch.Size([2, 3])      Note) x.shape
print(y.dtype)              # torch.int32             Note) x.dtype

# 2. Create a tensor using initializers
p = torch.rand(3, 2)        # Try zeros, ones, eyes, empty, arange, linspace,
q = torch.zeros_like(p)     #     and their ..._like
print(p.dtype)              # torch.float32
print(q.shape)              # torch.Size([3, 2])

# 3. Interpret as a tensor (generating only a view)
z = torch.as_tensor(x)      # Or torch.from_numpy(x)  Note) np.asarray()
x[-1,-1] = 86
print(z[-1])                # tensor([10, 18, 86], dtype=torch.int32)
```

Practice) Creating a Tensor (2/2)

```
# 4. Access elements
print(y[:,1])               # tensor([29, 18])
print(y[0,0])               # tensor(3)              Note) x[0,0] == 3
print(y[0,0].item())        # 3

# 5. CUDA tensors
if torch.cuda.is_available():
    print(y.device)         # 'cpu'
    y_cuda = y.cuda()       # Or y.to('cuda')
    print(y_cuda.device)    # 'cuda:0'
    y_cpu = y_cuda.cpu()    # Or y.cuda.to('cpu')
    print(y_cpu.device)     # 'cpu'

    x_cpu = y_cpu.numpy()   # Or np.array(y_cpu)
    x_cuda = y_cuda.numpy() # Error!
```

Practice) Reshaping a Tensor (1/2)

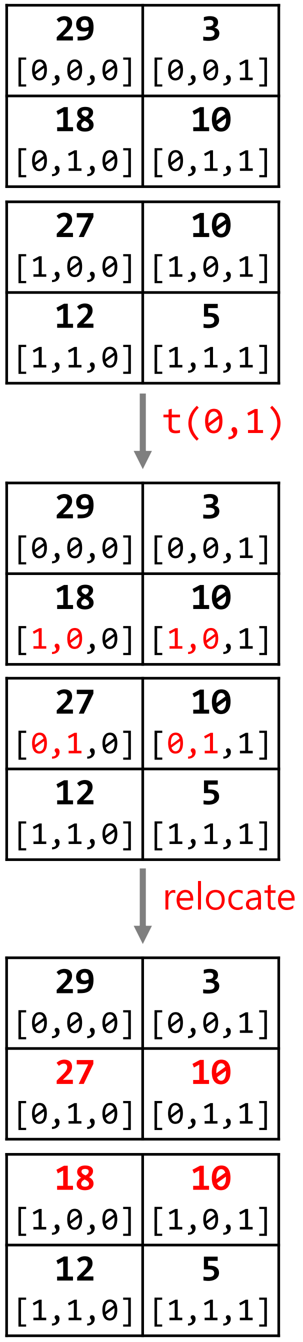
29 [0,0,0]	3 [0,0,1]
18 [0,1,0]	10 [0,1,1]

27 [1,0,0]	10 [1,0,1]
12 [1,1,0]	5 [1,1,1]

```
import numpy as np
import torch

x = np.array([[[29, 3], [18, 10]], [[27, 10], [12, 5]]])
y = torch.tensor(x)
print(y.ndim)          # 3
print(y.shape)         # torch.Size([2, 2, 2])

p = y.view(-1)         # tensor([29, 3, 18, 10, 27, 10, 12, 5])
print(p.shape, p)      # torch.Size([8])
q = y.view(1, -1)      # tensor([[29, 3, 18, 10, 27, 10, 12, 5]])
print(q.shape, q)      # torch.Size([1, 8])
r = y.view(2, -1)      # tensor([[29, 3, 18, 10], [27, 10, 12, 5]])
print(r.shape, r)      # torch.Size([2, 4])
# Of course, 'reshape' is also supported and the same as 'view'.
s = y.reshape(2, -1, 1) # tensor([[[29], [3], [18], [10]], [[27], [10], [12], [5]]])
print(s.shape, s)      # torch.Size([2, 4, 1])

ss = s.squeeze(2)      # Note) s.squeeze(0) and s.squeeze(1) have no effect.
print(ss)              # tensor([[29, 3, 18, 10], [27, 10, 12, 5]])
print(ss.shape)        # torch.Size([2, 4])
u0 = ss.unsqueeze(0)   # tensor([[[29, 3, 18, 10], [27, 10, 12,  5]]])
print(u0.shape, u0)    # torch.Size([1, 2, 4])
u1 = ss.unsqueeze(1)   # tensor([[[29, 3, 18, 10]], [[27, 10, 12, 5]]])
print(u1.shape, u1)    # torch.Size([2, 1, 4])
u2 = ss.unsqueeze(2)   # tensor([[[29], [3], [18], [10]], [[27], [10], [12], [5]]])
print(u2.shape, u2)    # torch.Size([2, 4, 1])
```

Practice) Reshaping a Tensor (2/2)

```
# Switch indices each other, (i, j) to (j, i)
t_021 = y.transpose(1, 2)                # tensor([[[29, 18],
print(t_021, t_021.is_contiguous()) #              [ 3, 10]], ... ]) False
c_021 = t_021.contiguous()               # tensor([[[29, 18],
print(c_021, c_021.is_contiguous()) #              [ 3, 10]], ... ]) True
t_102 = y.transpose(0, 1)                # tensor([[[29,  3],
print(t_102, t_102.is_contiguous()) #               27, 10]], ... ]) False

# Assign indices
t_201 = y.permute(2, 0, 1)               # tensor([[[29, 18],
print(t_201, t_201.is_contiguous()) #              [27, 12]], ... ]) False

# Note) Reshaping does not copy contents.
y[0,0,0] = 27
print(p)                                 # tensor([27, 3, 18, 10, 27, 10, 12, 5])
print(t_021)                             # tensor([[[27, 18], ...], ...])
print(c_021)                             # tensor([[[29, 18], ...], ...])

# Copy a tensor and detach it from its connected computational graph
z = y.clone().detach()                   # Note) x.clone()
y[0,0,0] = 1
print(y)                                 # tensor([[[ 1, 3], ...], ...])
print(z)                                 # tensor([[[27, 3], ...], ...])
```



Practice) Line Fitting from Two Points

- Find a line which passes through two points, $(1,4)$ and $(4,2)$
  - The tensor class, `torch.Tensor`, includes member functions of tensor operation and manipulation, and linear algebra (in contrast to `numpy.array`).
  - Note) [PyTorch APIs for `torch.Tensor`]()
    - Please remember that a function with ending with an underscore(_) means an in-place function.

```python
import torch

A = torch.tensor([[1., 1.], [4., 1.]])
b = torch.tensor([[4.], [2.]])
A_inv = A.inverse()        # Note) np.linalg.inv(A)
print(A_inv.mm(b))         # Note) np.matmul(A_inv, b)
```

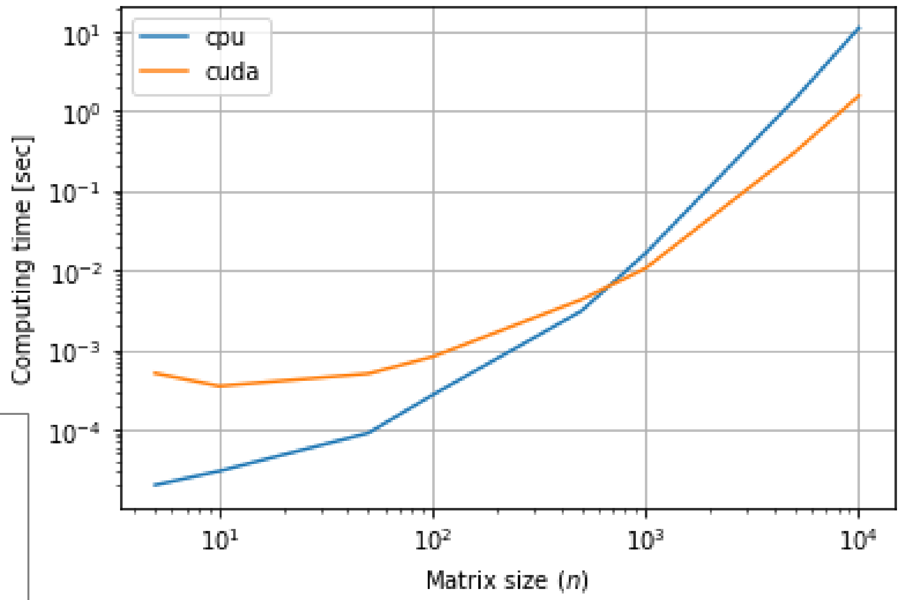
- Note) Line fitting with NumPy

```python
import numpy as np

A = np.array([[1., 1.], [4., 1.]])
b = np.array([[4.], [2.]])
A_inv = np.linalg.inv(A)
print(np.matmul(A_inv, b)) # [[-0.66666667], [ 4.66666667]]
```

Practice) CPU vs. GPU-acceleration

- Computing time on my laptop
  - cpu : **1.4** [sec] @ Intel i7-7700HQ 2.8GHz
  - cuda: **0.3** [sec] @ NVIDIA GTX 1060

```
import torch
import time

dev_name = 'cuda' if torch.cuda.is_available() else 'cpu' # Try 'cpu'
n = 5000

A = torch.rand(n, n, device=dev_name)
B = torch.rand(n, n, device=dev_name)
start = time.time()
C = A.inverse() * B
elapse = time.time() - start
print(f'Computing time by {dev_name}: {elapse:.3f} [sec]')
```



Practice) Automatic Differentiation

- A derivative value, `torch.Tensor.grad`, is available after setting `requires_grad=True` and its forward calculation and backward propagation (a.k.a. backpropagation).

```
import torch

x = torch.tensor([2.], requires_grad=True)
y = 0.1*x**3 - 0.8*x**2 - 1.5*x + 5.4
y.backward()
print(x.grad) # Derivative: tensor([-3.5000])
```

- Note) Symbolic differentiation with SymPy

```
import sympy as sp

x, y = sp.symbols('x y')
y = 0.1*x**3 - 0.8*x**2 - 1.5*x + 5.4
yd = sp.diff(y, x)
print(yd)                      # 0.3*x**2 - 1.6*x - 1.5
print(float(yd.subs({x: 2}))) # -3.5
```

Practice) Automatic Differentiation – More Analysis

- Given) $y(x) = x^3$, $z(y) = \log y$
- Q) A derivative value $\frac{\partial z}{\partial x}$ at $x = 5$?

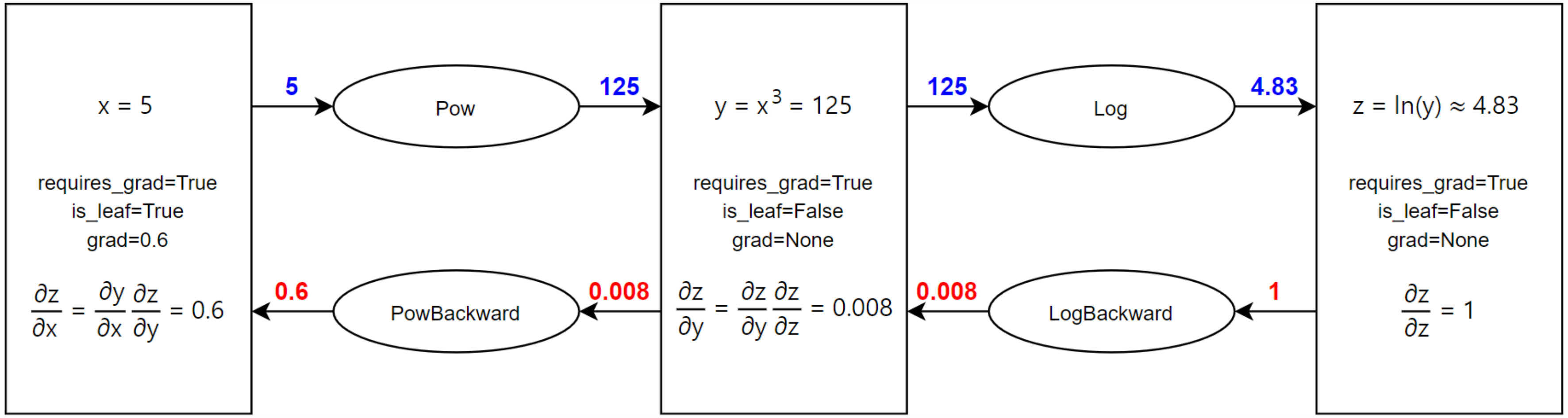
```python
import torch

def get_tensor_info(tensor):
  info = []
  for name in ['requires_grad', 'is_leaf', 'retains_grad', 'grad']:
    info.append(f'{name}({getattr(tensor, name, None)})')
  info.append(f'tensor({str(tensor)})')
  return ' '.join(info)

x = torch.tensor(5., requires_grad=True)
y = x ** 3
z = torch.log(y)
print("### Before 'z.backward()'")
print('* x:', get_tensor_info(x))
print('* y:', get_tensor_info(y))
print('* z:', get_tensor_info(z))

y.retain_grad()
z.retain_grad()
z.backward()
print("### After 'z.backward()'")
print('* x:', get_tensor_info(x))
print('* y:', get_tensor_info(y))
print('* z:', get_tensor_info(z))
```

Code/Image: [Dable Tech Blog]()

Practice) Automatic Differentiation – More Analysis

- Given) $y(x) = x^3$, $z(y) = \log y$
- Q) A derivative value $\frac{\partial z}{\partial x}$ at $x = 5$?

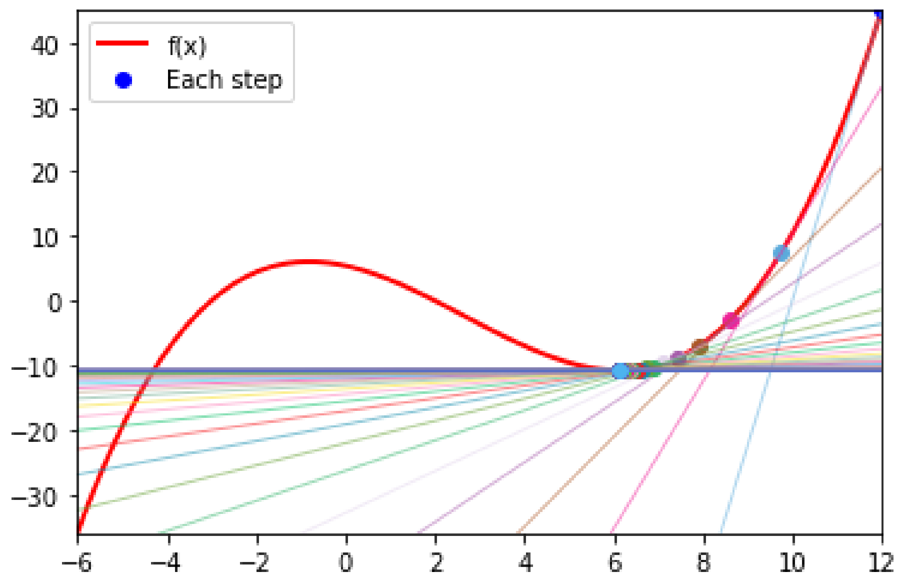
```
### Before 'z.backward()'
* x: requires_grad(True) is_leaf(True)  retains_grad(None) grad(None)  tensor(tensor(5.,   requires_grad=True))
* y: requires_grad(True) is_leaf(False) retains_grad(None) grad(None)  tensor(tensor(125., grad_fn=<PowBackward0>))
* z: requires_grad(True) is_leaf(False) retains_grad(None) grad(None)  tensor(tensor(4.83, grad_fn=<LogBackward>))

### After 'z.backward()'
* x: requires_grad(True) is_leaf(True)  retains_grad(None) grad(0.6)   tensor(tensor(5.,   requires_grad=True))
* y: requires_grad(True) is_leaf(False) retains_grad(True) grad(0.008) tensor(tensor(125., grad_fn=<PowBackward0>))
* z: requires_grad(True) is_leaf(False) retains_grad(True) grad(1.)    tensor(tensor(4.83, grad_fn=<LogBackward>))
```



Code/Image: [Dable Tech Blog]

Practice) Gradient Descent by Hands

```
f = lambda x: 0.1*x**3 - 0.8*x**2 - 1.5*x + 5.4
viz_range = torch.FloatTensor([-6, 12])
learn_rate =  0.1
max_iter = 100
min_tol = 1e-6
x_init = 12.

# Prepare visualization
xs = torch.linspace(*viz_range, 100)
plt.plot(xs, f(xs), 'r-', label='f(x)', linewidth=2)
plt.plot(x_init, f(x_init), 'b.', label='Each step', markersize=12)
plt.axis((*viz_range, *f(viz_range)))
plt.legend()

x = torch.tensor(x_init, requires_grad=True)
for i in range(max_iter):
    # Derive gradient with Autograd
    if x.grad != None:
        x.grad.zero_()              # Reset gradient tracking
    y = f(x)                        # Calculate the function (forward)
    y.backward()                    # Calculate the gradient (backward)

    # Run the gradient descent
    xp = x.clone().detach()         # Note) xp = x
    with torch.no_grad():           # Disable gradient tracking
        x -= learn_rate*x.grad      # Note) x = x - learn_rate*fd(x) is an original code.
                                    #       x = x - learn_rate*x.grad() does not work!

    # Update visualization for each iteration
    print(f'Iter: {i}, x = {xp:.3f} to {x:.3f}, f(x) = {f(xp):.3f} to {f(x):.3f} (f\'(x) = {x.grad:.3f})')
    lcolor = torch.rand(3).tolist()
    approx = x.grad*(xs-xp) + f(xp)
    plt.plot(xs, approx, '-', linewidth=1, color=lcolor, alpha=0.5)
    xc = x.clone().detach() # Copy 'x' for plotting
    plt.plot(xc, f(xc), '.', color=lcolor, markersize=12)

    # Check the terminal condition
    if abs(x - xp) < min_tol:
        break;
plt.show()
```



Practice) Gradient Descent by torch.optim

```
f = lambda x: 0.1*x**3 - 0.8*x**2 - 1.5*x + 5.4
viz_range = torch.FloatTensor([-6, 12])
learn_rate =  0.1
max_iter = 100
min_tol = 1e-6
x_init = 12.

# Prepare visualization
xs = torch.linspace(*viz_range, 100)
plt.plot(xs, f(xs), 'r-', label='f(x)', linewidth=2)
plt.plot(x_init, f(x_init), 'b.', label='Each step', markersize=12)
plt.axis((*viz_range, *f(viz_range)))
plt.legend()

x = torch.tensor(x_init, requires_grad=True)
optimizer = torch.optim.SGD([x], lr=learn_rate)
for i in range(max_iter):
    # Run the gradient descent with the optimizer
    optimizer.zero_grad()       # Reset gradient tracking
    y = f(x)                    # Calculate the function (forward)
    y.backward()                # Calculate the gradient (backward)
    xp = x.clone().detach()     # Note) xp = x
    optimizer.step()            # Update 'x'

    # Update visualization for each iteration
    print(f'Iter: {i}, x = {xp:.3f} to {x:.3f}, f(x) = {f(xp):.3f} to {f(x):.3f} (f\'(x) = {x.grad:.3f})')
    lcolor = torch.rand(3).tolist()
    approx = x.grad*(xs-xp) + f(xp)
    plt.plot(xs, approx, '-', linewidth=1, color=lcolor, alpha=0.5)
    xc = x.clone().detach() # Copy 'x' for plotting
    plt.plot(xc, f(xc), '.', color=lcolor, markersize=12)

    # Check the terminal condition
    if abs(x - xp) < min_tol:
        break;
plt.show()
```

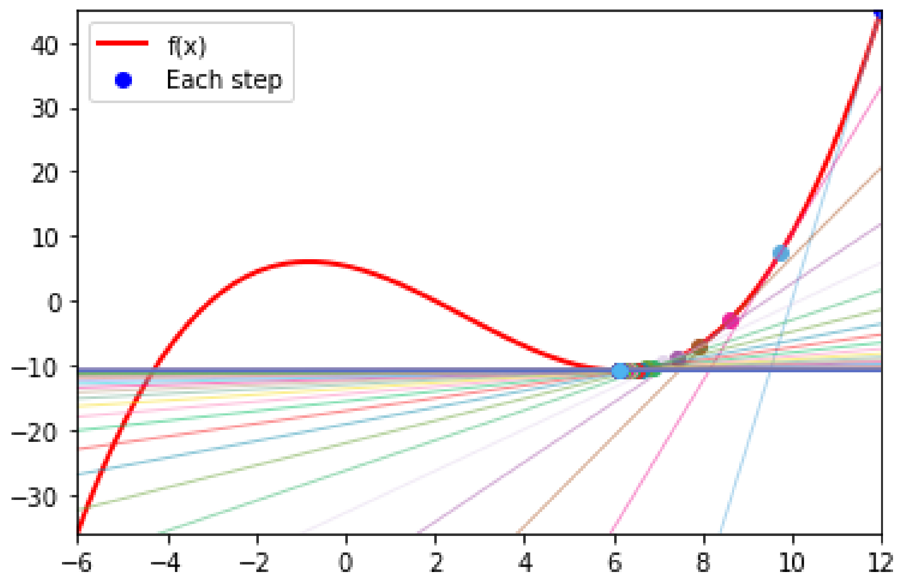
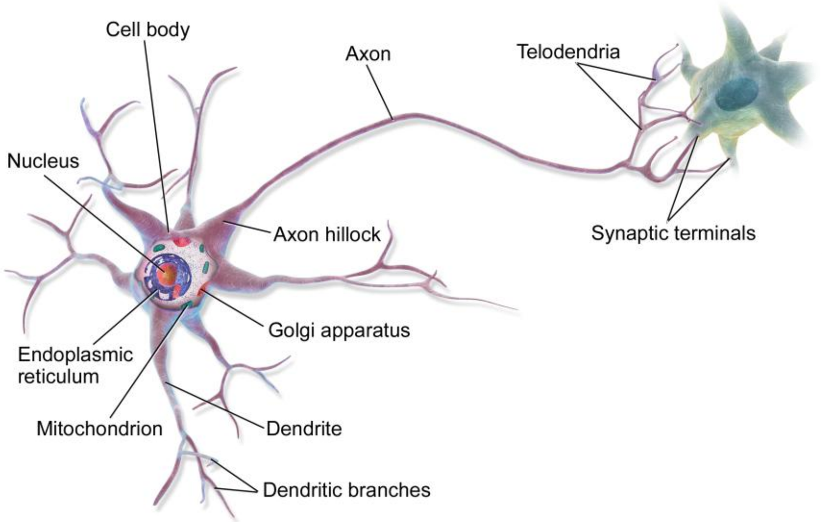


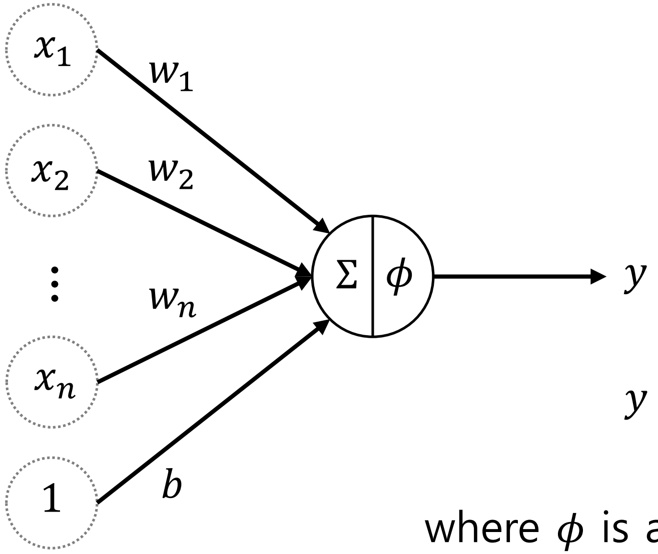
Table of Contents

Neural Network

- An *artificial neural network* (shortly *neural network*, *NN*) is a collection of perceptrons (a.k.a. artificial neurons) and their connection with weights.
  - Inspired by the biological neural networks.
- **Neuron vs. Perceptron**



vs.



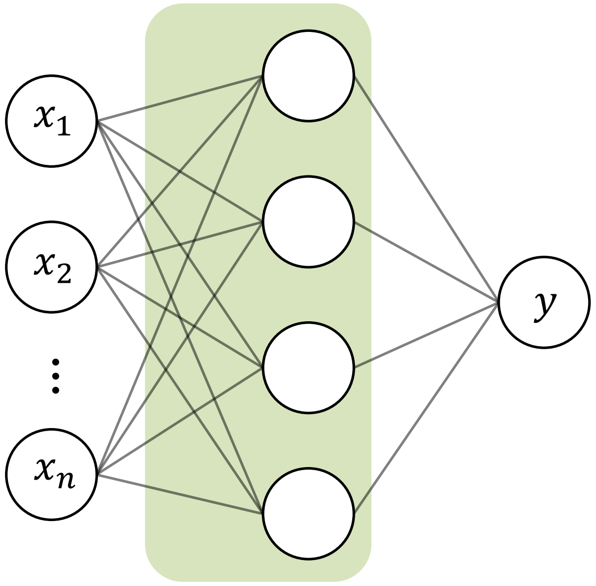
$$y = \phi\left(\sum_{i=1}^{n} w_i x_i + b\right)$$

where ϕ is an activation function (for nonlinearity) and b is bias.

Image: Wikipedia

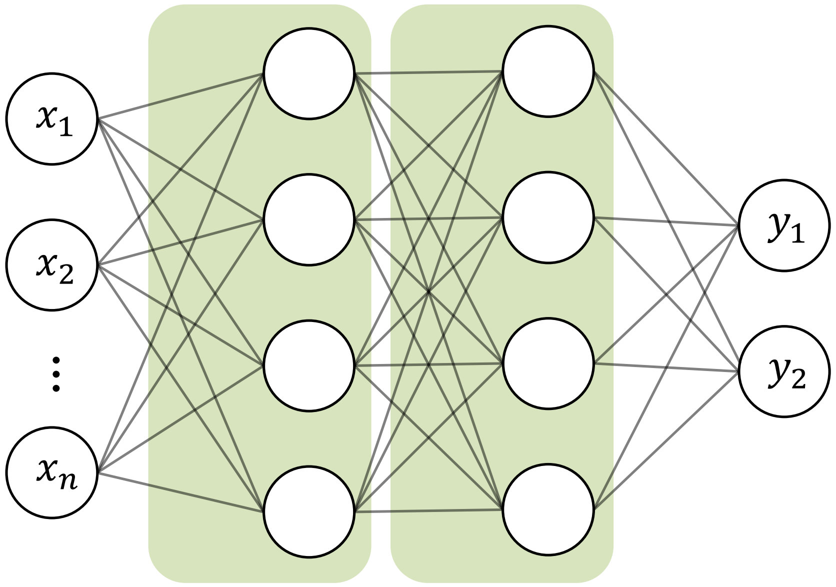
Neural Network

- **Multi-layer perceptrons (MLP)**
  - Note) Fully-connected (shortly FC) layer



2-layer NN

(1 x hidden layer, 1 x output)

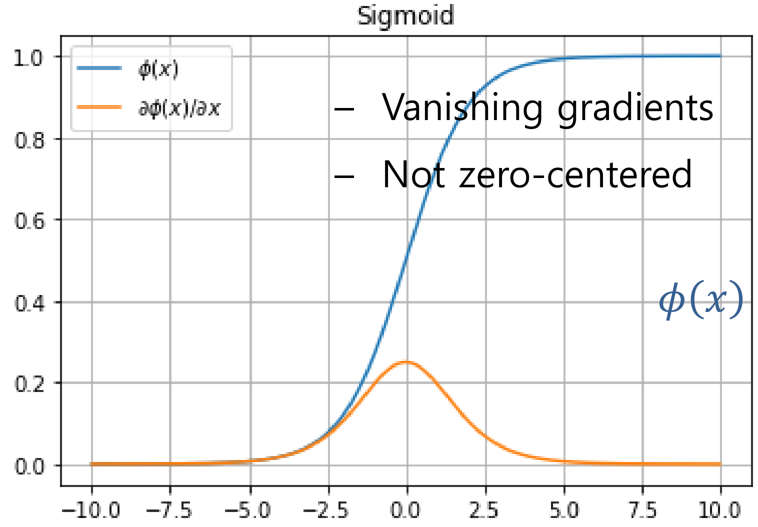


3-layer NN

(2 x hidden layer, 2 x output)

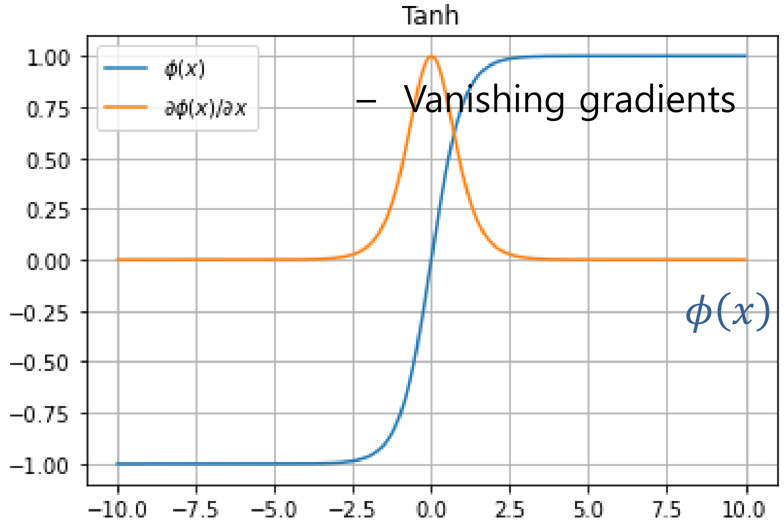
Activation Function

- **Activation function** imposes **nonlinearity** to a neural network.



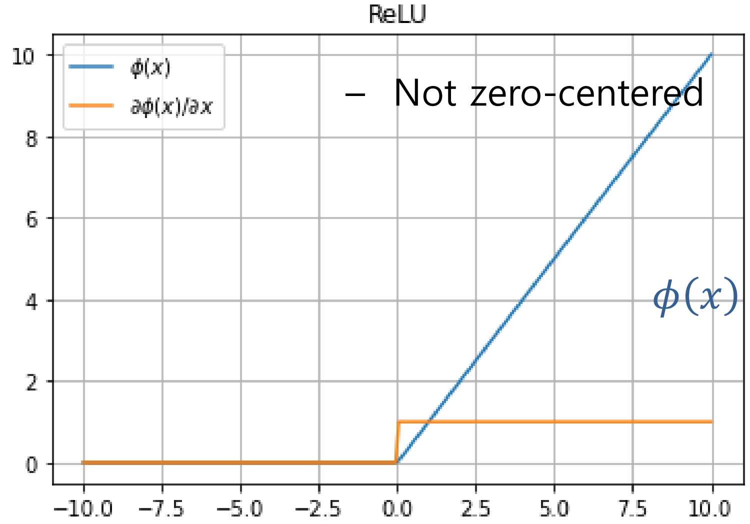
– Vanishing gradients
– Not zero-centered

$$\phi(x) = \frac{1}{1 + e^{-x}}$$



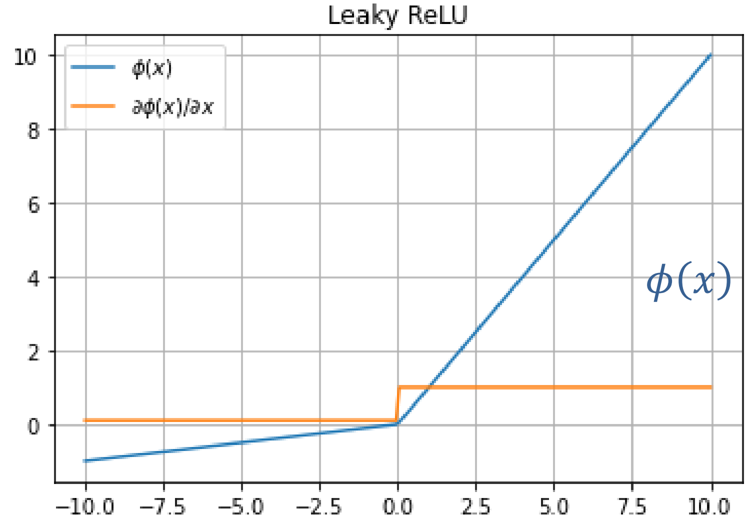
– Vanishing gradients

$$\phi(x) = \tanh x = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$



– Not zero-centered

$$\phi(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ x & \text{if } x > 0 \end{cases}$$



$$\phi(x) = \begin{cases} 0.01x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

Practice) Visualizing Activation Functions

- Visualize activation functions and their derivative functions
  - Note) PyTorch APIs for activation functions

```
import torch
import torch.nn as nn
import matplotlib.pyplot as plt

activation_funcs = [
    {'name': 'Sigmoid',    'func': nn.Sigmoid()},
    {'name': 'Tanh',       'func': nn.Tanh()},
    {'name': 'ReLU',       'func': nn.ReLU()},
    {'name': 'Leaky ReLU', 'func': nn.LeakyReLU(0.1)},
    {'name': 'ELU',        'func': nn.ELU()},
    # Try more activation functions
]

for act in activation_funcs:
    x = torch.linspace(-10, 10, 200, requires_grad=True)
    y = act['func'](x)
    y.sum().backward()

    plt.title(act['name'])
    x_np, y_np, grad = x.detach().numpy(), y.detach().numpy(), x.grad.numpy()
    plt.plot(x_np, y_np, label='$\phi(x)$')
    plt.plot(x_np, grad, label='$\partial \phi(x) / \partial x$')
    plt.grid()
    plt.legend()
    plt.show()
```

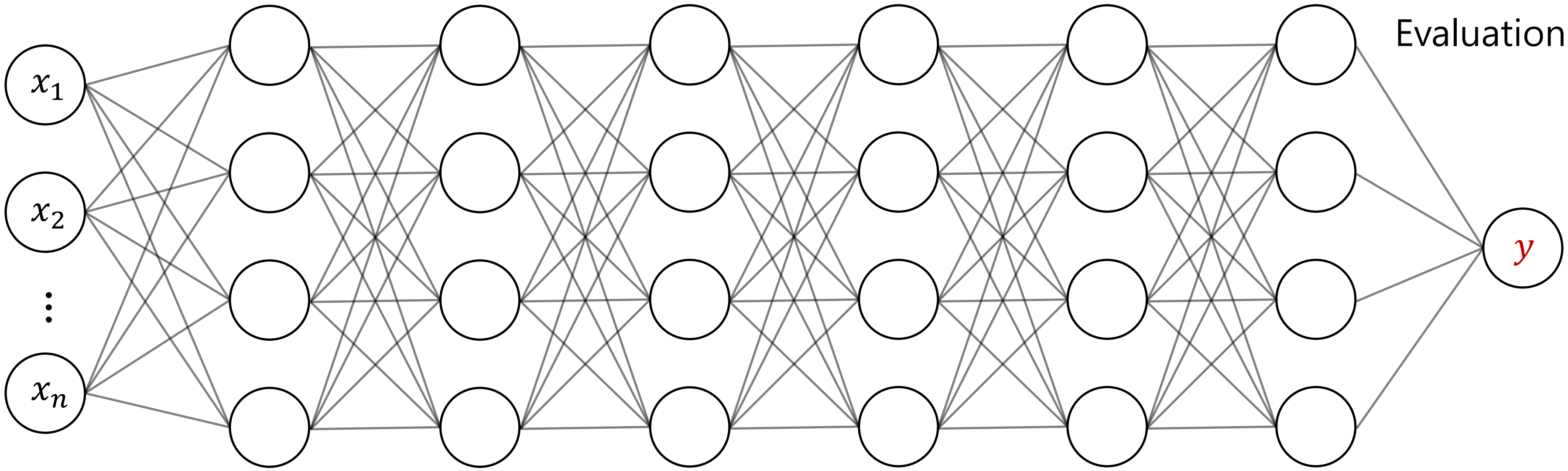
Backpropagation

- **Training a neural network (~ optimization)**

  Finding weight variables which minimize a cost function as

$$\boldsymbol{w}^{*} = \underset{\boldsymbol{w}}{\operatorname{argmin}} \frac{1}{N} \sum_{d=1}^{N} l(y^{d}, \hat{y}^{d})$$

where l is a loss function, N is the number of data, and $\hat{y}^{d}$ is the d-th target value.



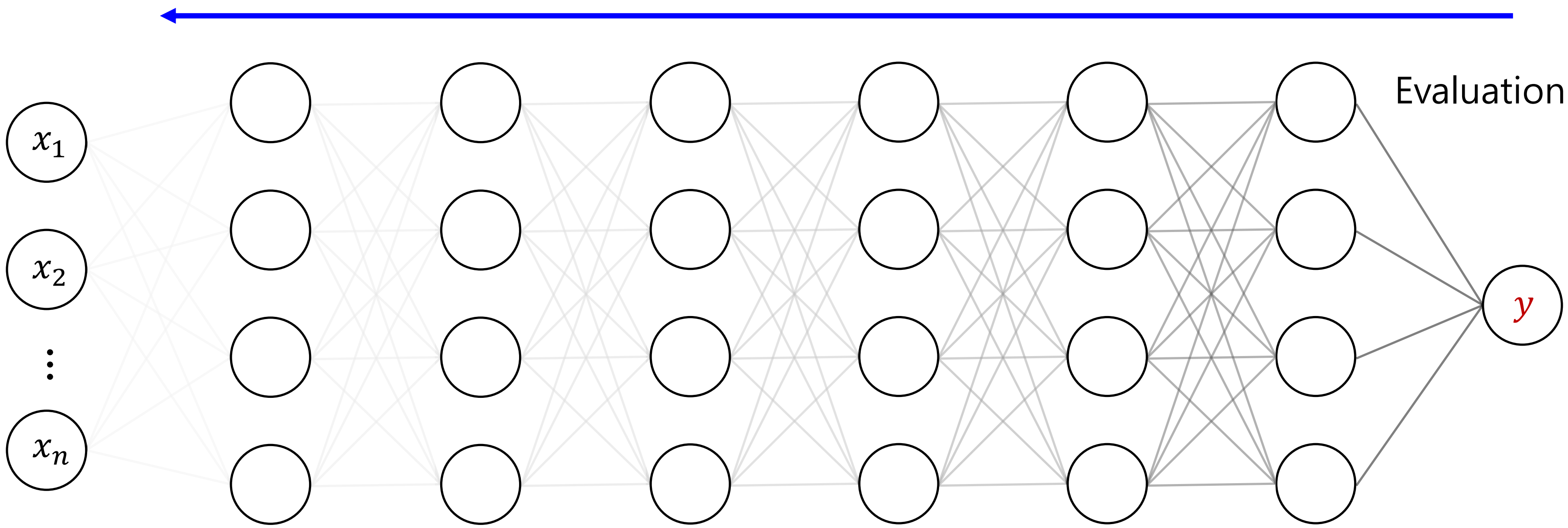
Evaluation with mean squared error

$$MSE = \frac{1}{N} \sum_{d=1}^{N} \left(\hat{y}^{d} - y^{d}\right)^{2}$$

where $l\left(y^{d}, \hat{y}^{d}\right) = \left(\hat{y}^{d} - y^{d}\right)^{2}$

Backpropagation

- **Backpropagation** is an algorithm for training a neural network.
  - It tries to find the optimal **weight variables** of a NN by **gradient descent**
- **Vanishing gradient problem**
  - During backpropagation, gradient values of a deep NN become close to 0.



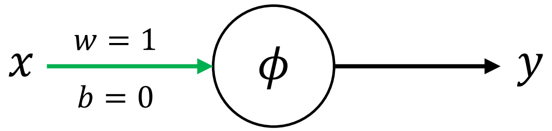
Evaluation with mean squared error

$$MSE = \frac{1}{N}\sum_{d=1}^{N}\left(\hat{y}^d - y^d\right)^2$$

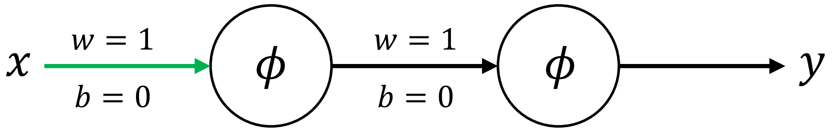
where $l\left(y^d, \hat{y}^d\right) = \left(\hat{y}^d - y^d\right)^2$

Practice) Observing Vanishing Gradients

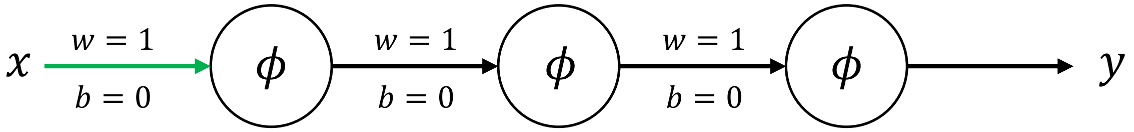
- Gradients of a single-node multi-layer NN
  - 1-layer forward: $y = \phi(wx + b) = \phi(x)$ **if $w = 1$ and $b = 0$**
  - 1-layer backward: $\frac{\partial y}{\partial x} = \phi'(x)$



  - 2-layer forward: $y = \phi(\phi(x))$
  - 2-layer backward: $\frac{\partial y}{\partial x} = \phi'\big(\phi(x)\big)\phi'(x)$ **$\because$ chain rule**



  - 3-layer forward: $y = \phi(\phi(\phi(x)))$
  - 3-layer backward: $\frac{\partial y}{\partial x} = \phi'\Big(\phi\big(\phi(x)\big)\Big)\phi'\big(\phi(x)\big)\phi'(x)$

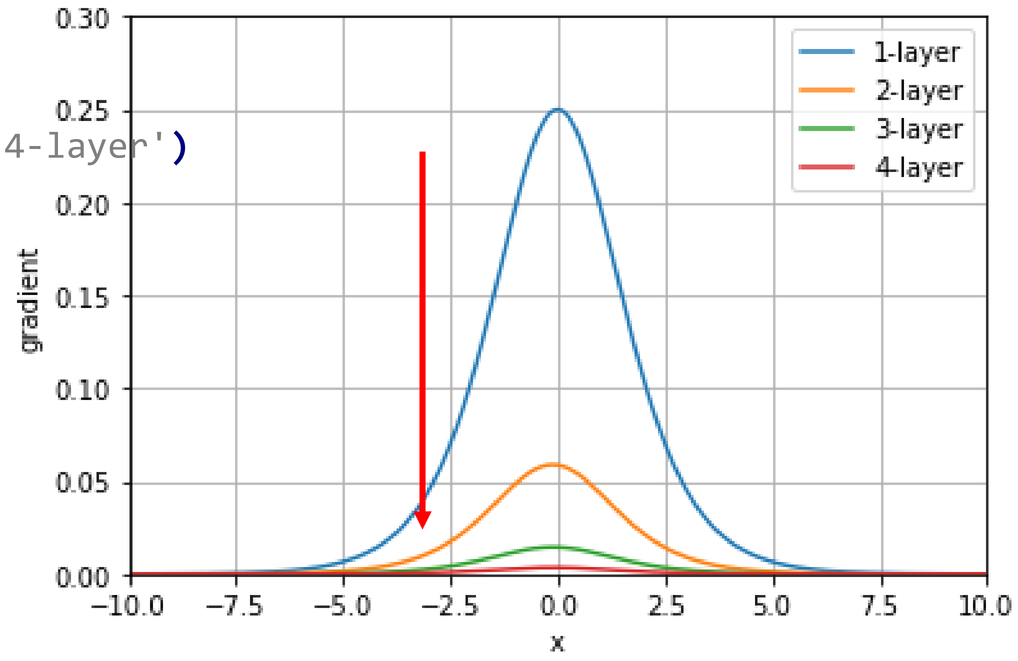


  - …

Practice) Observing Vanishing Gradients

- Gradients of a single-node multi-layer NN with the **sigmoid** function
  - Note) $\phi'(x) = \phi(x)\big(1 - \phi(x)\big)$

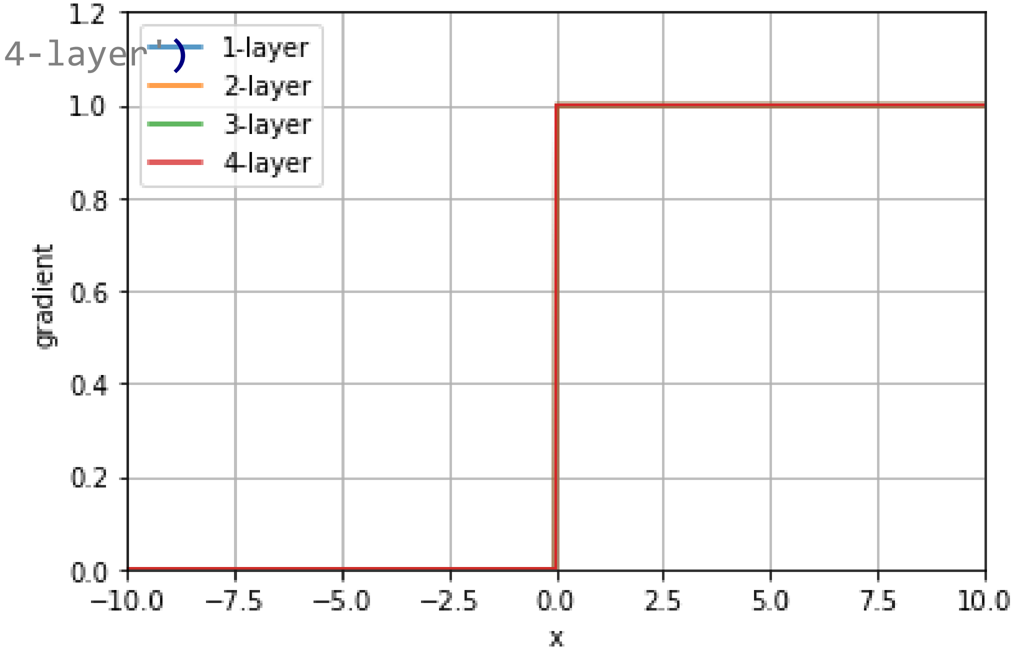
```
import numpy as np
import matplotlib.pyplot as plt

f  = lambda x: 1 / (1 + np.exp(-x))
df = lambda x: f(x) * (1 - f(x))

x = np.linspace(-10, 10, 1000)

plt.plot(x, df(x), label='1-layer')
plt.plot(x, df(f(x))*df(x), label='2-layer')
plt.plot(x, df(f(f(x)))*df(f(x))*df(x), label='3-layer')
plt.plot(x, df(f(f(f(x))))*df(f(f(x)))*df(f(x))*df(x), label='4-layer')
plt.axis((-10, 10, 0, 0.3))
plt.grid()
plt.legend()
plt.show()
```



Practice) Observing Vanishing Gradients

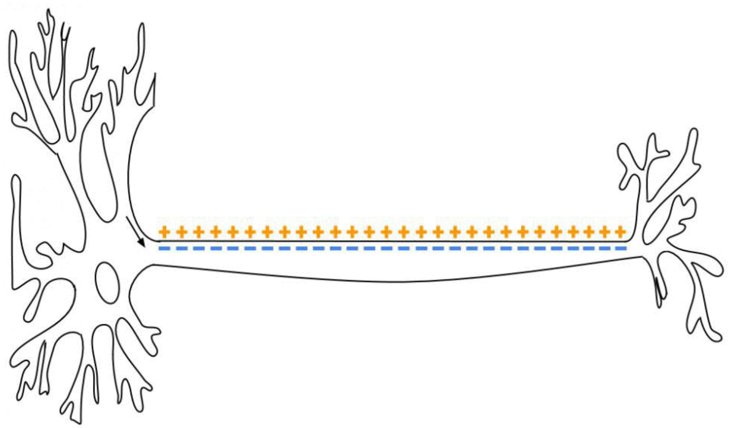
- Gradients of a single-node multi-layer NN with the **ReLU** function

```
import numpy as np
import matplotlib.pyplot as plt

f  = lambda x: x * (x >= 0)
df = lambda x: x >= 0

x = np.linspace(-10, 10, 1000)

plt.plot(x, df(x), label='1-layer')
plt.plot(x, df(f(x))*df(x), label='2-layer')
plt.plot(x, df(f(f(x)))*df(f(x))*df(x), label='3-layer')
plt.plot(x, df(f(f(f(x))))*df(f(f(x)))*df(f(x))*df(x), label='4-layer')
plt.axis((-10, 10, 0, 0.3))
plt.grid()
plt.legend()
plt.show()
```



Activation Function (Revisited)

- **Why ReLU?**
  - It can mitigate the vanishing gradient problem.
  - In addition, there is a biological analogue such as neural action potential.



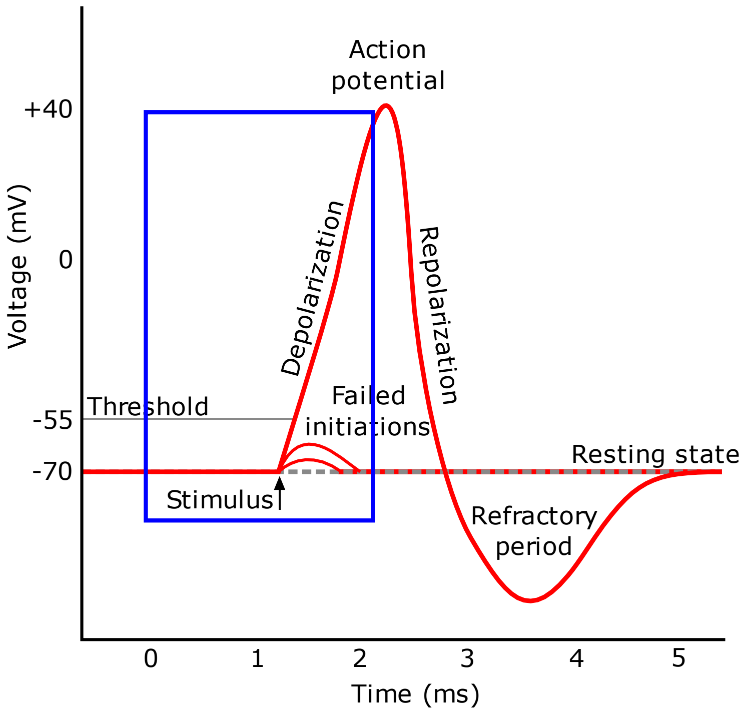


Image: Wikipedia

Loss Function

- **Training a neural network (~ optimization)**

  Finding weight variables which minimize a cost function as

$$\boldsymbol{w}^* = \operatorname*{argmin}_{\boldsymbol{w}} \frac{1}{N} \sum_{d=1}^{N} l(y^d, \hat{y}^d)$$

  where l is a loss function, N is the number of data, and $\hat{y}^d$ is the d-th target value.

- **Loss function**
  - A *loss function* quantifies the gap between the ground truth and prediction.
  - e.g. **Mean squared error** (usually for *regression*)

$$l(y, \hat{y}) = (\hat{y} - y)^2$$

    where $\hat{y}$ is the ground truth, y is prediction, and N is the number of data.

  - e.g. **Binary cross entropy error** (usually for *binary classification*)

$$l(y, \hat{y}) = -\hat{y} \log y - (1 - \hat{y}) \log(1 - y)$$

    In general, $-\sum \hat{y}_i \log y_i$ for *multi-class classification*

Loss Function

- **Loss function**
  - e.g. **Cross entropy error** (usually for *multi-class classification*)

$$l(\mathbf{y}, \hat{\mathbf{y}}) = -\sum_{i=1}^{C} \hat{y}_i \log y_i$$

where $\hat{y}_i$ is the one-hot-encoded truth, y_i is the predicted (softmax) confidence

	$\hat{y}_1$	$\hat{y}_2$	$\hat{y}_3$
$\hat{\boldsymbol{y}}$:	0	1	0

	y_1	y_2	y_3
$\boldsymbol{y}$:	0.012	0.929	0.058

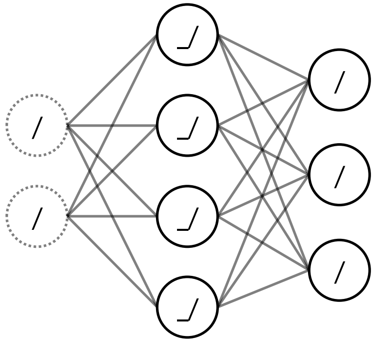
Softmax: $y_i = \exp z_i \,/\, \sum_{j=1}^{C} \exp z_j$

NN output $\boldsymbol{z}$	-1.03	3.29	0.52

  - Note) PyTorch APIs for loss functions

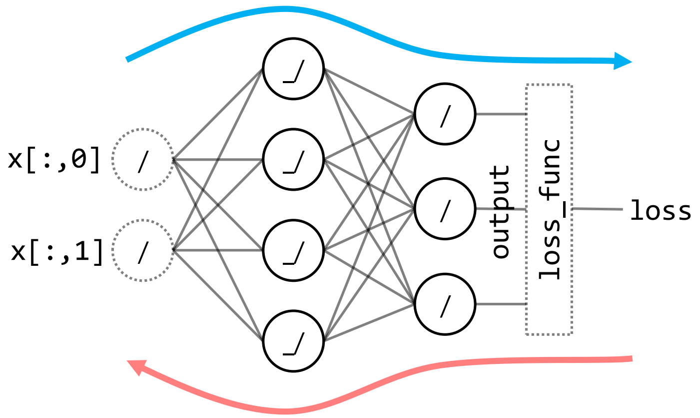
Practice) Iris Flower Classification (1/3)

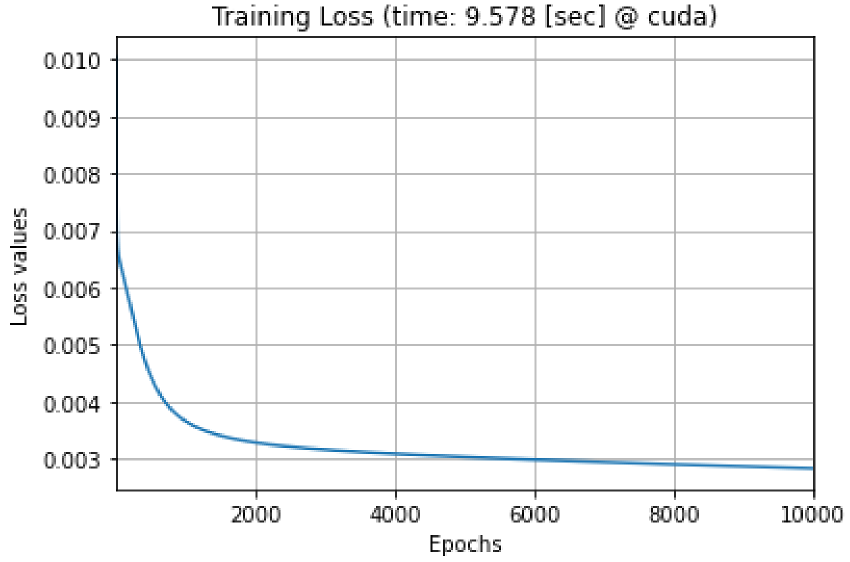
```
import torch
import torch.nn as nn
import numpy as np
import matplotlib.pyplot as plt
from sklearn import (datasets, metrics)
from matplotlib.colors import ListedColormap
import time

# 1.1. Load a dataset partially
iris = datasets.load_iris()
iris.data = iris.data[:,0:2]
iris.feature_names = iris.feature_names[0:2]
iris.color = np.array([(1, 0, 0), (0, 1, 0), (0, 0, 1)])

# 1.2. Load the dataset as tensors
dev_name = 'cuda' if torch.cuda.is_available() else 'cpu' # Try 'cpu'
x = torch.tensor(iris.data,   device=dev_name).float()
y = torch.tensor(iris.target, device=dev_name).long()

# 2. Define a model
# - Try the different number of hidden layers
# - Try less or more layers with different transfer functions
input_size, output_size = len(iris.feature_names), len(iris.target_names)
model = nn.Sequential(
    nn.Linear(input_size, 4),
    nn.ReLU(),
    nn.Linear(4, output_size),
).to(dev_name)
```



Practice) Iris Flower Classification (2/3)

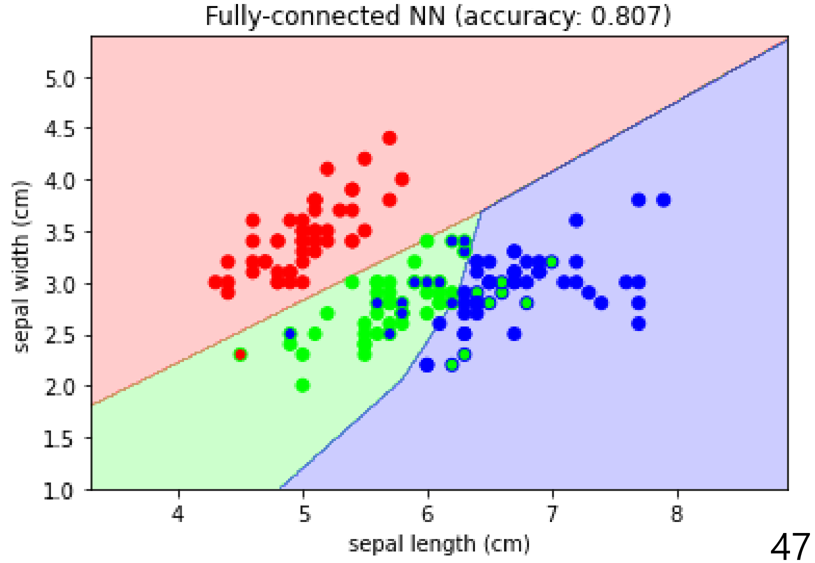
```python
# 3. Train the model
loss_func = nn.CrossEntropyLoss()
optimizer = torch.optim.SGD(model.parameters(), lr=0.01) # Try other optimizers
epoch_max = 10000
loss_list = []
start = time.time()
for i in range(epoch_max):
    # Train one iteration
    optimizer.zero_grad()
    output = model(x)
    loss = loss_func(output, y)
    loss.backward()
    optimizer.step()

    # Record the loss
    loss_list.append(loss / len(x))
elapse = time.time() - start

# 4.1. Visualize the training loss curve
plt.title(f'Training Loss (time: {elapse/60:.2f} [min] @ {dev_name})')
plt.plot(range(1, epochs + 1), loss_list)
plt.xlabel('Epochs')
plt.ylabel('Loss values')
plt.xlim((1, epochs))
plt.grid()
plt.show()
```





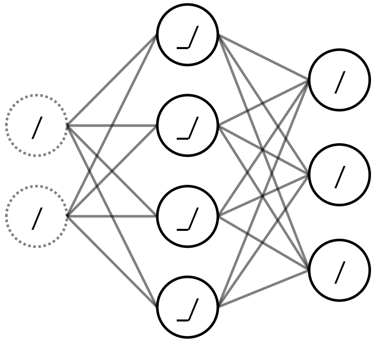
Practice) Iris Flower Classification (3/3)

```
# 4.2. Visualize training results (decision boundaries)
x_min, x_max = iris.data[:, 0].min() - 1, iris.data[:, 0].max() + 1
y_min, y_max = iris.data[:, 1].min() - 1, iris.data[:, 1].max() + 1
xx, yy = np.meshgrid(np.arange(x_min, x_max, 0.01), np.arange(y_min, y_max, 0.01))
xy = np.vstack((xx.flatten(), yy.flatten())).T
xy_tensor = torch.from_numpy(xy).float().to(dev_name)
zz = torch.argmax(model(xy_tensor), dim=1).cpu().detach().numpy()
plt.contourf(xx, yy, zz.reshape(xx.shape), cmap=ListedColormap(iris.color), alpha=0.2)

# 4.3. Visualize data with their classification
predict = torch.argmax(model(x), dim=1).cpu().detach().numpy()
accuracy = metrics.balanced_accuracy_score(iris.target, predict)
plt.title(f'Fully-connected NN (accuracy: {accuracy:.3f})')
plt.scatter(iris.data[:,0], iris.data[:,1], c=iris.color[iris.target], edgecolors=iris.color[predict])
plt.xlabel(iris.feature_names[0])
plt.ylabel(iris.feature_names[1])
plt.show()
```



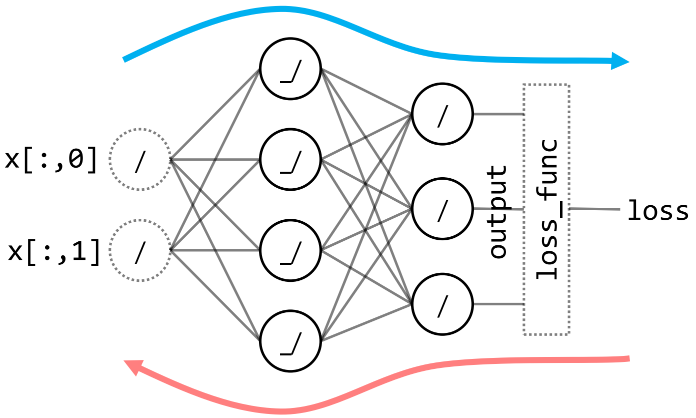
Practice) Iris Flower Classification – My Style (1/5)

```python
import torch
import torch.nn as nn
import torch.nn.functional as F
import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets
from matplotlib.colors import ListedColormap
import time

# Define hyperparameters
EPOCH_MAX = 10000
EPOCH_LOG = 1000
OPTIMIZER_PARAM = {'lr': 0.01}
DATA_LOADER_PARAM = { 'batch_size': 50, 'shuffle': True }
USE_CUDA = torch.cuda.is_available()
RANDOM_SEED = 777

# A two-layer NN model
class MyDNN(nn.Module):
    def __init__(self, input_size=2, output_size=3):
        super(MyDNN, self).__init__()
        self.fc1 = nn.Linear(input_size, 4)
        self.fc2 = nn.Linear(4, output_size)

        nn.init.xavier_uniform_(self.fc1.weight)
        nn.init.xavier_uniform_(self.fc2.weight)

    def forward(self, x):
        x = F.relu(self.fc1(x))
        x = self.fc2(x)
        return x
```



Practice) Iris Flower Classification – My Style (2/5)

```
# Train a model with the given batches
def train(model, batch_data, loss_func, optimizer):
    model.train() # Notify layers (e.g. DropOut, BatchNorm) that it’s now training
    train_loss, n_data = 0, 0
    dev = next(model.parameters()).device
    for batch_idx, (x, y) in enumerate(batch_data):
        x, y = x.to(dev), y.to(dev)
        optimizer.zero_grad()
        output = model(x)
        loss = loss_func(output, y)
        loss.backward()
        optimizer.step()

        train_loss += loss.item()
        n_data += len(y)
    return train_loss / n_data

# Evaluate a model with the given batches
def evaluate(model, batch_data, loss_func):
    model.eval() # Notify layers (e.g. DropOut, BatchNorm) that it’s now testing
    test_loss, n_correct, n_data = 0, 0, 0
    with torch.no_grad():
        dev = next(model.parameters()).device
        for x, y in batch_data:
            x, y = x.to(dev), y.to(dev)
            output = model(x)
            loss = loss_func(output, y)
            y_pred = torch.argmax(output, dim=1)

            test_loss += loss.item()
            n_correct += (y == y_pred).sum().item()
            n_data += len(y)
    return test_loss / n_data, n_correct / n_data
```



Practice) Iris Flower Classification – My Style (3/5)

```
if __name__ == '__main__':
    # 0. Preparation
    torch.manual_seed(RANDOM_SEED)
    if USE_CUDA:
        torch.cuda.manual_seed_all(RANDOM_SEED)
    dev = torch.device('cuda' if USE_CUDA else 'cpu')

    # 1.1. Load the Iris dataset partially
    iris = datasets.load_iris()
    iris.data = iris.data[:,0:2]
    iris.feature_names = iris.feature_names[0:2]
    iris.color = np.array([(1, 0, 0), (0, 1, 0), (0, 0, 1)])

    # 1.2. Wrap the dataset with torch.utils.data.DataLoader
    x = torch.tensor(iris.data, dtype=torch.float32, device=dev)
    y = torch.tensor(iris.target, dtype=torch.long, device=dev)
    data_train = torch.utils.data.TensorDataset(x, y)
    loader_train = torch.utils.data.DataLoader(data_train, **DATA_LOADER_PARAM)
```

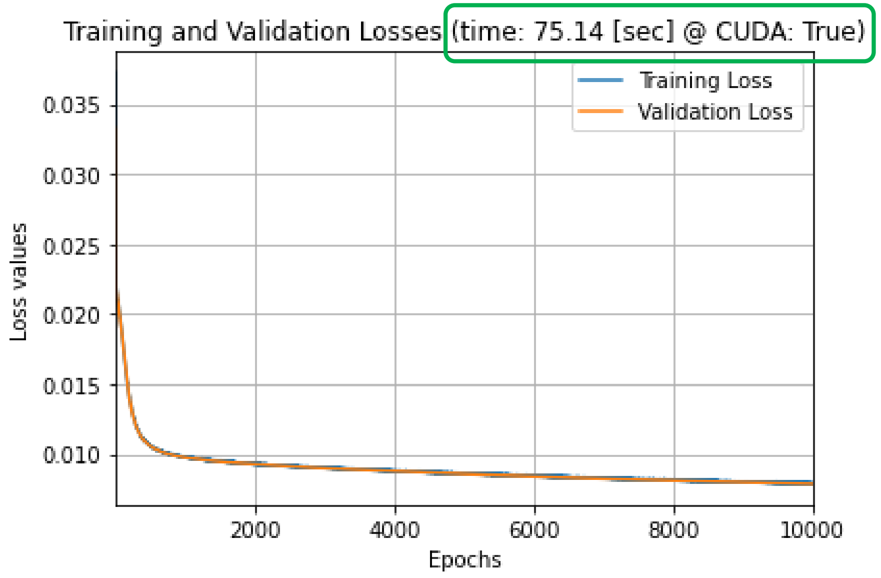
Practice) Iris Flower Classification – My Style (4/5)

```
# 2. Instantiate a model, loss function, and optimizer
model = MyDNN().to(dev)
loss_func = F.cross_entropy
optimizer = torch.optim.SGD(model.parameters(), **OPTIMIZER_PARAM)

# 3. Train the model
loss_list = []
start = time.time()
for epoch in range(1, EPOCH_MAX + 1):
    train_loss = train(model, loader_train, loss_func, optimizer)
    valid_loss, valid_accuracy = evaluate(model, loader_train, loss_func)

    loss_list.append([epoch, train_loss, valid_loss, valid_accuracy])
    if epoch % EPOCH_LOG == 0:
        elapse = time.time() - start
        print(f'{epoch:>6} ({elapse:>6.2f} sec), TrLoss={train_loss:.6f}, VaLoss={valid_loss:.6f}, VaAcc={valid_accuracy:.3f}')
elapse = time.time() - start

# 4.1. Visualize the loss curves
# 4.2. Visualize training results (decision boundaries)
# 4.3. Visualize data with their classification
```

Practice) Iris Flower Classification – My Style (5/5)

```
 1000 (  7.71 sec), TrLoss=0.009773, VaLoss=0.009769, VaAcc=0.707
 2000 ( 15.17 sec), TrLoss=0.009300, VaLoss=0.009281, VaAcc=0.747
 3000 ( 22.85 sec), TrLoss=0.009022, VaLoss=0.009007, VaAcc=0.780
 4000 ( 30.41 sec), TrLoss=0.008800, VaLoss=0.008786, VaAcc=0.800
 5000 ( 37.85 sec), TrLoss=0.008606, VaLoss=0.008594, VaAcc=0.813
 6000 ( 45.49 sec), TrLoss=0.008433, VaLoss=0.008420, VaAcc=0.813
 7000 ( 53.06 sec), TrLoss=0.008274, VaLoss=0.008262, VaAcc=0.813
 8000 ( 60.31 sec), TrLoss=0.008138, VaLoss=0.008118, VaAcc=0.807
 9000 ( 67.91 sec), TrLoss=0.008008, VaLoss=0.007990, VaAcc=0.800
10000 ( 75.14 sec), TrLoss=0.007928, VaLoss=0.007880, VaAcc=0.800
```

Discussion) CPU vs. GPU
Data as raw tensors vs. Data as `DataLoader`



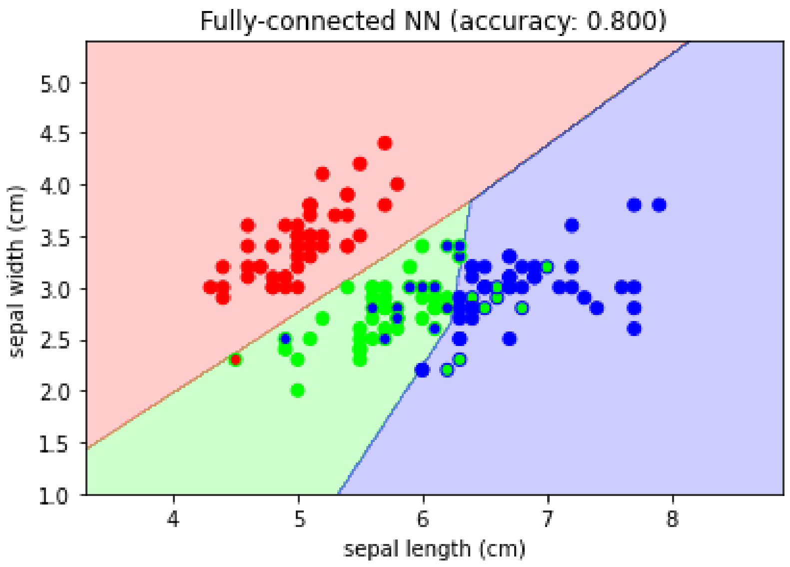


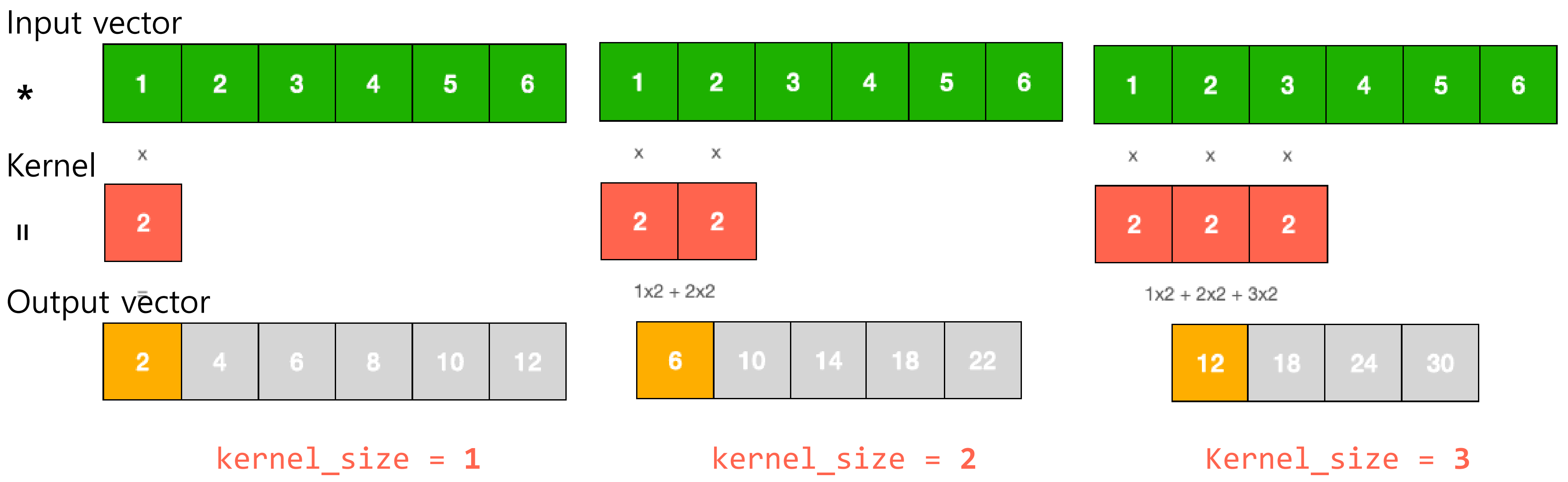
Table of Contents

Convolution

- **Discrete convolution**

$$(f * g)(k) = \sum_{i=-\infty}^{\infty} f(i)g(k-i)$$

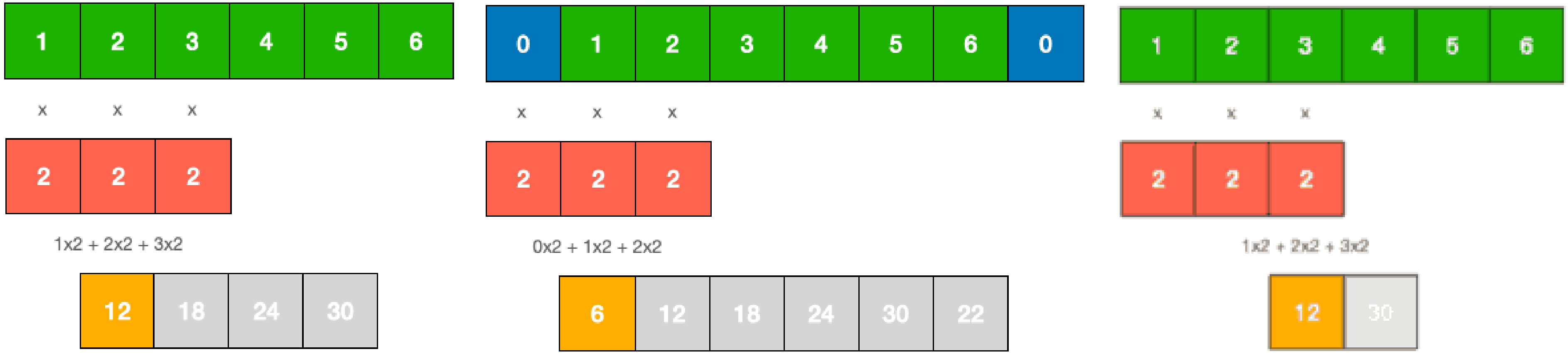
- 1D discrete convolution (e.g. voice)



Animation: [Towards Data Science article (by Jingles)]

Convolution

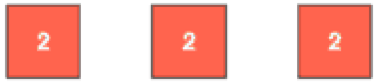
- 1D discrete convolution (continued)




```
kernel_size = 3
padding = 0
stride = 1
dilation = 1
```

```
kernel_size = 3
padding = 1
stride = 1
dilation = 1
```

```
kernel_size = 3
padding = 0
stride = 3
dilation = 1
```




```
dilation = 2
```

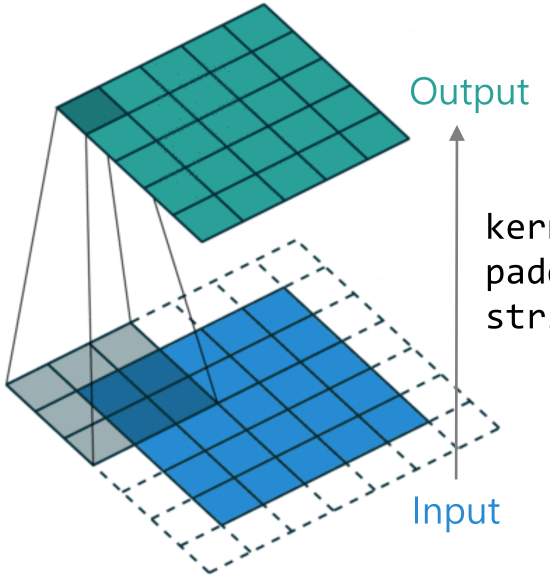
- The size of output vectors

$$\text{output\_size} = \left\lfloor \frac{\text{input\_size} + 2 \times \text{padding} - (\text{dilation} \times (\text{kernel}_{\text{size}} - 1) + 1)}{\text{stride}} + 1 \right\rfloor$$

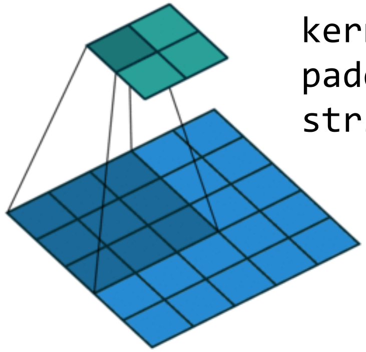
Animation: [Towards Data Science article (by Jingles)]

Convolution

- 2D discrete convolution (e.g. image)

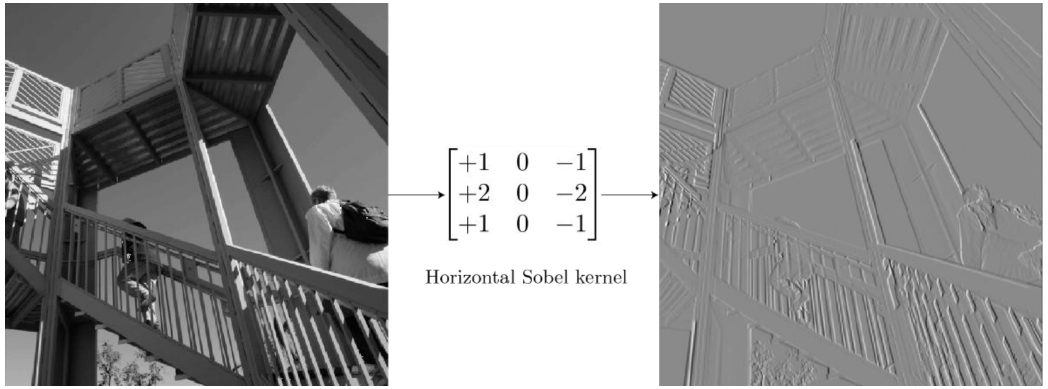



```
kernel_size = 3
padding = 1
stride = 1
```



```
kernel_size = 3
padding = 0
stride = 2
```

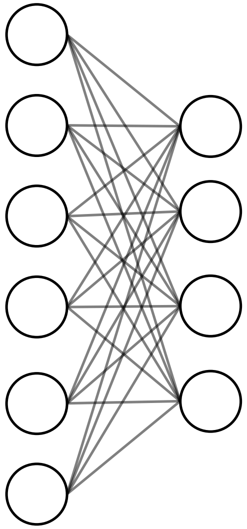
  – A kernel and its output are also called a *filter* and *feature map* (or *activation map*), respectively.
    - CNN visualization examples: ConvNetJS, ~~CNN Explainer~~
    - e.g. When a kernel is a horizontal Sobel filter,

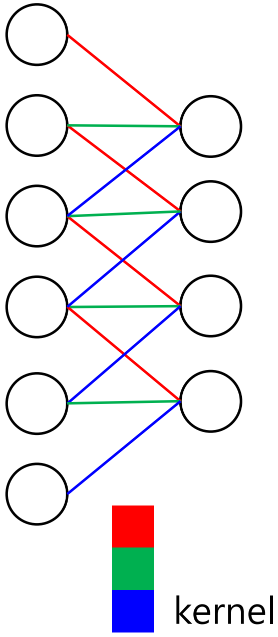


Convolutional Layer

- A **convolutional layer** (shortly *conv layer*) is a NN layer which uses *convolution* as its <u>feedforward propagation</u> and *kernel* as <u>weight variables</u>.
- (In contrast to a FC layer) A convolutional layer has two important points, **weight sharing** and **local connectivity**.
  - e.g. `input_size = 6`, `output_size = 4`, `kernel_size = 3`

    <u>FC layer</u> has 24 weight variables, but <u>conv layer</u> has only 3 weight variables.

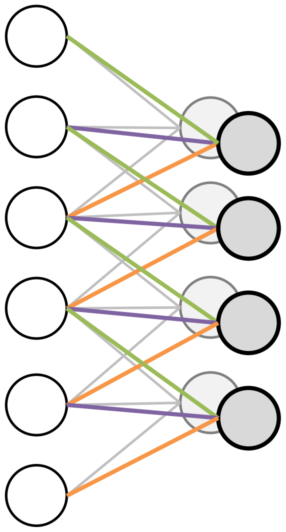




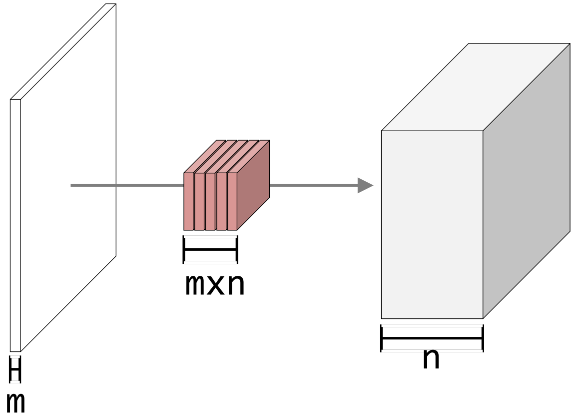
Why it works?

Convolutional Layer

- A convolutional layer can have multiple kernels so that its output will be multiple *channels*.
  - e.g. `in_channel_size = 1`, `out_channel_size = 2`, `kernel_size = 3`
  - e.g. `in_channel_size = m`, `out_channel_size = n`, `kernel_size = 3`



# of weights (w/o bias): 3 x 2

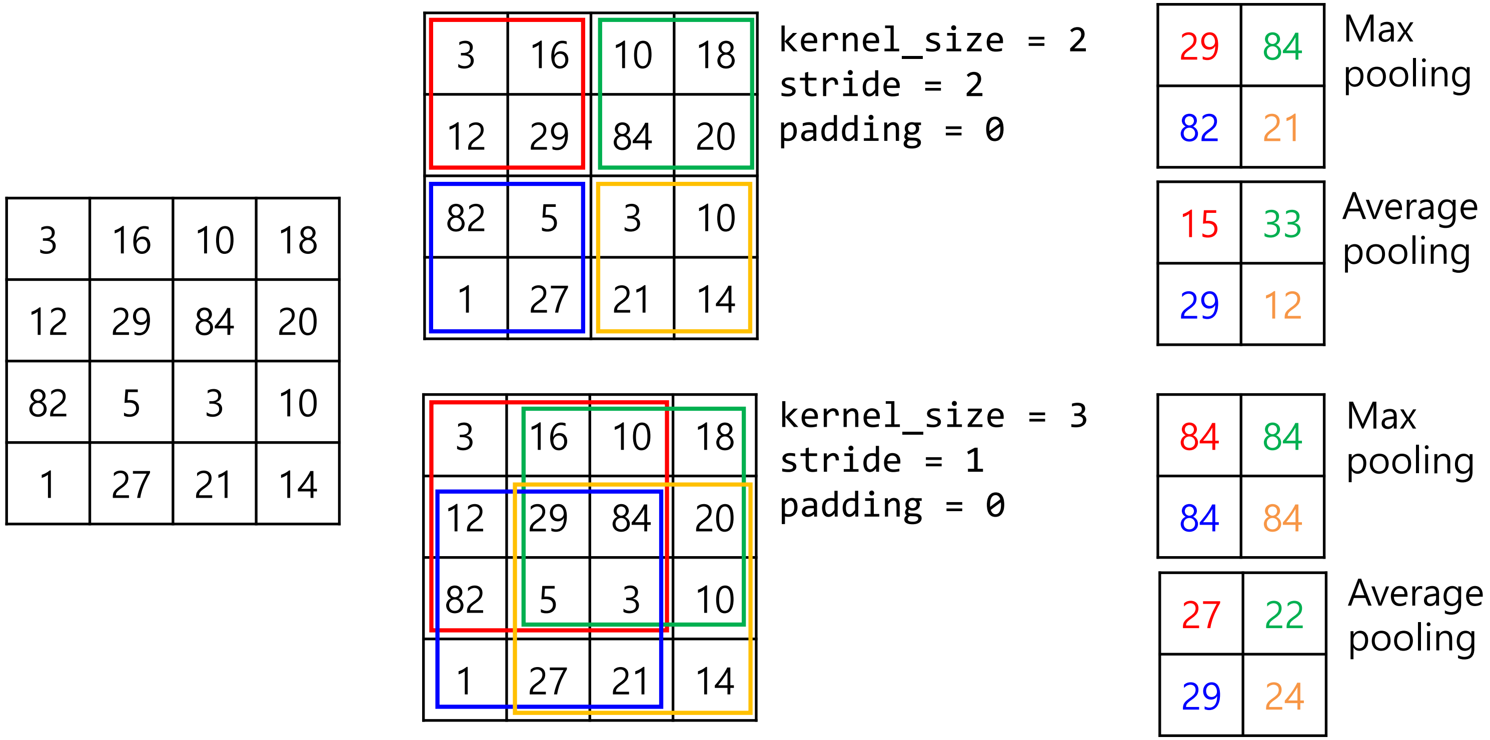


# of weights (w/o bias): 3 x 3 x m x n

  - Note) The number of weights is independent from input and output size in a convolutional layer.
    - Note) The number of weight (w/o bias) in a FC layer = input size x output size
      - e.g. The left example: 6 x 8 = 48

Pooling Layer

- A **pooling layer** performs a non-linear down-sampling operation.
- Especially max pooling is common.
  - Note) [PyTorch APIs for pooling layers]
- Its parameters and working is similar to a convolutional layer.
  - In PyTorch, `stride` is assigned as `kernel_size` if it is not given.



Pooling Layer

- Why is pooling important?
  - It reduces the network size. → Less time/space complexity
  - It provide larger receptive fields with a limited size of kernels.

bag

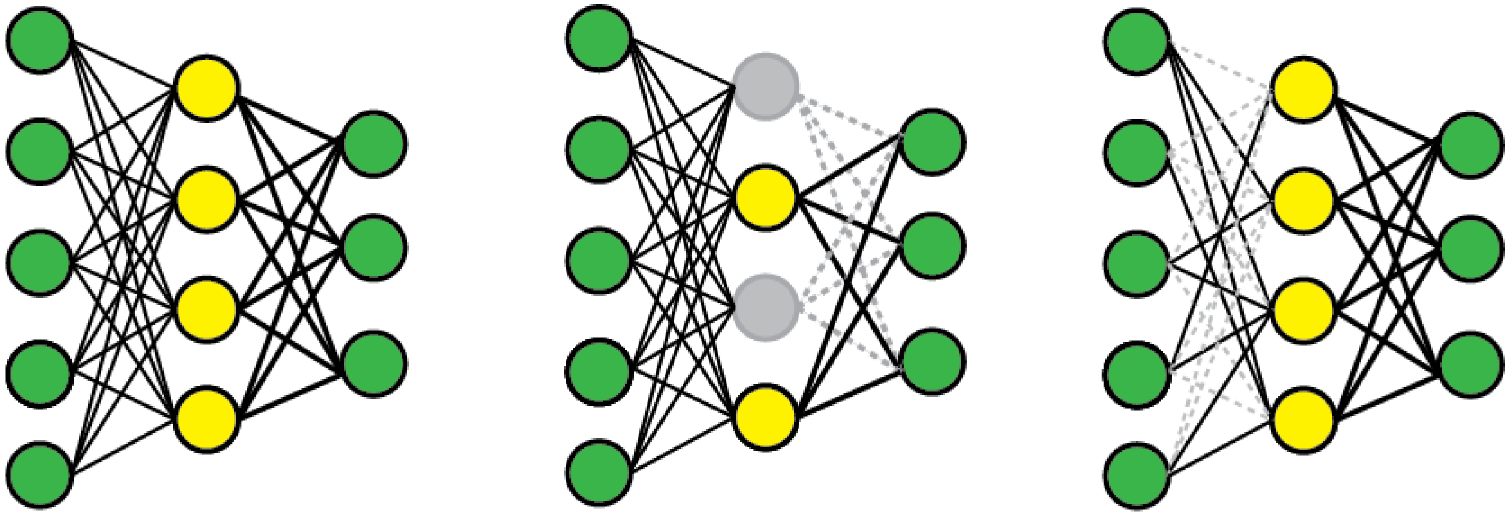
human upper body

human full body

bridge

Dropout

- **Dropout** is one of regularization methods against overfitting.
- It randomly drops a ratio of hidden units during training.
  - It prevents hidden units from co-adaptation.
  - e.g. When some hidden units has high weights, the others in the same layers have little contribution (low weight values, rarely training) to their output.



Standard   Dropout   DropConnect

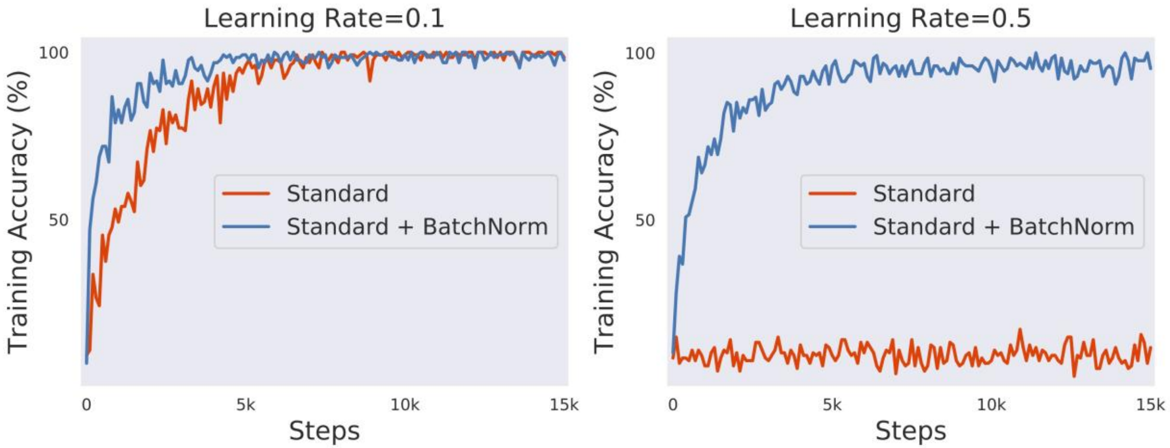
Image: Stack Exchange comment (by Matt Krause)

Batch Normalization

- **Batch normalization (BatchNorm; BN)** is a technique for making neural network training more stable and smooth.
- It normalizes activations using the mean and variance of a mini-batch, and then learns how to scale and shift them. (N: batch size)

$$\mathrm{BN}(x_{n,c,h,w}) = \gamma_c \cdot \left(\frac{x_{n,c,h,w} - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}}\right) + \beta_c \quad \text{where} \quad \mu_c = \frac{1}{HWN}\sum_{n=1}^{N}\sum_{h=1}^{H}\sum_{w=1}^{W} x_{n,c,h,w} \quad \text{and} \quad \sigma_c^2 = \frac{1}{HWN}\sum_{i=1}^{N}\sum_{h=1}^{H}\sum_{w=1}^{W}(x_{n,c,h,w} - \mu_c)^2$$

  – It often allows the network to use larger learning rates and converge faster.



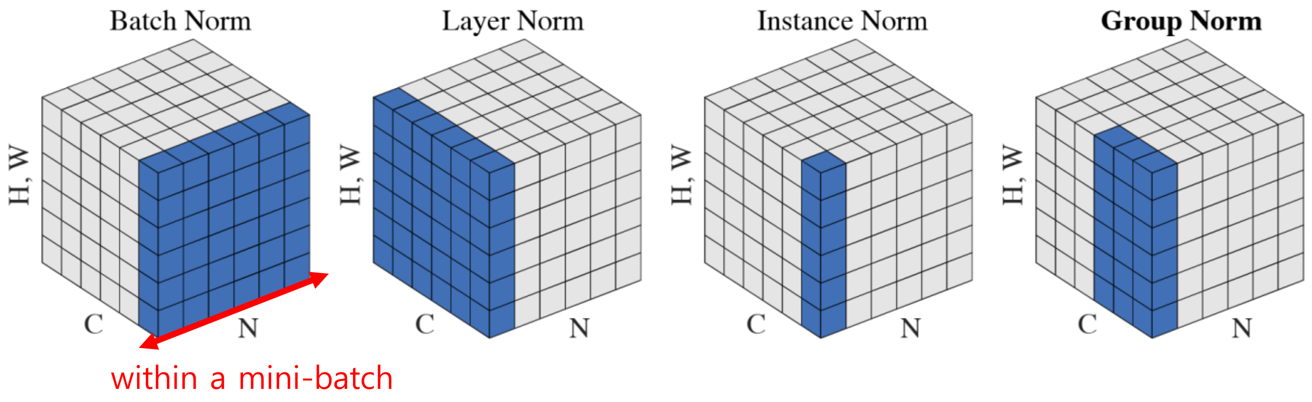
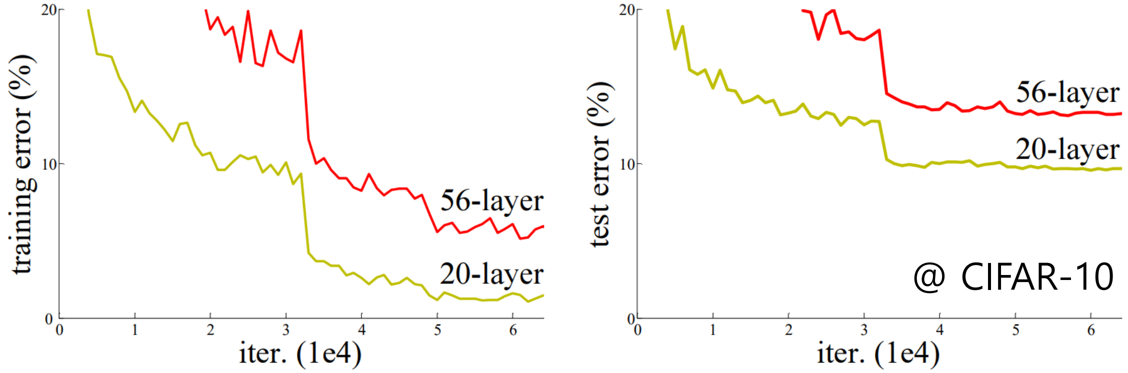


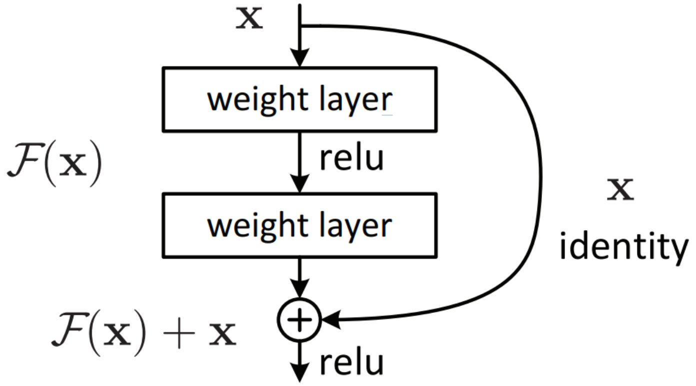
Image: gradient science / Wu and He, “Group Normalization”, ECCV, 2018 DOI arXiv

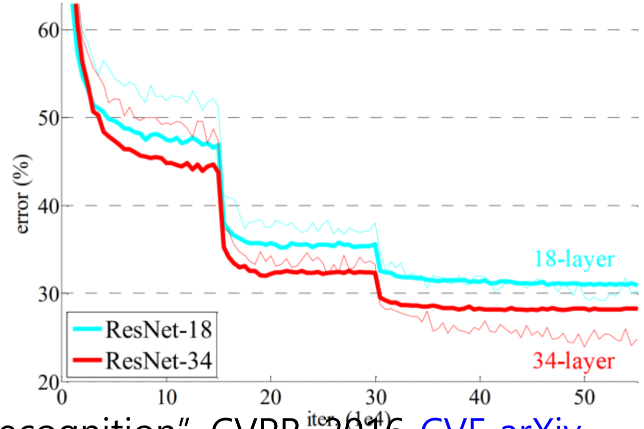
Skip Connection

- Motivation) Why is a deeper network worse?



- A **skip connection** is a shortcut connection that bypasses several layers which contain nonlinearities (e.g. ReLU) and batch normalization.
  - Alias: *Residual connection* (used in *residual neural network*, *ResNet*)
  - It resolves the vanishing gradient problem.





gradient highway

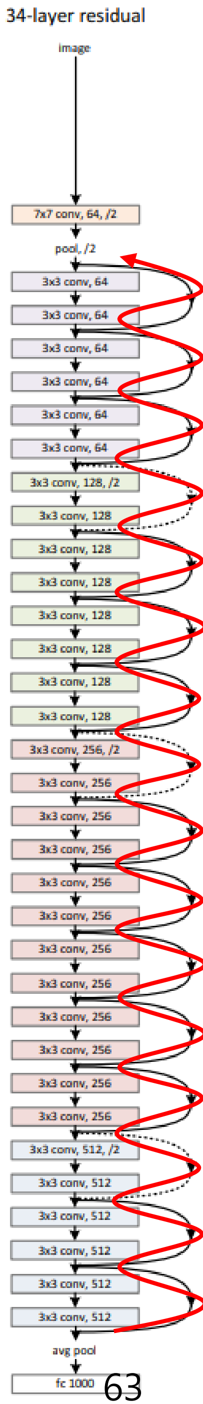
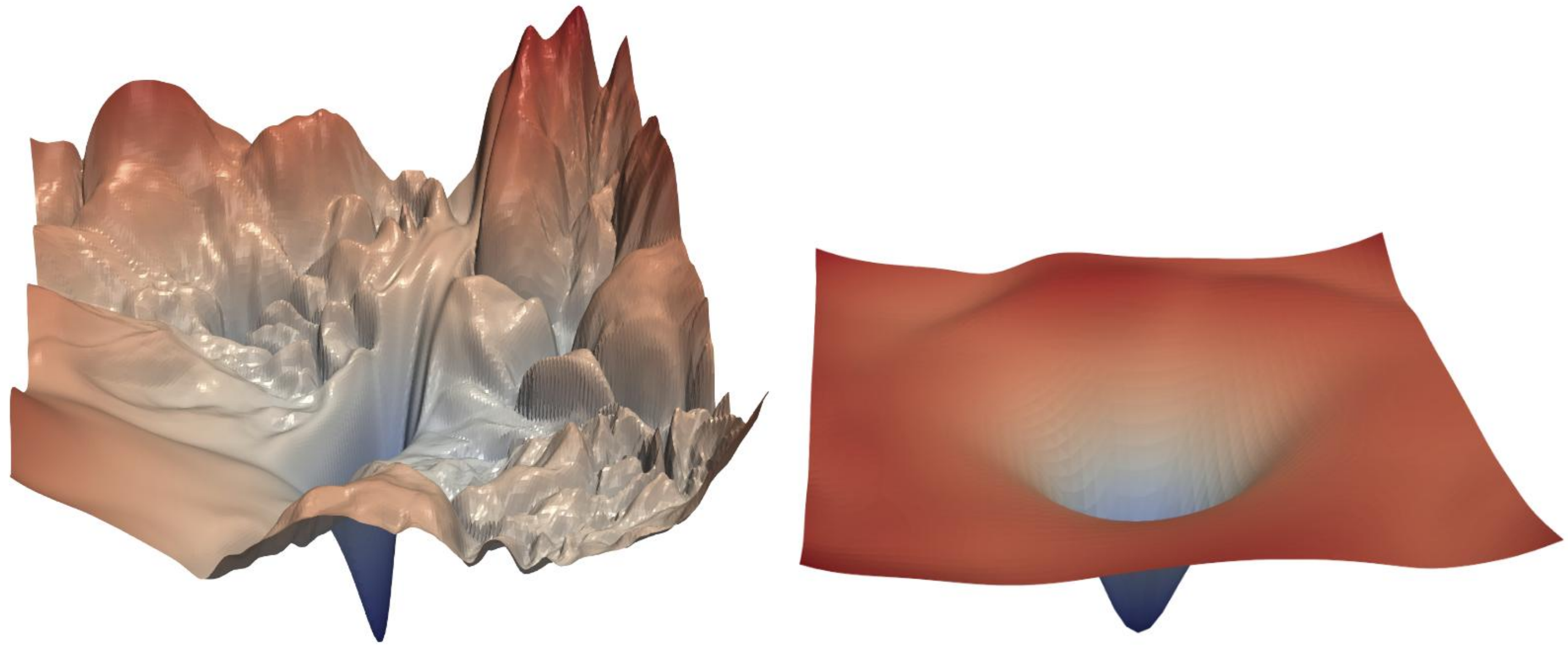


Image: He et al., "Deep Residual Learning for Image Recognition", CVPR, 2016 CVF arXiv

Skip Connection

- Visualizing the loss landscape



(a) without skip connections (b) with skip connections

Figure 1: The loss surfaces of ResNet-56 with/without skip connections. The proposed filter normalization scheme is used to enable comparisons of sharpness/flatness between the two figures.

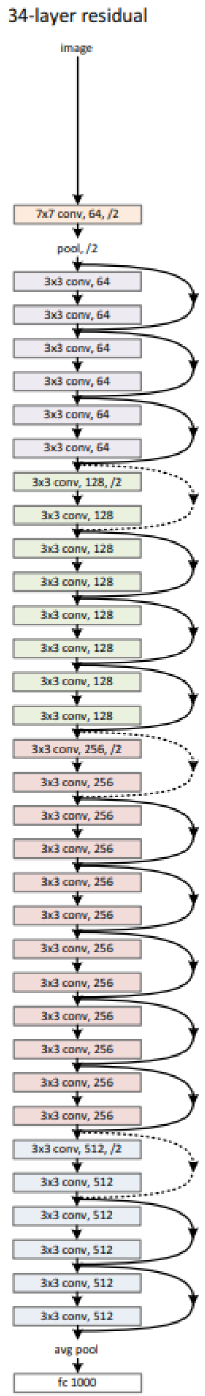
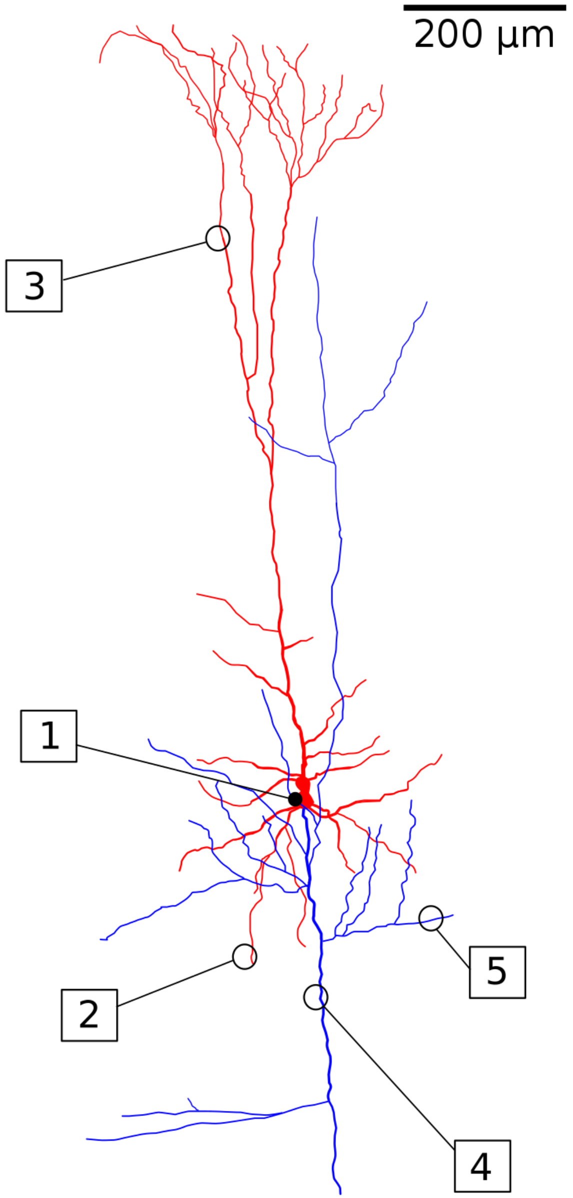


Image: Li et al., “Visualizing the Loss Landscape of Neural Nets”, NIPS, 2018 [Github](#) [arXiv](#)

Skip Connection

- Biological analogue [Wikipedia]
  - Cortical layer VI neurons @ the cerebral cortex



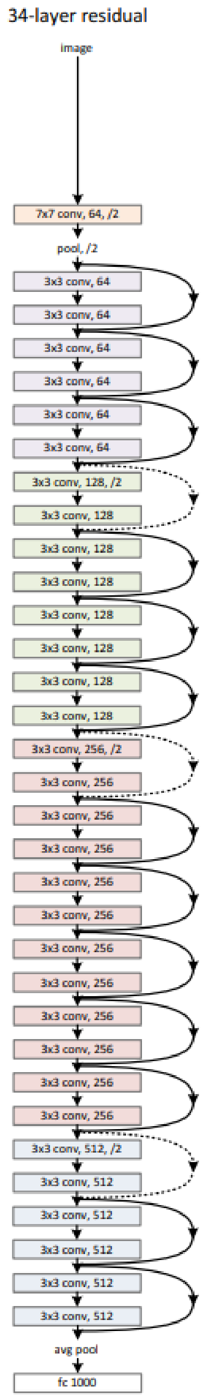


Image: Wikipedia

MNIST Dataset

- The MNIST dataset is a large database of handwritten digits. [Wikipedia]
- It was constructed by “re-mixing” the NIST’s original datasets.
  - Full name: *Modified* National Institute of Standards and Technology Database
- Specification
  - Classes: 10 (0, 1, 2, …, 9)
  - Images: 28 x 28 (8-bit gray scale)
  - The number of data: 60,000 for training and 10,000 for test

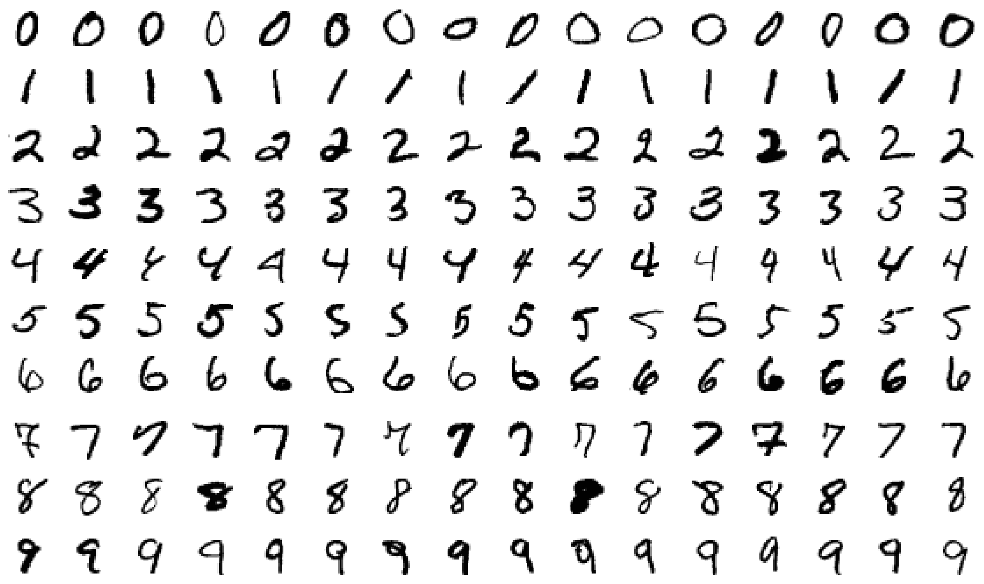
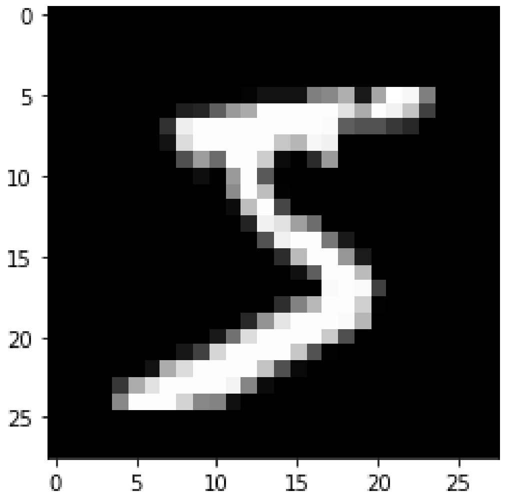


Image: Wikipedia

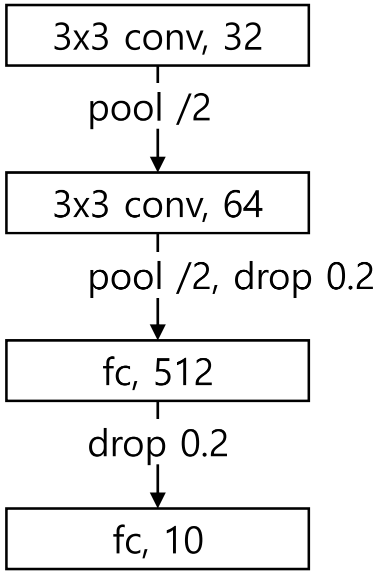
Practice) Loading the MNIST Dataset

```python
import torchvision
import matplotlib.pyplot as plt

# Note) You can download the MNIST dataset through its mirror.
# - Reference: https://stackoverflow.com/questions/66577151/http-error-when-trying-to-download-mnist-data
torchvision.datasets.MNIST.resources = [
    ('https://ossci-datasets.s3.amazonaws.com/mnist/train-images-idx3-ubyte.gz', 'f68b3c2dcbeaaa9fbdd348bbdeb94873'),
    ('https://ossci-datasets.s3.amazonaws.com/mnist/train-labels-idx1-ubyte.gz', 'd53e105ee54ea40749a09fcbcd1e9432'),
    ('https://ossci-datasets.s3.amazonaws.com/mnist/t10k-images-idx3-ubyte.gz',  '9fb629c4189551a2d022fa330f9573f3'),
    ('https://ossci-datasets.s3.amazonaws.com/mnist/t10k-labels-idx1-ubyte.gz',  'ec29112dd5afa0611ce80d1b7f02629c')
]

# Load the MNIST dataset
DATA_PATH = './data'
data_train = torchvision.datasets.MNIST(DATA_PATH, train=True, download=True)
data_valid = torchvision.datasets.MNIST(DATA_PATH, train=False)

# Look inside of the dataset
print(data_train)                   # ... 60000 ...
print(data_valid)                   # ... 10000 ...
print(data_train.data.shape)        # torch.Size([60000, 28, 28])
print(data_train.data.dtype)        # torch.uint8
print(data_train.data[0,:,:])       # tensor([[0, 0, ...], ..., [..., 166, 255, 247, ...], ...])
plt.imshow(data_train.data[0,:,:], cmap='gray')
plt.show()
print(data_train.targets[0])        # Guess and check it!
```



Practice) Digit Classification with the MNIST Dataset (1/4)

```
# A four-layer CNN model
# - Try more or less layers, channels, and kernel size
# - Try to apply batch normalization (e.g. 'nn.BatchNorm' and 'nn.BatchNorm2d')
# - Try to apply skip connection (used in ResNet)
class MyCNN(nn.Module):
    def __init__(self):
        super(MyCNN, self).__init__()
        # Notation:    (batch_size, channel, height, width)
        # Input :      (batch_size,  1, 28, 28)
        # Layer1: conv (batch_size, 32, 28, 28)
        #         pool (batch_size, 32, 14, 14)
        self.conv1 = nn.Conv2d(1, 32, kernel_size=3, stride=1, padding=1)
        self.pool1 = nn.MaxPool2d(kernel_size=2)

        # Layer2: conv (batch_size, 64, 14, 14)
        #         pool (batch_size, 64,  7,  7)
        self.conv2 = nn.Conv2d(32, 64, kernel_size=3, stride=1, padding=1)
        self.pool2 = nn.MaxPool2d(kernel_size=2)
        self.drop2 = nn.Dropout(0.2)

        # Input :      (batch_size, 64*7*7)
        # Layer3: fc   (batch_size, 512)
        self.fc3   = nn.Linear(64*7*7, 512)
        self.drop3 = nn.Dropout(0.2)

        # Layer4: fc   (batch_size, 10)
        self.fc4   = nn.Linear(512, 10)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        x = self.pool1(x)

        x = F.relu(self.conv2(x))
        x = self.pool2(x)
        x = self.drop2(x)
        x = torch.flatten(x, 1)

        x = F.relu(self.fc3(x))
        x = self.drop3(x)

        x = F.log_softmax(self.fc4(x), dim=1)
        return x
```



Practice) Digit Classification with the MNIST Dataset (2/4)

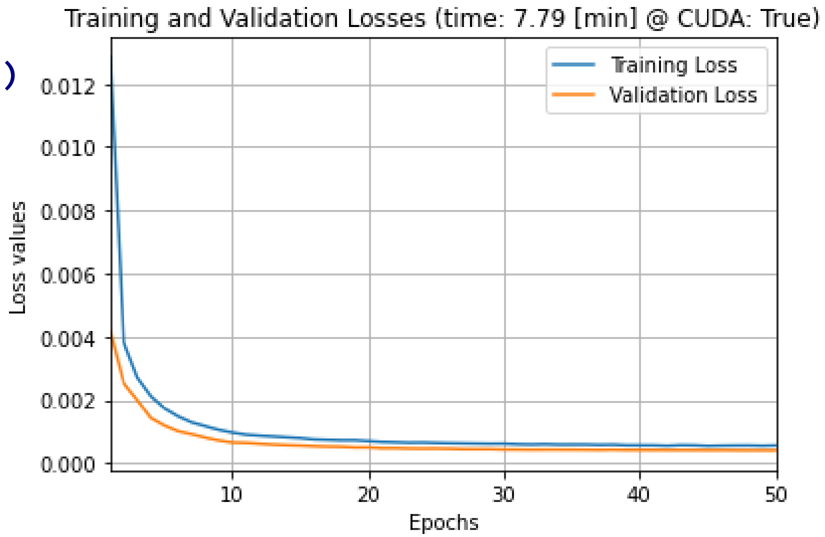
```python
if __name__ == '__main__':
    # 0. Preparation
    torch.manual_seed(RANDOM_SEED)
    if USE_CUDA:
        torch.cuda.manual_seed_all(RANDOM_SEED)
    dev = torch.device('cuda' if USE_CUDA else 'cpu')

    # 1. Load the MNIST dataset
    preproc = torchvision.transforms.ToTensor()
    data_train = torchvision.datasets.MNIST(DATA_PATH, train=True,  download=True, transform=preproc)
    data_valid = torchvision.datasets.MNIST(DATA_PATH, train=False, transform=preproc)
    loader_train = torch.utils.data.DataLoader(data_train, **DATA_LOADER_PARAM)
    loader_valid = torch.utils.data.DataLoader(data_valid, **DATA_LOADER_PARAM)

    # 2. Instantiate a model, loss function, and optimizer
    model = MyCNN().to(dev)
    loss_func = F.cross_entropy
    optimizer = torch.optim.SGD(model.parameters(), **OPTIMIZER_PARAM)
    scheduler = torch.optim.lr_scheduler.StepLR(optimizer, **SCHEDULER_PARAM)
```

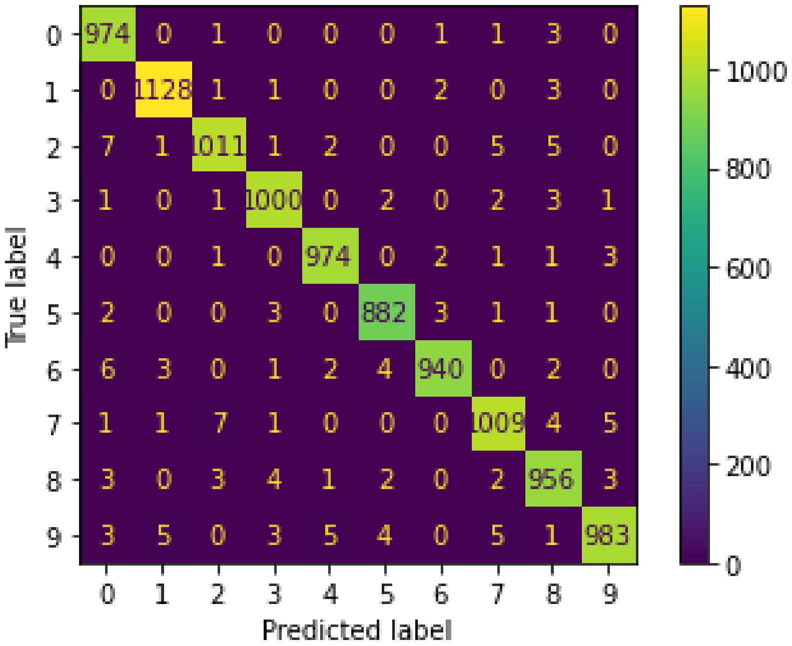
Note) `SCHEDULER_PARAM = { 'step_size': 10, 'gamma': 0.5 }`

```python
    # 3.1. Train the model
    loss_list = []
    start = time.time()
    for epoch in range(1, EPOCH_MAX + 1):
        train_loss = train(model, loader_train, loss_func, optimizer)
        valid_loss, valid_accuracy = evaluate(model, loader_valid, loss_func)
        scheduler.step()

        loss_list.append([epoch, train_loss, valid_loss, valid_accuracy])
        if epoch % EPOCH_LOG == 0:
            elapse = (time.time() - start) / 60
            print(f'{epoch:>6} ({elapse:>6.2f} min), TrLoss={train_loss:.6f}, VaLoss={valid_loss:.6f}, VaAcc={valid_accuracy:.3f}, lr={scheduler.get_last_lr()}')
    elapse = (time.time() - start) / 60

    # 3.2. Save the trained model if necessary
    if SAVE_MODEL:
        torch.save(model.state_dict(), SAVE_MODEL)
```

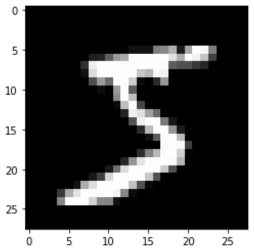
Practice) Digit Classification with the MNIST Dataset (3/4)

```python
# 4.1. Visualize the loss curves
plt.title(f'Training and Validation Losses (time: {elapse:.2f} [min] @ CUDA: {USE_CUDA})')
loss_array = np.array(loss_list)
plt.plot(loss_array[:,0], loss_array[:,1], label='Training Loss')
plt.plot(loss_array[:,0], loss_array[:,2], label='Validation Loss')
plt.xlabel('Epochs')
plt.ylabel('Loss values')
plt.xlim(loss_array[0,0], loss_array[-1,0])
plt.grid()
plt.legend()
plt.show()

# 4.2. Visualize the confusion matrix
predicts = [predict(datum, model) for datum in data_valid.data]
conf_mat = metrics.confusion_matrix(data_valid.targets, predicts)
conf_fig = metrics.ConfusionMatrixDisplay(conf_mat)
conf_fig.plot()

# 5. Test your image
print(predict(data_train.data[0], model)) # 5
with PIL.Image.open('data/cnn_mnist_test.png').convert('L') as image:
    print(predict(image, model))          # 3
```

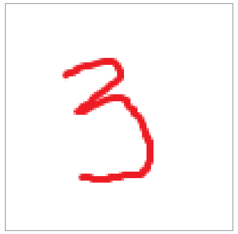




data_train.data[0]



'data/cnn_mnist_test.png'



Practice) Digit Classification with the MNIST Dataset (4/4)

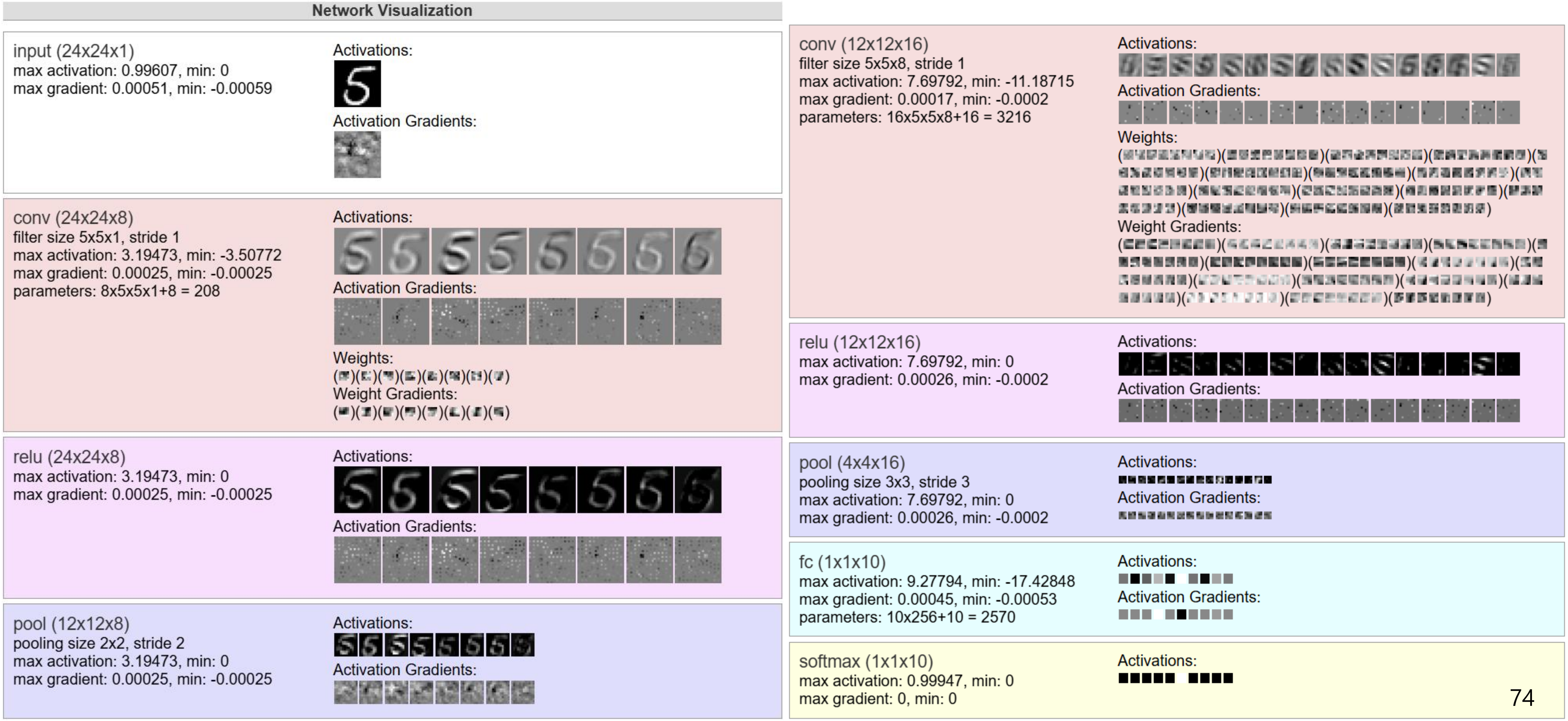
```
# Predict a digit using the given model
def predict(image, model):
    model.eval()
    with torch.no_grad():
        # Convert the given image to its 1 x 1 x 28 x 28 tensor
        if type(image) is torch.Tensor:
            tensor = image.type(torch.float) / 255  # Normalize to [0, 1]
        else:
            tensor = 1 - TF.to_tensor(image)        # Invert (white to black)
        if tensor.ndim < 3:
            tensor = tensor.unsqueeze(0)
        if tensor.shape[0] == 3:
            tensor = TF.rgb_to_grayscale(tensor)    # Make grayscale
        tensor = TF.resize(tensor, 28)              # Resize to 28 x 28
        dev = next(model.parameters()).device
        tensor = tensor.unsqueeze(0).to(dev)        # Add one more dim

        output = model(tensor)
        digit = torch.argmax(output, dim=1)
        return digit.item()
```

Practice) Loading My Network and Testing My Image

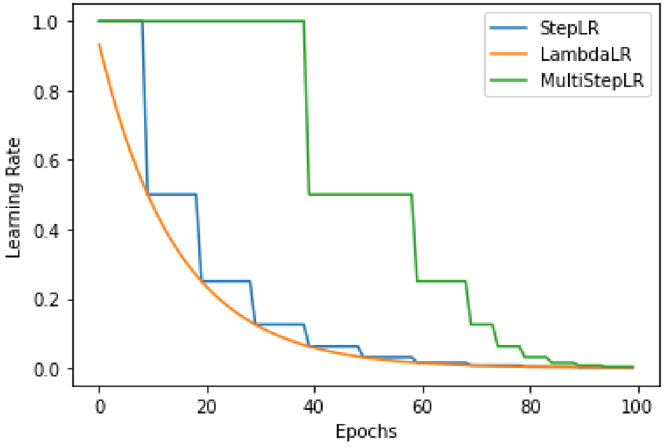
```
import torch, PIL
from cnn_mnist import MyCNN, predict

# Load a model
model = MyCNN()
model.load_state_dict(torch.load('cnn_mnist.pt'))

# Test the model
with PIL.Image.open('data/cnn_mnist_test.png').convert('L') as image:
    print(predict(image, model))
```

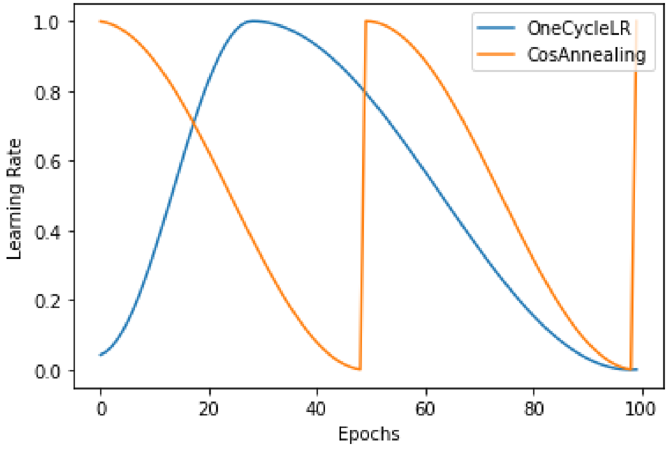
Practice) Digit Classification with the MNIST Dataset (Visualization)

- Example) ConvNetJS MNIST demo



Practice) Visualizing Learning Rate Schedulers (1/2)






```python
import torch
import torch.optim.lr_scheduler as lr_scheduler
import matplotlib.pyplot as plt

base_lr = 1.
epoch_max = 100
schedulers = [
    {'name': 'StepLR',        'class': lr_scheduler.StepLR,
                              'param': {'step_size': 10, 'gamma': 0.5}},
    {'name': 'LambdaLR',      'class': lr_scheduler.LambdaLR,
                              'param': {'lr_lambda': lambda epoch: 0.5**(epoch / 10)}},
    {'name': 'MultiStepLR',   'class': lr_scheduler.MultiStepLR,
                              'param': {'milestones': [40, 60, 70, 75, 80, 85, 90, 95], 'gamma': 0.5}},
    {'name': 'OneCycleLR',    'class': lr_scheduler.OneCycleLR,
                              'param': {'max_lr': base_lr, 'total_steps': epoch_max}},
    {'name': 'CosAnnealing', 'class': lr_scheduler.CosineAnnealingWarmRestarts,
                              'param': {'T_0': 50}},
    # Try more learning rate schedulers
]
```

Practice) Visualizing Learning Rate Schedulers (2/2)

```
for sch in schedulers:
    x = torch.tensor(1., requires_grad=True)              # A dummy parameter
    optimizer = torch.optim.SGD([x], lr=base_lr)          # Instantiate an optimizer
    scheduler = sch['class'](optimizer, **sch['param']) # Instantiate a LR scheduler
    lr_values = []
    for i in range(epoch_max):
        optimizer.step()
        scheduler.step()
        lr_values.append(optimizer.param_groups[0]['lr'])
    plt.plot(range(epoch_max), lr_values, label=sch['name'])

plt.xlabel('Epochs')
plt.ylabel('Learning Rate')
plt.legend()
plt.show()
```

Practice) Different Styles for NN Classes (1/2)

My style

```
class MyCNN(nn.Module):
    def __init__(self):
        super(MyCNN, self).__init__()
        self.conv1 = nn.Conv2d(1, 32, 3, 1, 1)
        self.pool1 = nn.MaxPool2d(2)

        self.conv2 = nn.Conv2d(32, 64, 3, 1, 1)
        self.pool2 = nn.MaxPool2d(2)
        self.drop2 = nn.Dropout(0.2)

        self.fc3   = nn.Linear(64*7*7, 512)
        self.drop3 = nn.Dropout(0.2)

        self.fc4   = nn.Linear(512, 10)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        x = self.pool1(x)

        x = F.relu(self.conv2(x))
        x = self.pool2(x)
        x = self.drop2(x)
        x = torch.flatten(x, 1)

        x = F.relu(self.fc3(x))
        x = self.drop3(x)

        x = F.log_softmax(self.fc4(x), dim=1)
        return x
```

Functional-oriented style

```
class MyCNN_Functional(nn.Module):
    def __init__(self):
        super(MyCNN_FStyle, self).__init__()
        self.conv1 = nn.Conv2d( 1, 32, 3, 1, 1)
        self.conv2 = nn.Conv2d(32, 64, 3, 1, 1)
        self.fc1 = nn.Linear(64*7*7, 512)
        self.fc2 = nn.Linear(512, 10)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        x = F.max_pool2d(x, 2)

        x = F.relu(self.conv2(x))
        x = F.max_pool2d(x, 2)
        x = F.dropout(x, 0.2, self.training)
        x = torch.flatten(x, 1)

        x = F.relu(self.fc1(x))
        x = F.dropout(x, 0.2, self.training)

        x = F.log_softmax(self.fc2(x), dim=1)
        return x
```

Practice) Different Styles for NN Classes (2/2)

Object-oriented style

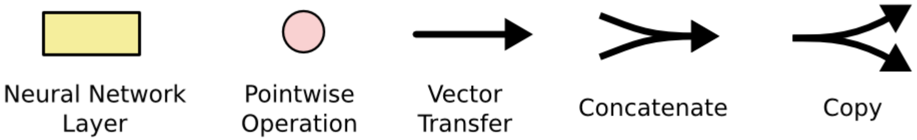
```python
class MyCNN_Object(nn.Module):
    def __init__(self):
        super(MyCNN_ObjStyle, self).__init__()
        self.conv1 = nn.Conv2d(1, 32, 3, 1, 1)
        self.relu1 = nn.ReLU()
        self.pool1 = nn.MaxPool2d(2)

        self.conv2 = nn.Conv2d(32, 64, 3, 1, 1)
        self.relu2 = nn.ReLU()
        self.pool2 = nn.MaxPool2d(2)
        self.drop2 = nn.Dropout(0.2)

        self.fc3   = nn.Linear(64*7*7, 512)
        self.relu3 = nn.ReLU()
        self.drop3 = nn.Dropout(0.2)

        self.fc4   = nn.Linear(512, 10)
        self.smax4 = nn.LogSoftmax(dim=1)

    def forward(self, x):
        x = self.conv1(x)
        x = self.relu1(x)
        x = self.pool1(x)

        x = self.conv2(x)
        x = self.relu2(x)
        x = self.pool2(x)
        x = self.drop2(x)
        x = torch.flatten(x, 1)

        x = self.fc3(x)
```

Layer-oriented style

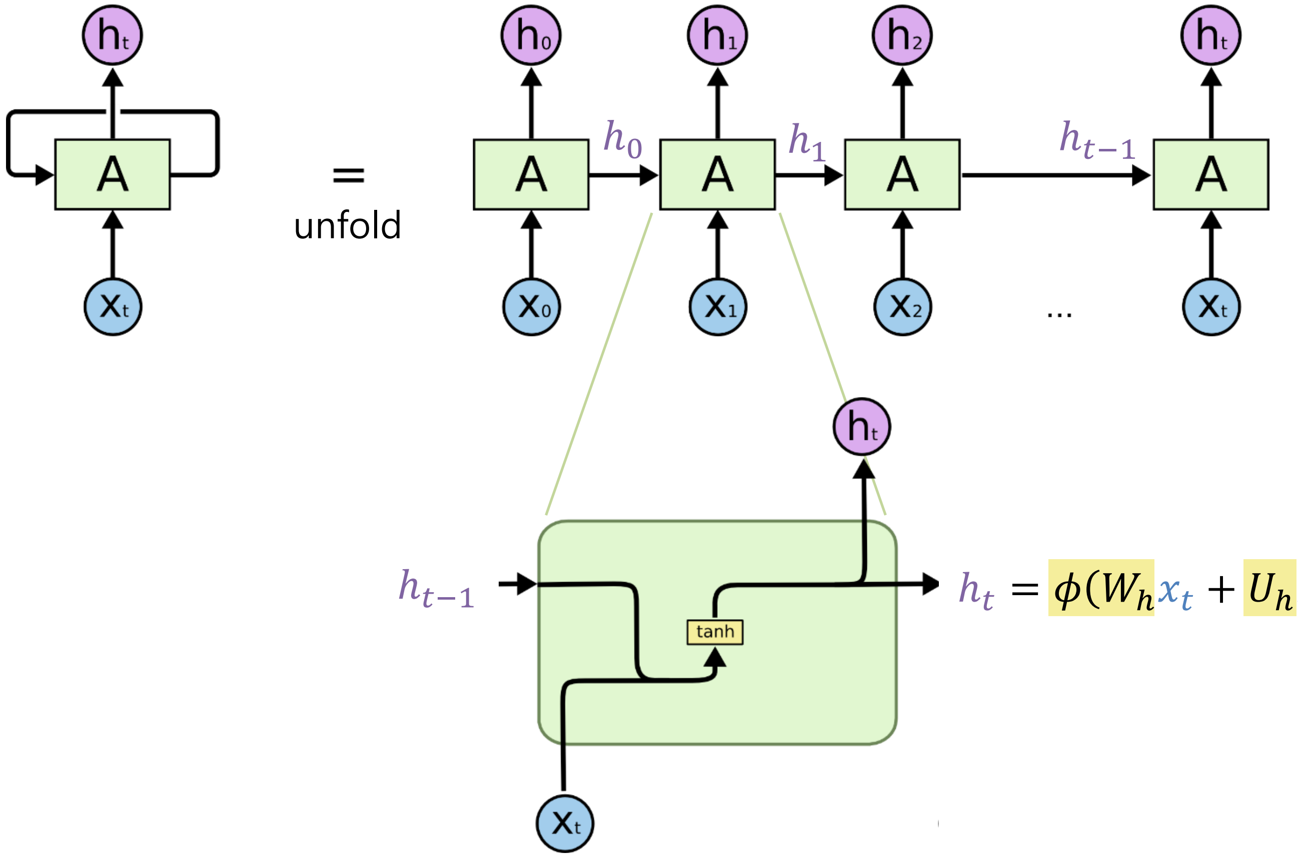
```python
class MyCNN_Layer(nn.Module):
    def __init__(self):
        super(MyCNN_SeqStyle, self).__init__()
        self.layer1 = nn.Sequential(
            nn.Conv2d(1, 32, 3, 1, 1),
            nn.ReLU(),
            nn.MaxPool2d(2))

        self.layer2 = nn.Sequential(
            nn.Conv2d(32, 64, 3, 1, 1),
            nn.ReLU(),
            nn.MaxPool2d(2),
            nn.Dropout(0.2))

        self.layer3 = nn.Sequential(
            nn.Linear(64*7*7, 512),
            nn.ReLU(),
            nn.Dropout(0.2))

        self.layer4 = nn.Sequential(
            nn.Linear(512, 10),
            nn.LogSoftmax(dim=1))

    def forward(self, x):
        x = self.layer1(x)
        x = self.layer2(x)
        x = torch.flatten(x, 1)
        x = self.layer3(x)
        x = self.layer4(x)
        return x
```

Table of Contents

Recurrent Neural Network (RNN)



- A **recurrent neural network** (shortly *RNN*) is a NN with a **loop**.
- A RNN can preserve a *memory* or *(hidden) state* or *information* inside.
  - Note) ~ sequential logic circuits (vs. combinatorial logic circuits) in digital circuits



$$h_t = \phi(W_h x_t + U_h h_{t-1} + b_h)$$

Note) All $\boxed{\mathbf{A}}$s have the same weight.

Image: [colah's blog]

Recurrent Neural Network (RNN)

- **Multi-layer RNNs**

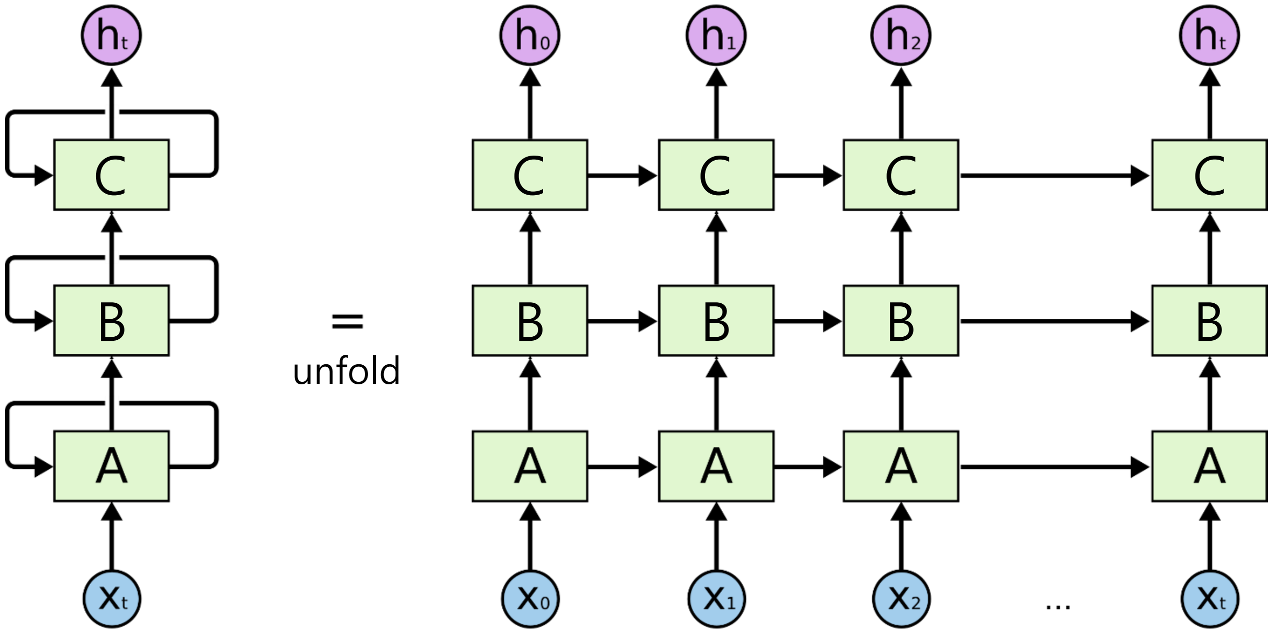
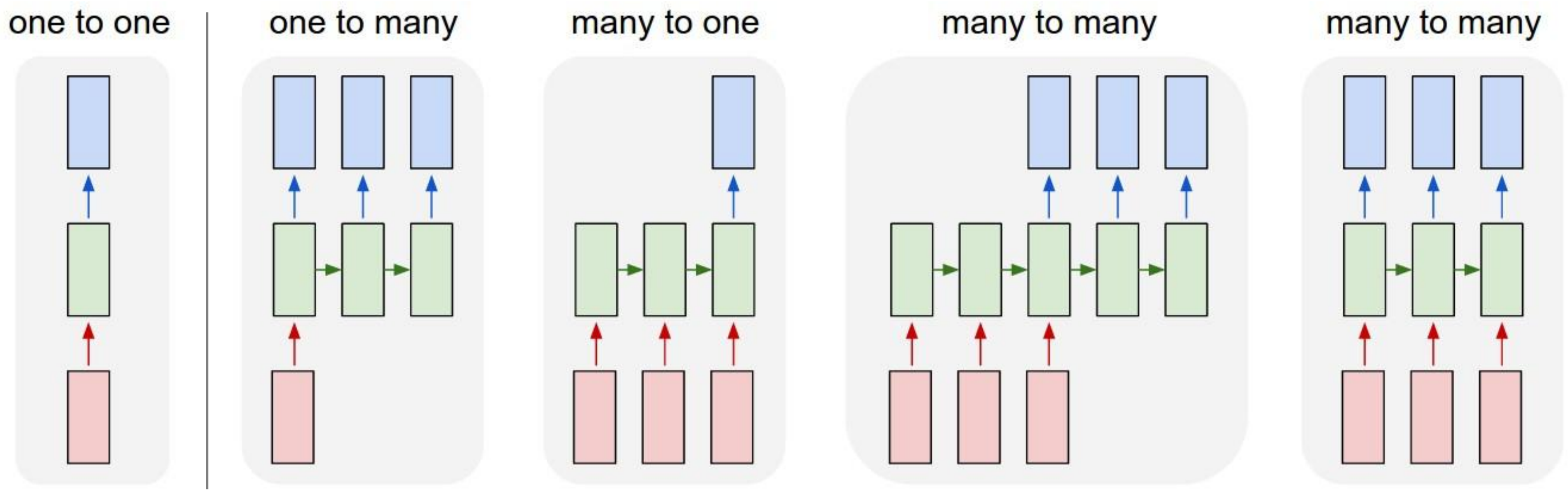


Image: [colah's blog]()

Recurrent Neural Network (RNN)

- Memory in a RNN allows it to deal with *sequential* or *temporal* data.
- A RNN can deal with **various lengths of input vectors** and **output vectors** by attaching more hidden/output layers at its end.



not RNN | image captioning | sentiment analysis (classification) | machine translation | video classification

Image: [Andrej Karpathy blog]

Recurrent Neural Network (RNN)

- The **vanishing gradient problem** in training a long sequence data
  - A vanilla RNN cannot deal with long-term dependency.
  - e.g. Guessing the last word, "I grew up in France … I speak fluent French."

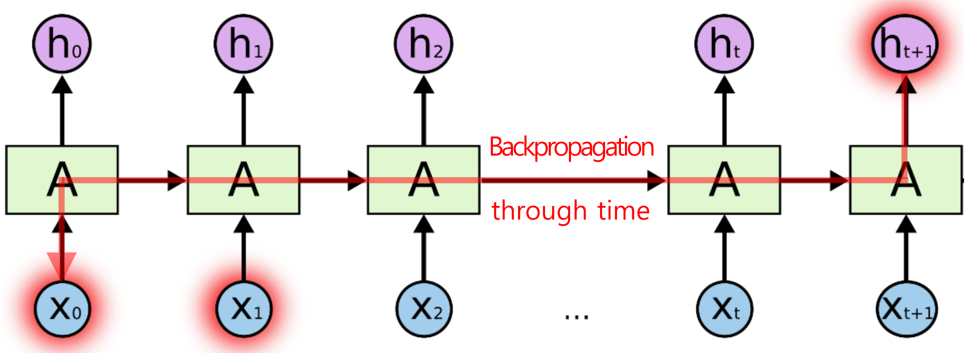


Image: colah's blog

Long Short-Term Memory (LSTM)

- A **long short-term memory** (shortly *LSTM*) is a recurrent neural network unit to deal with the vanishing gradient problem of a vanilla RNN. [Wikipedia]
- It is composed of a **forget gate**, an **input gate**, a **cell**, and an **output gate**.

forget gate	How much forget c_{t-1} to the cell state	$f_t = \sigma\left(W_f x_t + U_f h_{t-1} + b_f\right)$
input gate	How much write $\tilde{c}_t$ to the cell state	$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$
cell	The intermediate cell state	$\tilde{c}_t = \tanh(W_o x_t + U_o h_{t-1} + b_o)$
output gate	How much reveal c_t to the hidden state	$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$

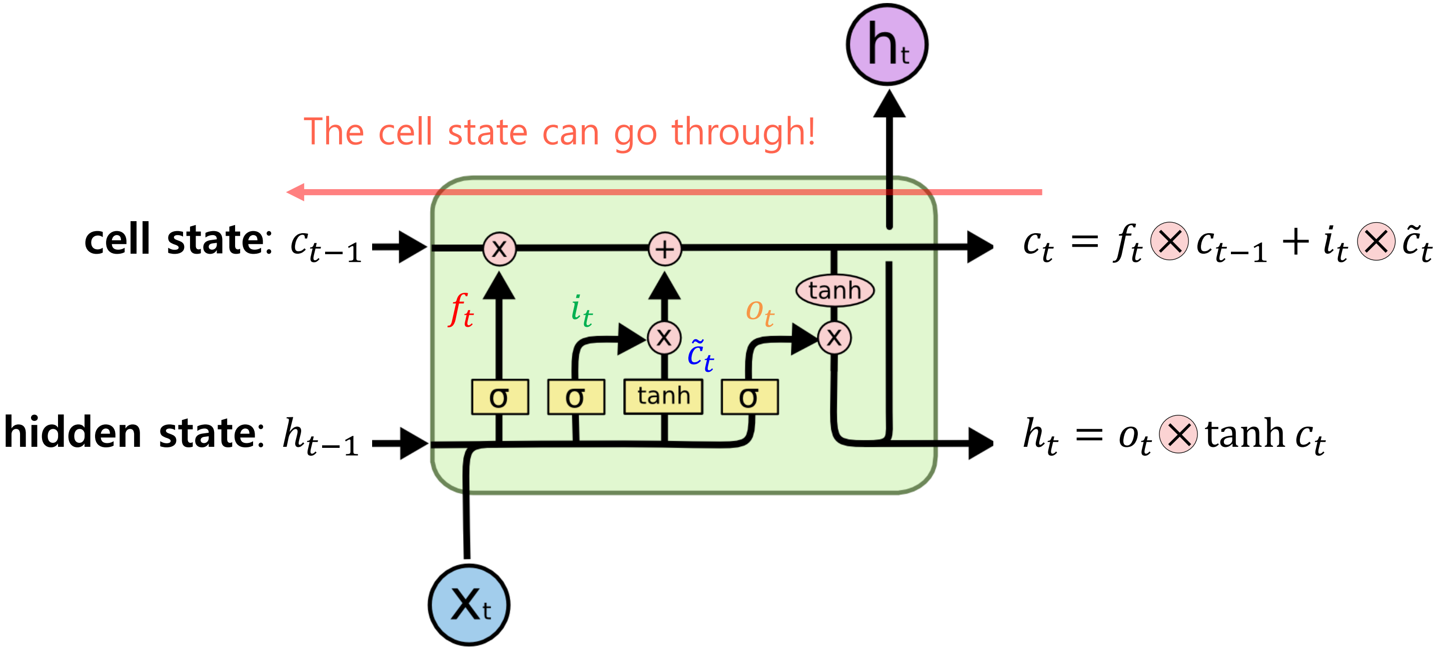
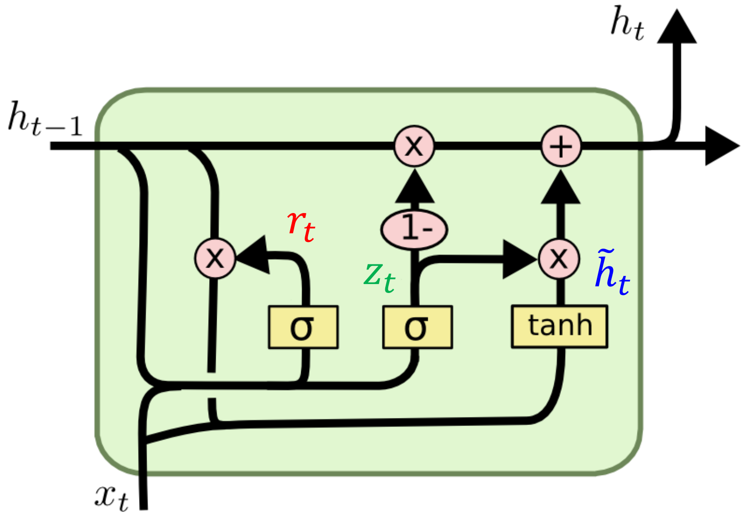


Image: colah's blog

Gated Recurrent Unit (GRU)

- A **gated recurrent unit** (shortly *GRU*) is a LSTM variant with a simplified structure, but it still has similar performance. [Wikipedia]

reset gate	How much forget h_{t-1} to the hidden state	$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r)$
update gate	How much write $\tilde{h}_t$ to the hidden state	$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z)$
	The intermediate hidden state	$\tilde{h}_t = \tanh(W_h x_t + U_h(r_t \otimes h_{t-1}) + b_h)$



$$h_t = (1 - z_t) \otimes h_{t-1} + z_t \otimes \tilde{h}_t$$

Image: colah's blog

Practice) Name2Lang Classification with a Character-level RNN (1/7)

- Motivation: A registration form on Internet

First Name Last Name

E-mail address

Country

- Input: **Name** (string; a sequence of characters)
  - e.g. 'Choi', 'Jane', 'Daniel', 'Chow', 'Tanaka', ...
- Classes: **18 languages** (integer; 0-17)
  - e.g. Arabic, Chinese, Czech, Dutch, English, French, German, Greek, Iris, Italian, Japanese, Korean, Polish, Portuguese, Russian, Scottish, Spanish, Vietnamese
- The dataset and bottom-up implementation are available on the PyTorch official tutorial.

Practice) Name2Lang Classification with a Character-level RNN (2/7)

- Input: **Name** (string; a sequence of characters)
  - e.g. 'Choi', 'Jane', 'Daniel', 'Chow', 'Tanaka', ...
- How to represent the **name**?
  - The name is represented using 57 characters (52 alphabets and 5 special letters).
    - e.g. `'a'` (0x61), `'b'` (0x62), `'c' (0x63)`, ..., `'-'` (0x45) in ASCII code
  - Each character is encoded as a 57-bit one-hot vector.
    - e.g.

a:	**1**	0	0	0	...
b:	0	**1**	0	0	...
c:	0	0	**1**	0	...

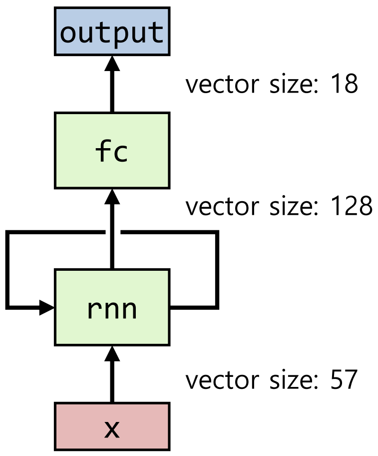
Why one-hot encoding?

- e.g. `'Choi'` represented as a 4 x 57 array

C:	0	...	0	0	0	...	0	0	...	**1**	0	...
h:	0	...	**1**	0	0	...	0	0	...	0	0	...
o:	0	...	0	0	0	...	**1**	0	...	0	0	...
i:	0	...	0	**1**	0	...	0	0	...	0	0	...
Index)			7	8			14			28		

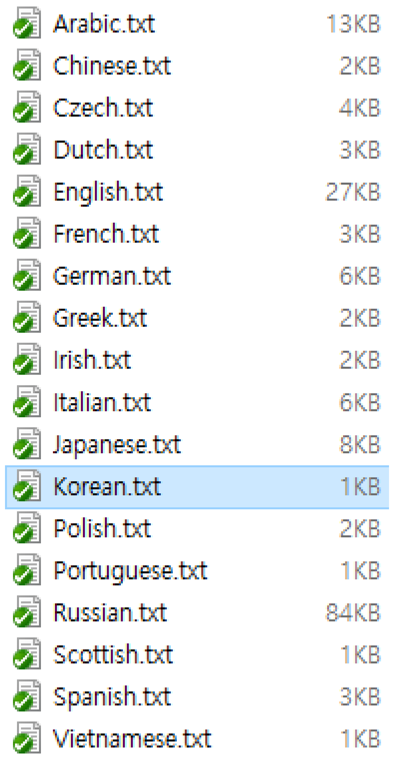
Practice) Name2Lang Classification with a Character-level RNN (3/7)

```python
# A simple RNN model
# - Try a different RNN unit such LSTM and GRU
# - Try less or more hidden units
# - Try more layers (e.g. 'num_layers=2') and dropout (e.g. 'dropout=0.4')
class MyRNN(nn.Module):
    def init(self, input_size, output_size):
        super(MyRNN, self).init()
        self.rnn = torch.nn.RNN(input_size, 128)
        self.fc = torch.nn.Linear(128, output_size)

    def forward(self, x):
        output, hidden = self.rnn(x)
        x = self.fc(output[-1])  # Use the output of the last sequence
        return x
```



Practice) Name2Lang Classification with a Character-level RNN (4/7)

```python
# Convert Unicode to ASCII
# e.g. Ślusàrski to Slusarski
def unicode2ascii(s):
    return ''.join(
        c for c in unicodedata.normalize('NFD', s)
        if unicodedata.category(c) != 'Mn' and c in LETTER_DICT)

# Read raw files which contain names belong to each language
# Note) Each filename is used as its target's name.
def load_name_dataset(files):
    data = []
    targets = []
    target_names = []
    for idx, filename in enumerate(files):
        lang = os.path.splitext(os.path.basename(filename))[0]
        names = open(filename, encoding='utf-8').read().strip().split('\n')

        data += [unicode2ascii(name) for name in names]
        targets += [idx] * len(names)
        target_names.append(lang)
    return data, targets, target_names

# Transform the given text to its one-hot encoded tensor
# Note) Tensor size: len(text) x 1 x len(LETTER_DICT)
#                    sequence_length x batch_size x input_size
def text2onehot(text, device='cpu'):
    tensor = torch.zeros(len(text), 1, len(LETTER_DICT), device=device)
    for idx, letter in enumerate(text):
        tensor[idx][0][LETTER_DICT.find(letter)] = 1
    return tensor
```

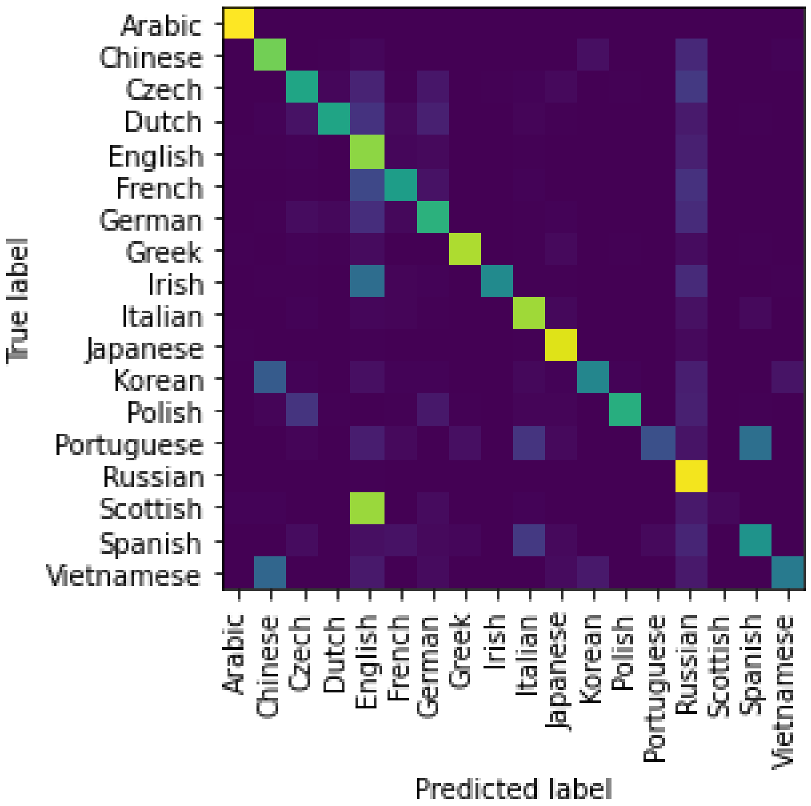


Practice) Name2Lang Classification with a Character-level RNN (5/7)

```
if __name__ == '__main__':
    # 0. Preparation
    # 1. Load the name2lang dataset
    data, targets, target_names = load_name_dataset(glob.glob(DATA_PATH))
    data_train = [(text2onehot(data[i], device=dev), torch.LongTensor(
        [targets[i]]).to(dev)) for i in range(len(data))]
    random.shuffle(data_train)

    # 2. Instantiate a model, loss function, and optimizer
    # 3.1. Train the model
    # 3.2. Save the trained model if necessary

    # 4.1. Visualize the loss curves
    # 4.2. Visualize the confusion matrix
    predicts = [predict(datum, model) for datum in data]
    conf_mat = sklearn.metrics.confusion_matrix(targets, predicts, normalize='true')
    plt.imshow(conf_mat)
    plt.xlabel('Predicted label')
    plt.ylabel('True label')
    plt.gca().set_xticklabels([''] + target_names, rotation=90)
    plt.gca().set_yticklabels([''] + target_names)
    plt.gca().xaxis.set_major_locator(ticker.MultipleLocator(1))
    plt.gca().yaxis.set_major_locator(ticker.MultipleLocator(1))
    plt.show()

    # 5. Test your texts
    report_predict('Choi', model, target_names)
    report_predict('Jane', model, target_names)
    report_predict('Daniel', model, target_names)
    report_predict('Chow', model, target_names)
    report_predict('Tanaka', model, target_names)
```

Practice) Name2Lang Classification with a Character-level RNN (6/7)

```
* Name: Choi
  1. Korean    : 50.5 %
  2. Chinese   : 36.1 %
  3. Vietnamese: 11.8 %
  4. Russian   :  0.9 %
  5. Arabic    :  0.5 %
* Name: Jane
  1. English   : 62.4 %
  2. German    : 10.5 %
  3. Korean    :  9.0 %
  4. Chinese   :  7.0 %
  5. Dutch     :  6.2 %
* Name: Daniel
  1. English   : 48.4 %
  2. French    : 20.6 %
  3. Czech     : 13.1 %
  4. Russian   :  6.6 %
  5. Portuguese:  3.1 %
* Name: Chow
  1. Korean    : 69.8 %
  2. Chinese   : 16.0 %
  3. English   :  4.7 %
  4. Vietnamese:  4.3 %
  5. Russian   :  3.1 %
* Name: Tanaka
  1. Japanese  : 84.4 %
  2. Russian   : 14.0 %
  3. Czech     :  1.3 %
  4. English   :  0.1 %
  5. Irish     :  0.1 %
```



Practice) Name2Lang Classification with a Character-level RNN (7/7)

```python
# Predict the best result of the given text
def predict(text, model):
    model.eval()
    with torch.no_grad():
        dev = next(model.parameters()).device
        text_tensor = text2onehot(text, dev) # Convert text to one-hot vectors
        output = model(text_tensor)[0]       # Get the last output
        lang = torch.argmax(output)          # Get the best among 18 classes
        return lang.item()

# Predict and report top-k results of the given text
def report_predict(text, model, target_names, n_predict=5):
    print(f'* Name: {text}')
    model.eval()
    with torch.no_grad():
        dev = next(model.parameters()).device
        text_tensor = text2onehot(text, dev)
        output = model(text_tensor)[0]
        prob = nn.functional.softmax(output, dim=0) # Make output as probability
        top_val, top_idx = prob.topk(n_predict)     # Get top-k among 18 classes
        for i in range(len(top_val)):
            print(f'  {i+1}. {target_names[top_idx[i].item()]:<10}: 
{top_val[i]*100:4.1f} %')
```

y_i: 0.012 | 0.929 | 0.058

Softmax: $y_i = \exp z_i \,/\, \sum_{j=1}^{C} \exp z_j$

NN output z_i: -1.03 | 3.29 | 0.52

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Word Representations

- **One-hot encoding** represents each word as a binary one-hot vector in which exactly one element is 1 and all other elements are 0.
  - e.g. `apple`: $[1, 0, 0, 0, \cdots]$, `bear`: $[0\ 1, 0, 0, \cdots]$, `cat`: $[0, 0, 1, 0, \cdots]$, `dog`: $[0, 0, 0, 1, \cdots]$, …
  - The vocabulary size is equal to the dimension size of the one-hot vector.
  - Limitations
    - **Sparse representation** → very high-dimensional feature space
      - Large memory and computational requirements
      - Potential risk of **overfitting**, especially when the training data are limited
    - **Equal distance** between any two distinct words
      - Difficulty in capturing meaningful relationships between words

Word Representations

- **Word embedding** represents a word as a low-dimensional, dense vector of real numbers.

$$\text{word} \to \boldsymbol{e}_{\text{word}} \in \mathbb{R}^d \quad \text{where} \quad d \text{ can be } 100, 128, 256, \ldots$$

  - Note) **Embedding** is a learned continuous vector representation in which geometric relationships reflect useful linguistic or semantic properties.
  - The key idea is to learn word representations from the **contexts** in which words occur.
    - Words appearing in similar contexts tend to have similar vector representations.
    - e.g. $|\texttt{dog} - \texttt{cat}| < |\texttt{dog} - \texttt{apple}|$
    - e.g. $\texttt{puppy} - \texttt{dog} + \texttt{cat} = \texttt{kitten}$
  - Approaches
    - Static word embedding: Word2Vec (2013), GloVe (Global Vectors; 2014), fastText (2016), …
    - Contextualized word embeddings: ELMo (bidirectional LSTM; 2018), BERT (bidirectional Transformer; 2018), …
      - e.g. I went to the bank to deposit some money. / We sat on the bank of the river. → different vectors

Word Embedding

- Online demo) [TensorFlow Embedding Projector](): Word retrieval and visualization

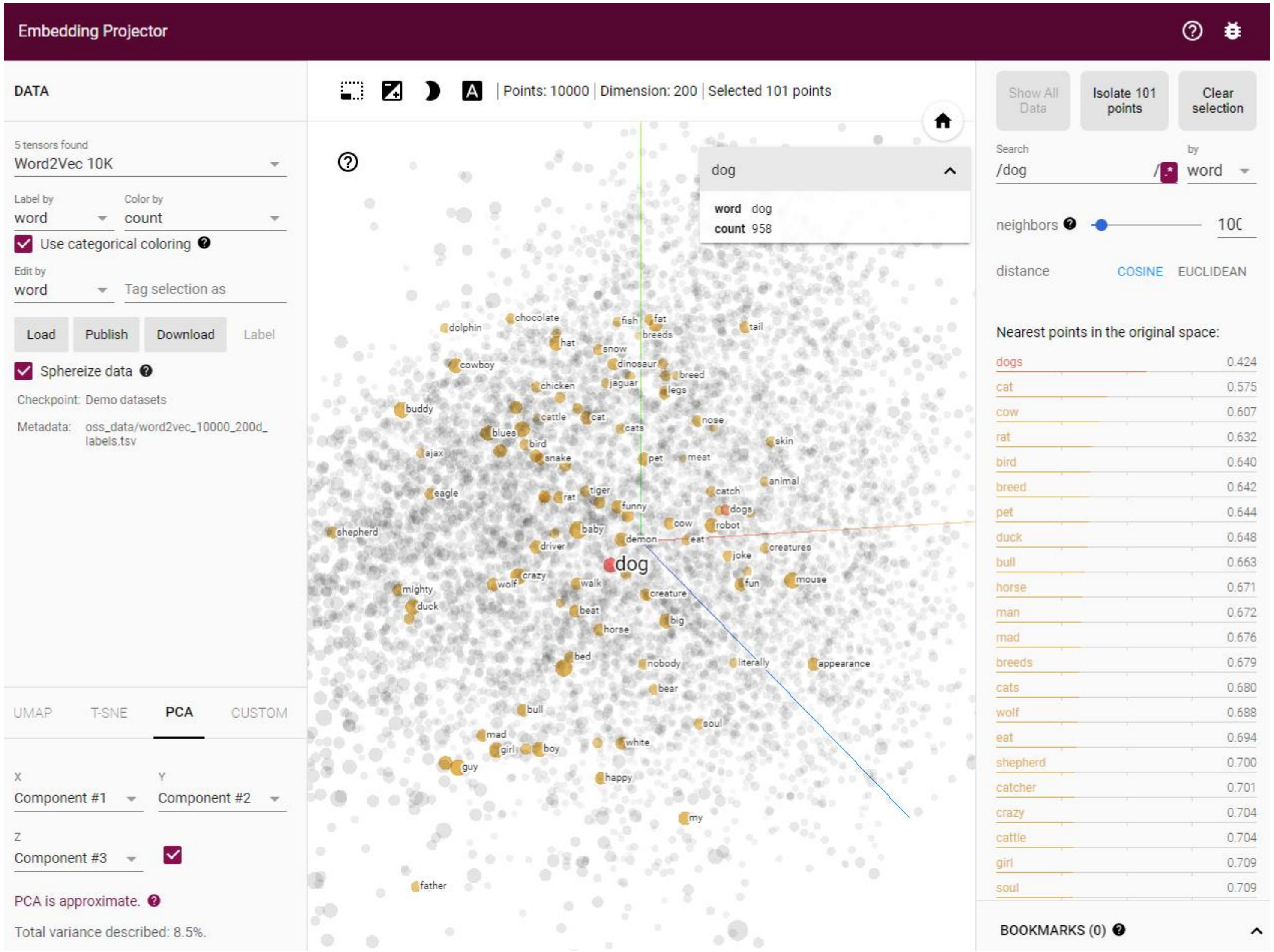


Image: [TensorFlow Embedding Projector]() (retrieved May 13th, 2021)

Word Embedding

- Online demo) WebVectors Semantic Calculator
  - e.g. puppy – dog + cat = kitten

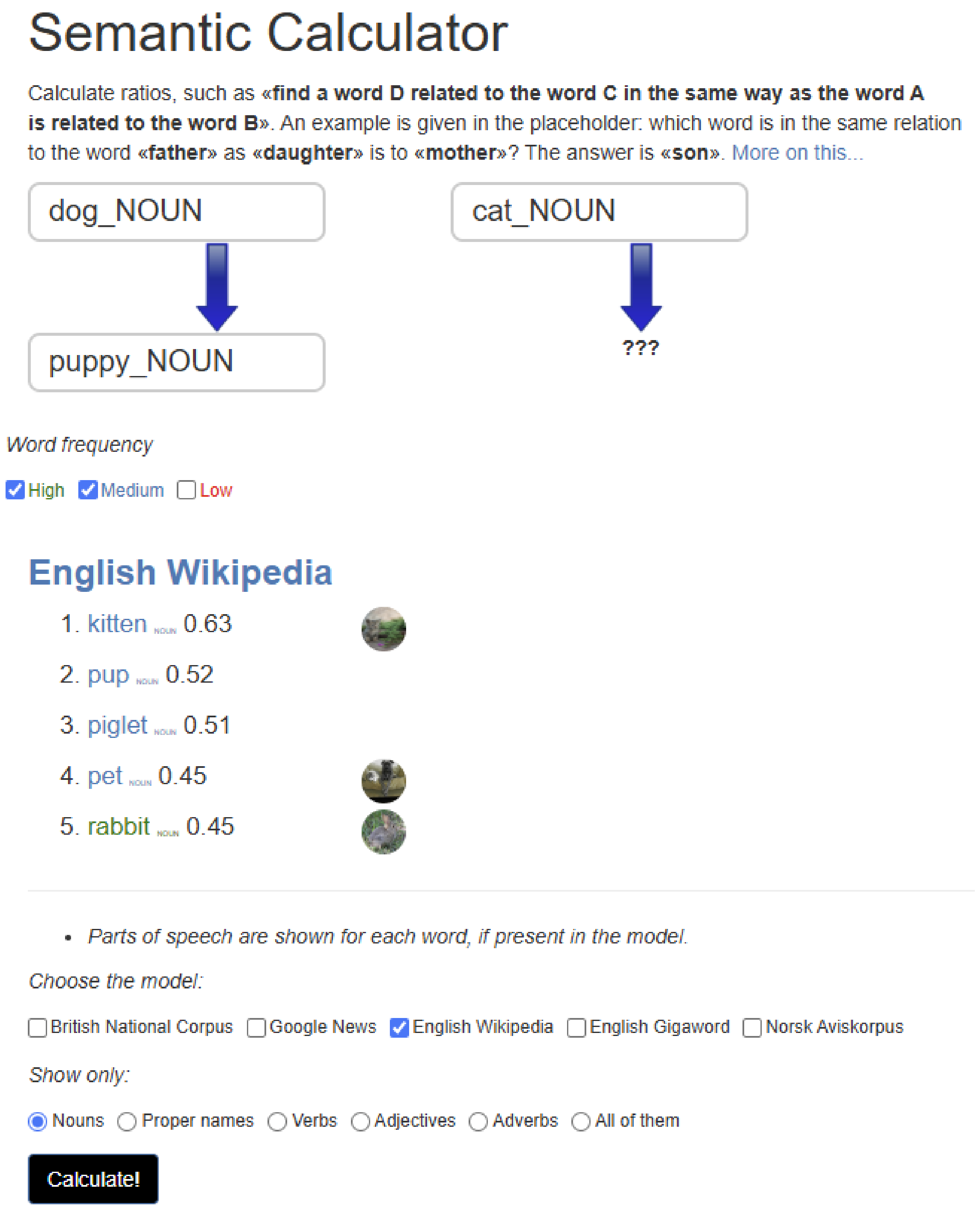


Image: WebVectors Semantic Calculator (retrieved September 7th, 2026)

Transformer

- Paper) Vaswani et al., "Attention Is All You Need", NIPS, 2017 arXiv
  - e.g. Attention is like highlighting important parts of a book.

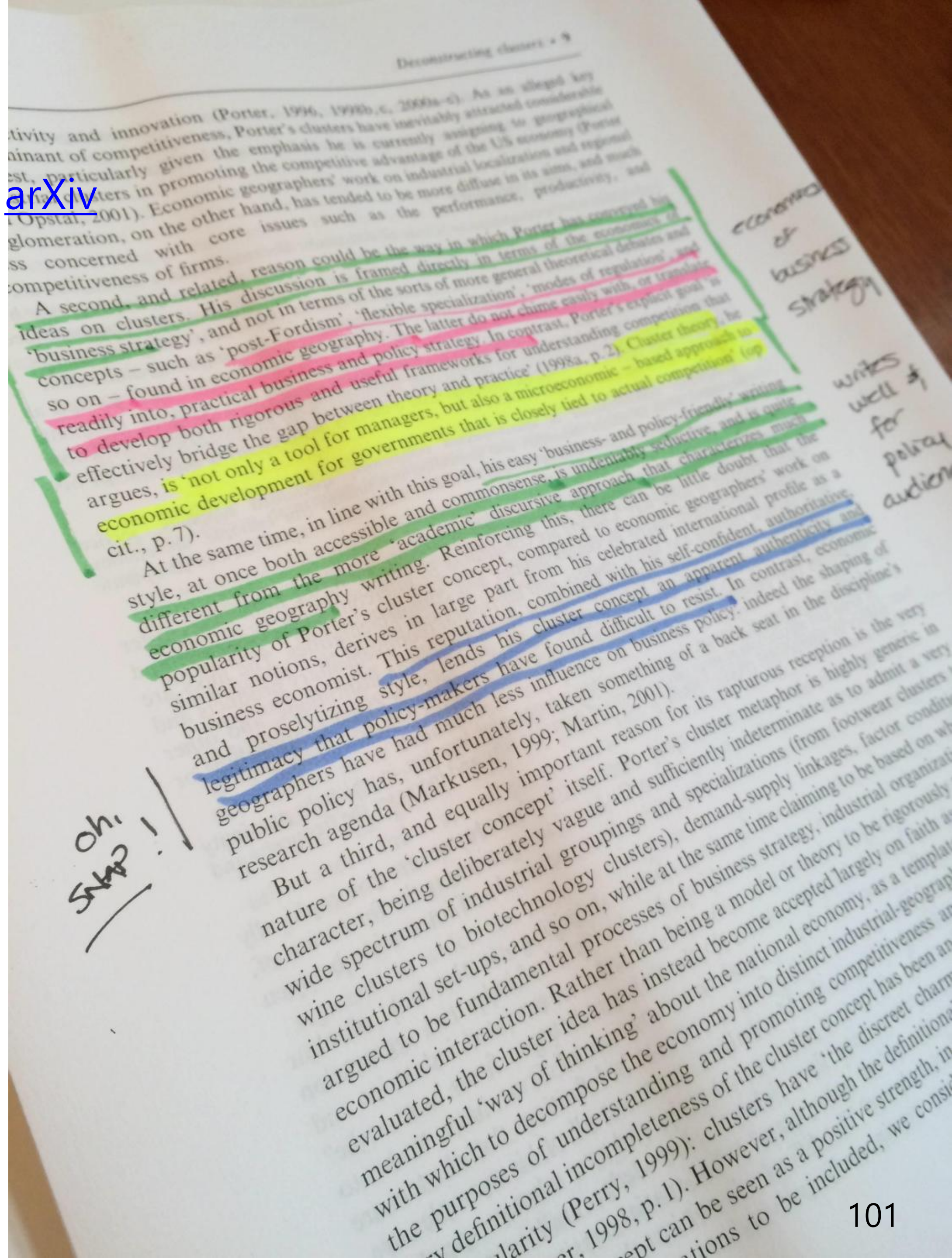


Image: Raul Pacheco-Vega

Transformer

- Paper) Vaswani et al., "Attention Is All You Need", NIPS, 2017 arXiv
  - e.g. Machine translation

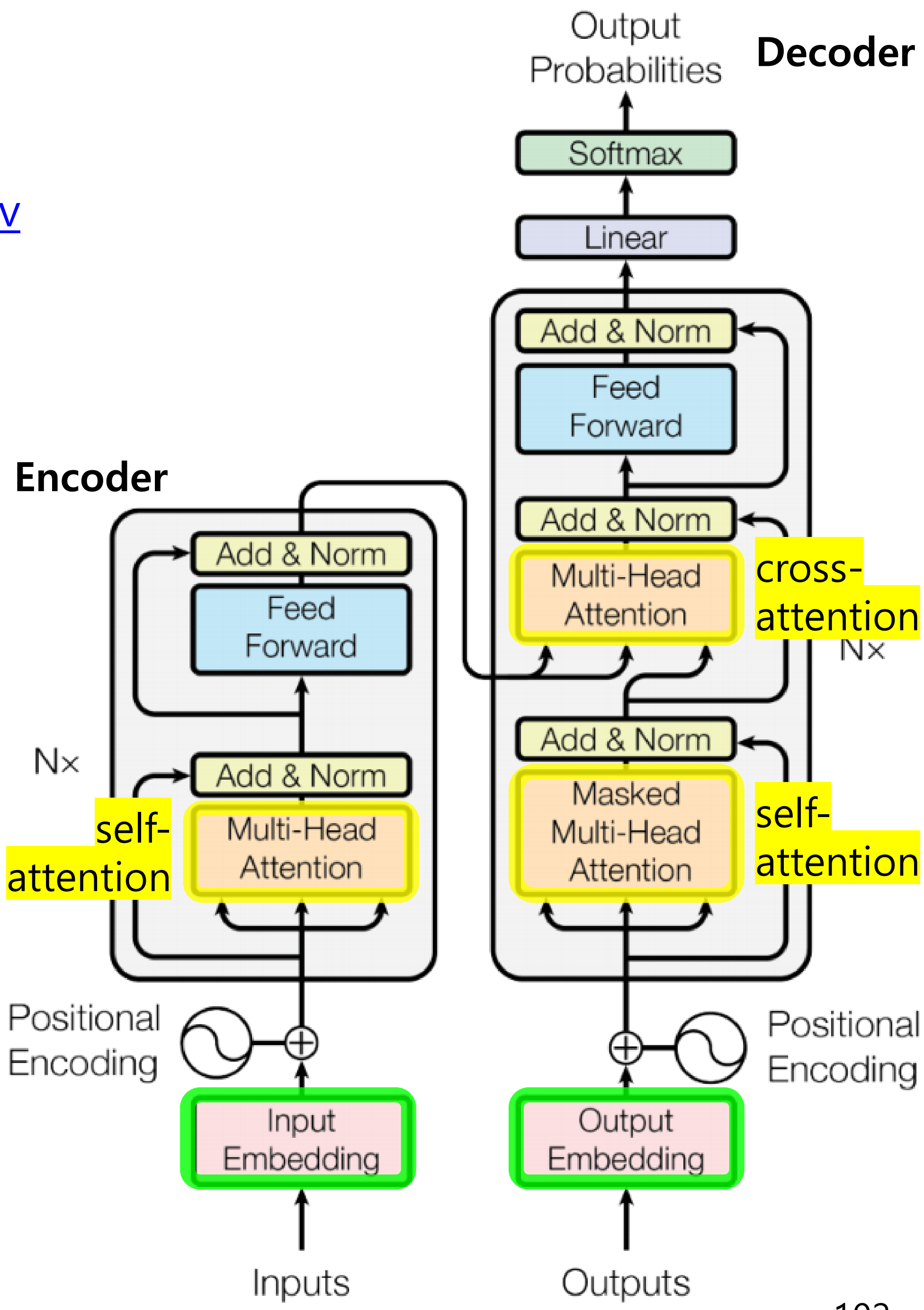


Image: Jakob Uszkoreit, "Transformer: A Novel Neural Network Architecture for Language Understanding", Google Research Blog, 2017 Blog

Transformer

- **Transformer** is a family of NN architectures based on the multi-head attention mechanism.
  - Originally proposed as an **encoder-decoder** architecture for sequence-to-sequence tasks such as machine translation.
    - The encoder and decoder can be used independently depending on the task.
      - e.g. Encoder-only architecture: **BERT** → text classification
      - e.g. Decoder-only architecture: **GPT** → text generation
  - During training, **tokens in a sequence can be processed in parallel**, rather than sequentially as in RNNs.
    - This high degree of **parallelization** enables significantly **faster training** and **better scalability** than RNNs and their variants.
  - It has been widely applied to many other domains including non-sequential tasks (e.g. image classification).

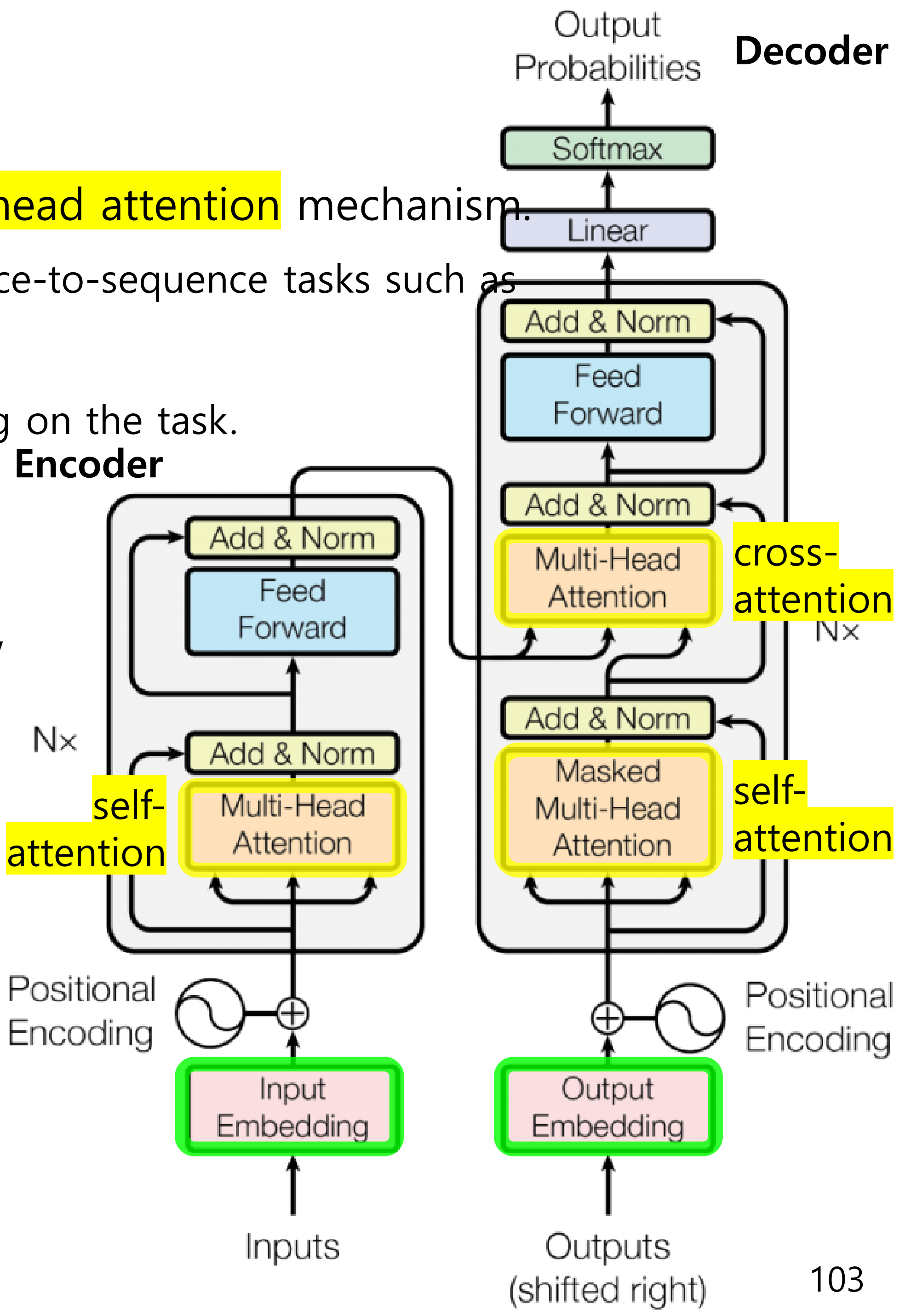


Image: Vaswani et al., "Attention Is All You Need", NIPS, 2017 [arXiv]

Transformer

- **Multi-head attention**
  - Multi-head? Use h parallel attention heads to capture different types of relationships among tokens
  - Input: **Q**uery (질의 in Korean), **K**ey (요약 in Korean), **V**alue (내용 in Korean)
  - **Scaled dot-product attention**
    - $\text{attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$
    - $\text{head}_i = \text{attention}\left(QW_i^Q, KW_i^K, VW_i^V\right)$
    - $\text{multihead}(Q, K, V) = \text{concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)\, W^O$

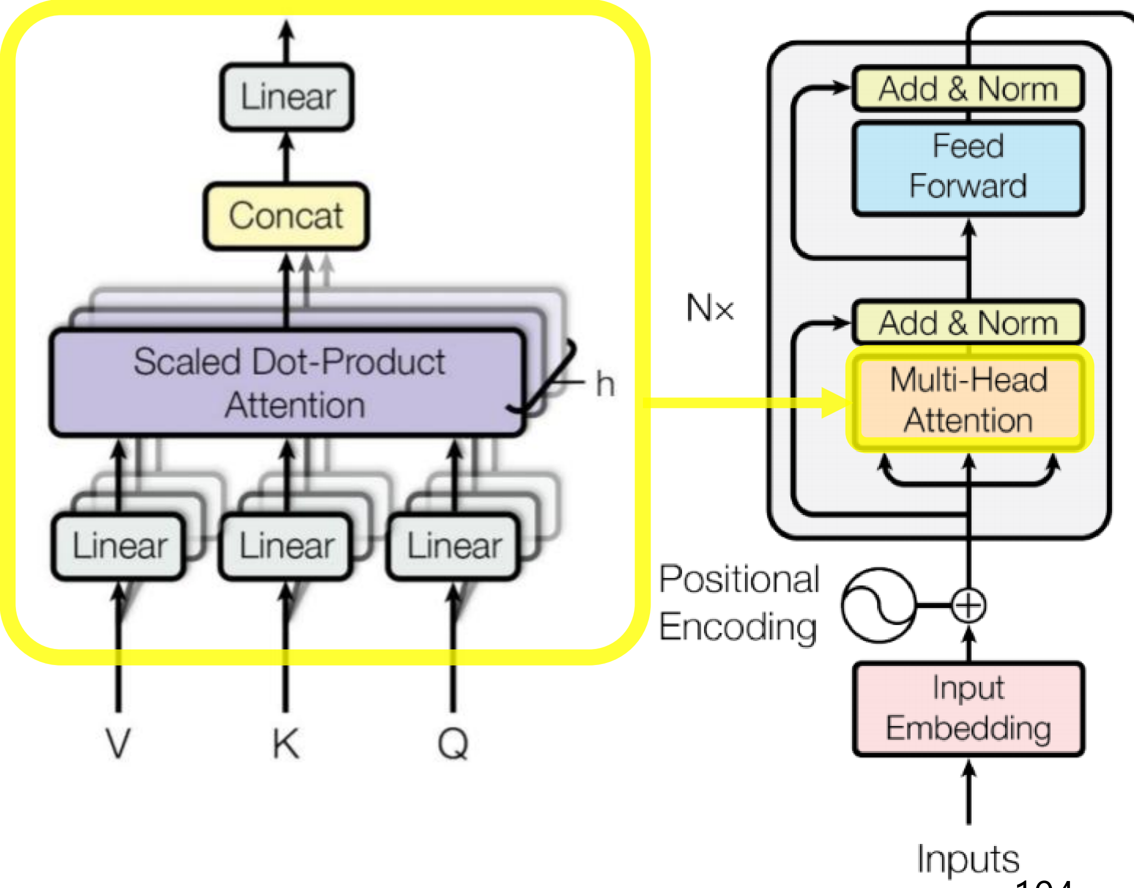


Image: Vaswani et al., "Attention Is All You Need", NIPS, 2017 arXiv

Transformer

- **Multi-head attention**
  - Multi-head? Use h parallel attention heads to capture different types of relationships among tokens
  - Input: Query (질의 in Korean), Key (요약 in Korean), Value (내용 in Korean)
    - Query represents what information a token is looking for
    - Key represents what information a token can be matched by
    - Value represents the actual information to be retrieved or aggregated
  - Analogue) Web searching

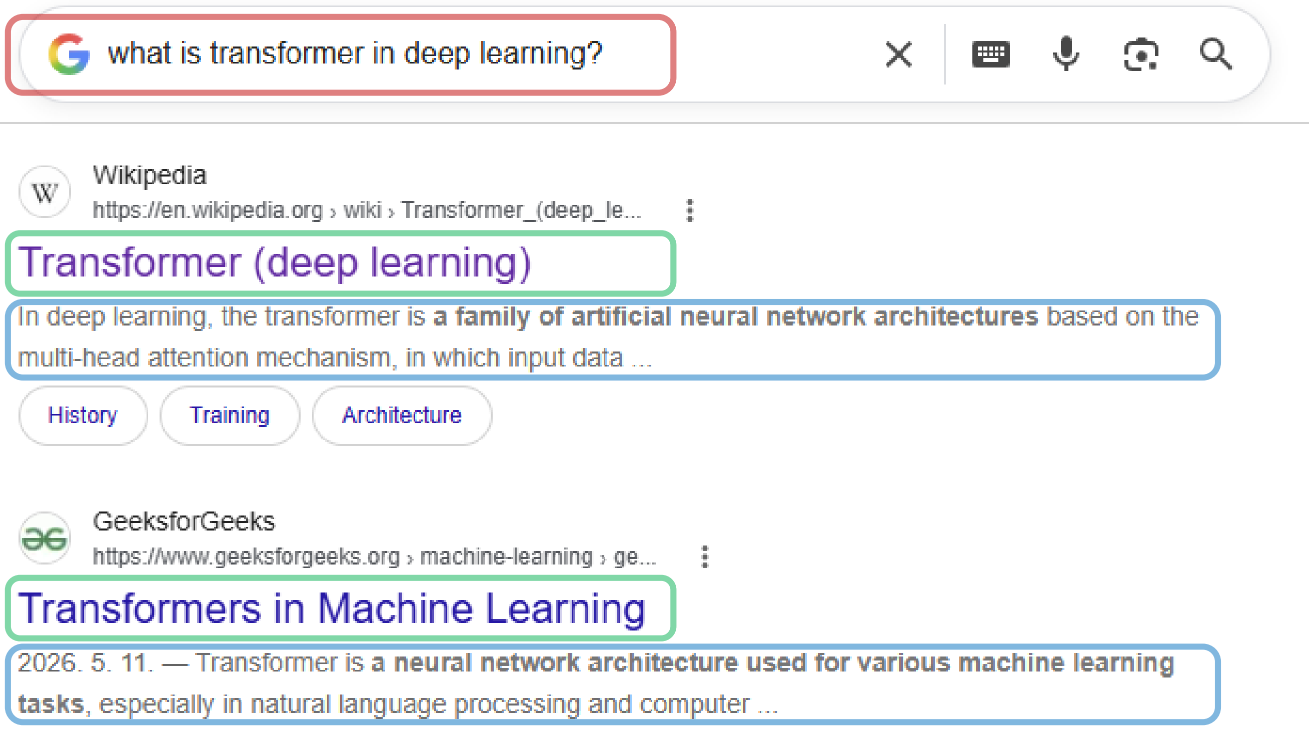
Query

Key

Value

Key

Value



Transformer

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  - **Scaled dot-product attention**
    - $\text{attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$
    - $\text{head}_i = \text{attention}(QW_i^Q, KW_i^K, VW_i^V)$
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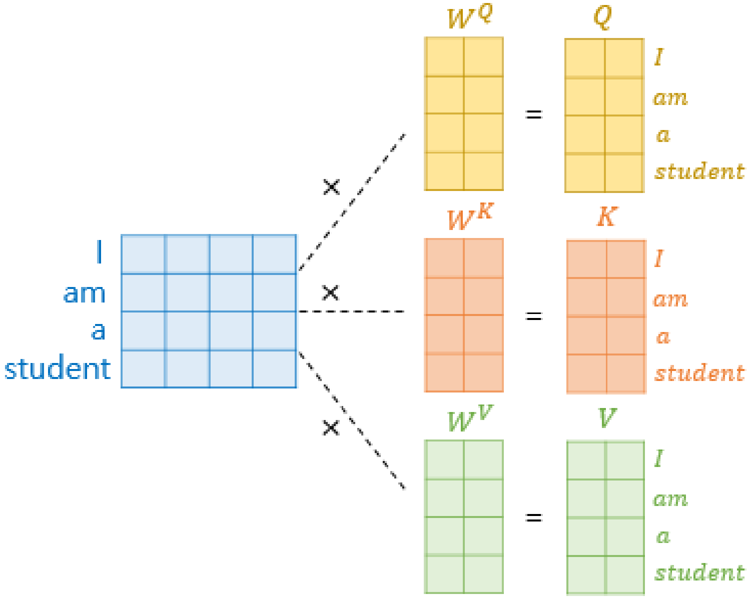

$d_{model} = 4$
$d_q = 2$
$d_k = 2$
$d_v = 2$

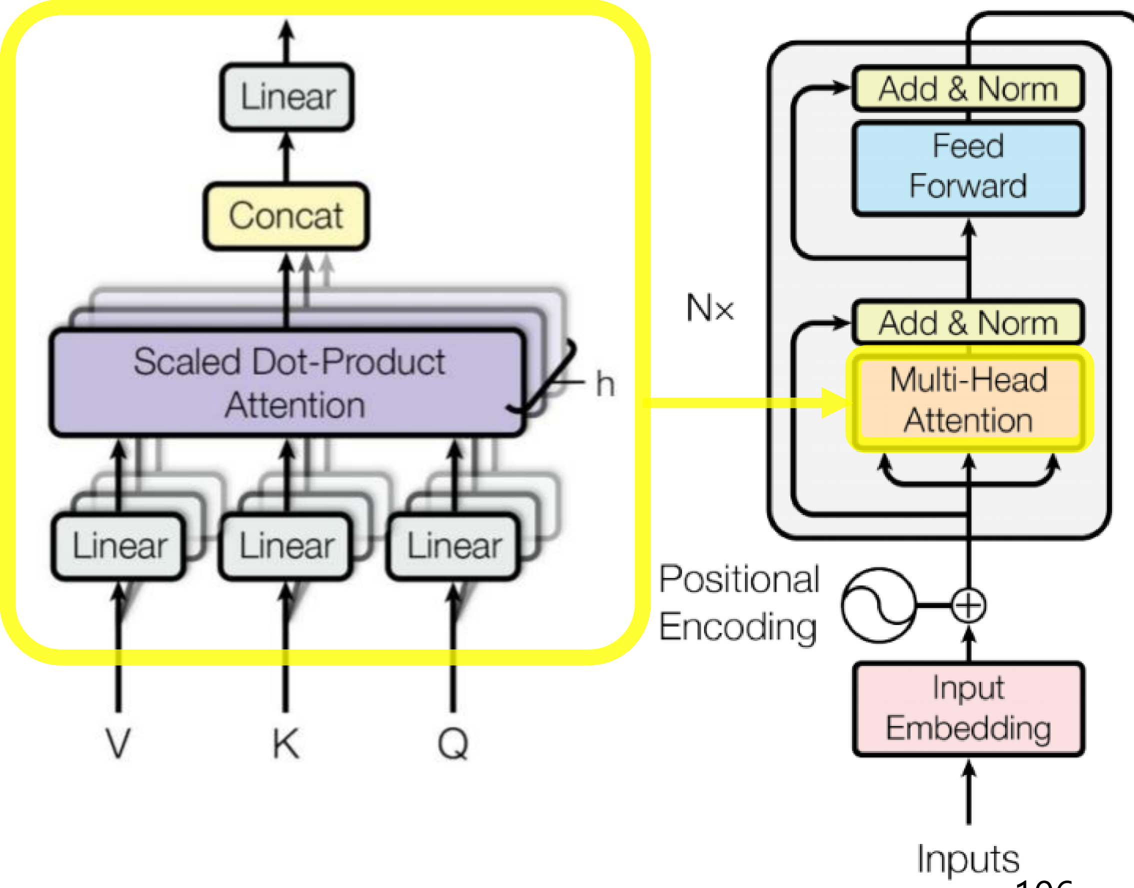

Image: 나동빈, 꼼꼼한 딥러닝 논문 리뷰와 코드 실습 / 유원준, "딥 러닝을 이용한 자연어 처리 입문", Wikidocs

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- **Multi-head attention**
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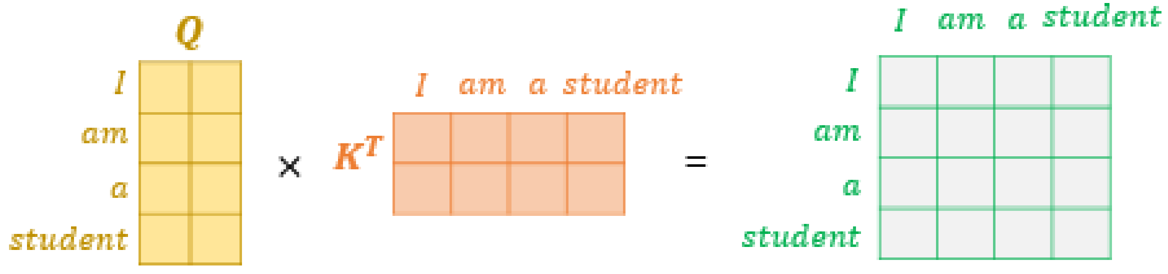
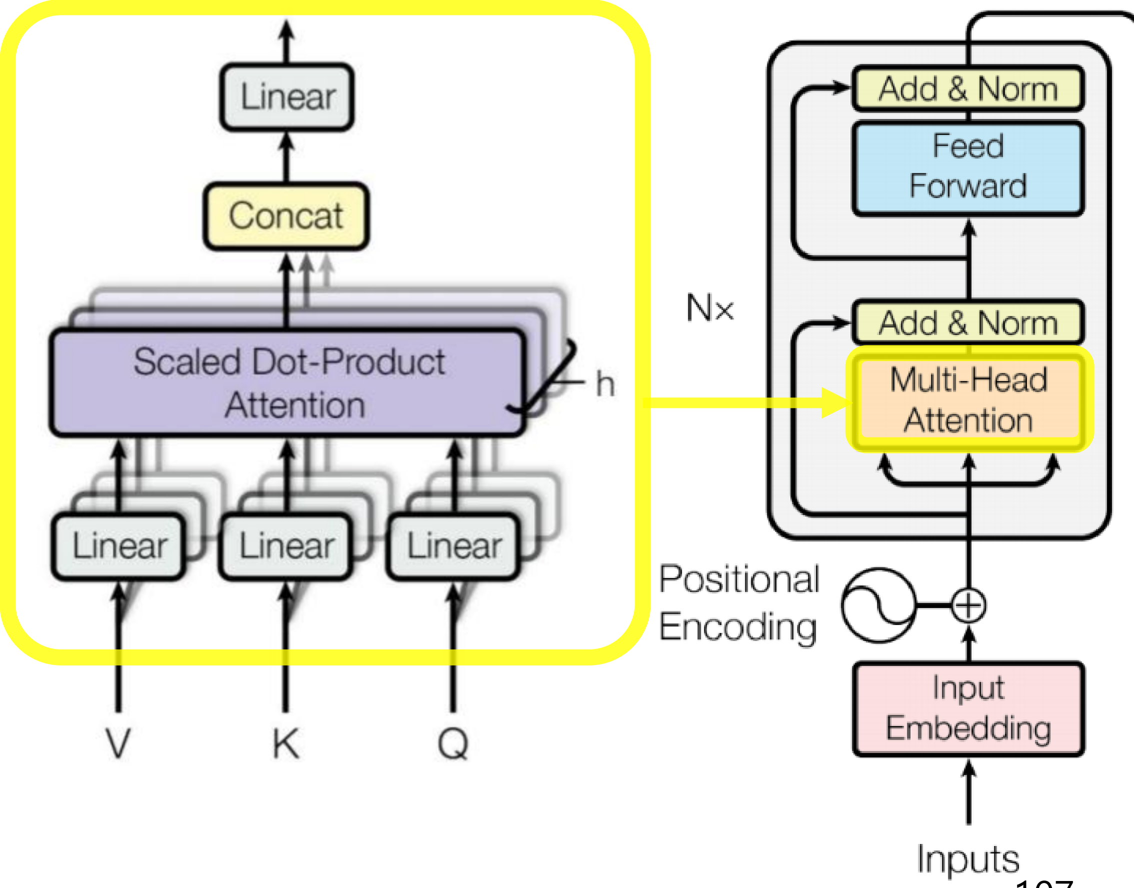
    - Q) Why not compute attention directly?
      (w/o queries and keys)





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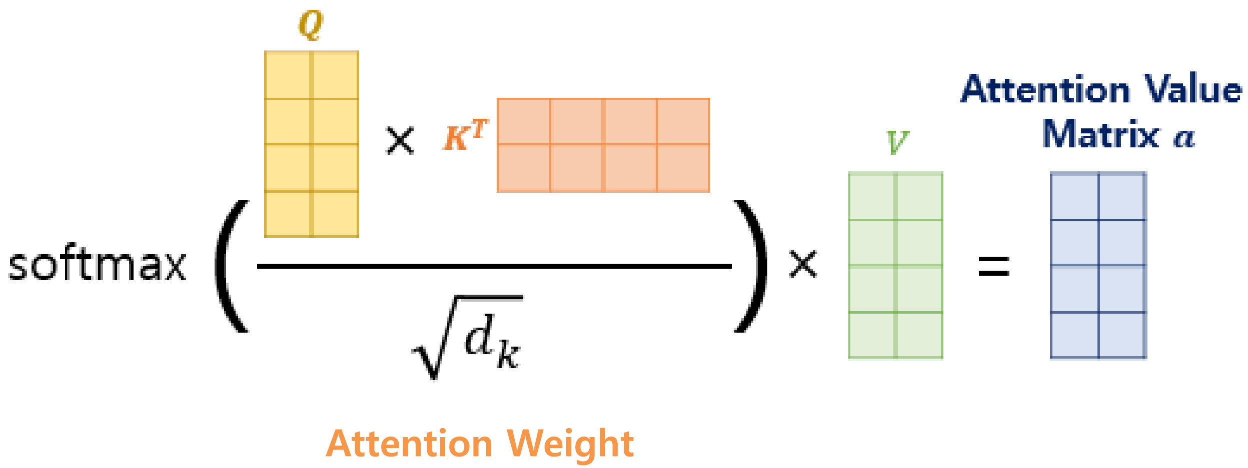

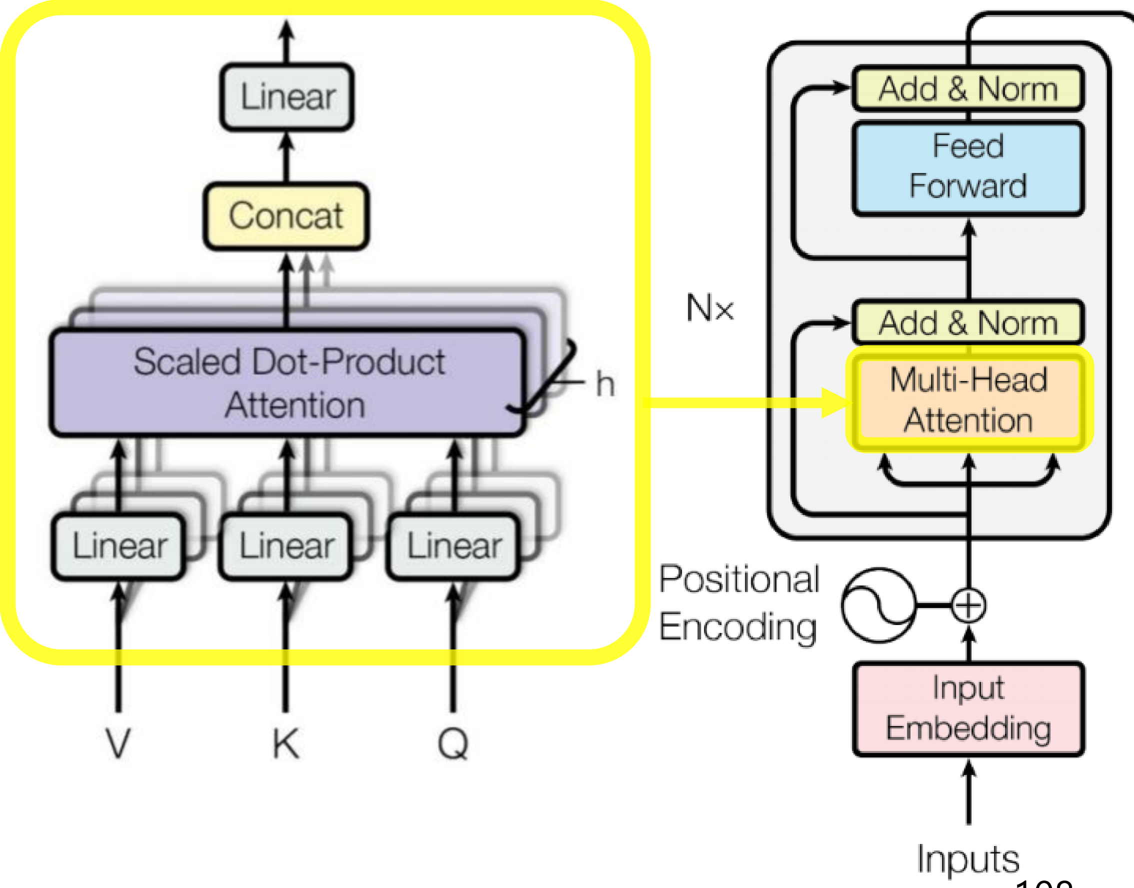

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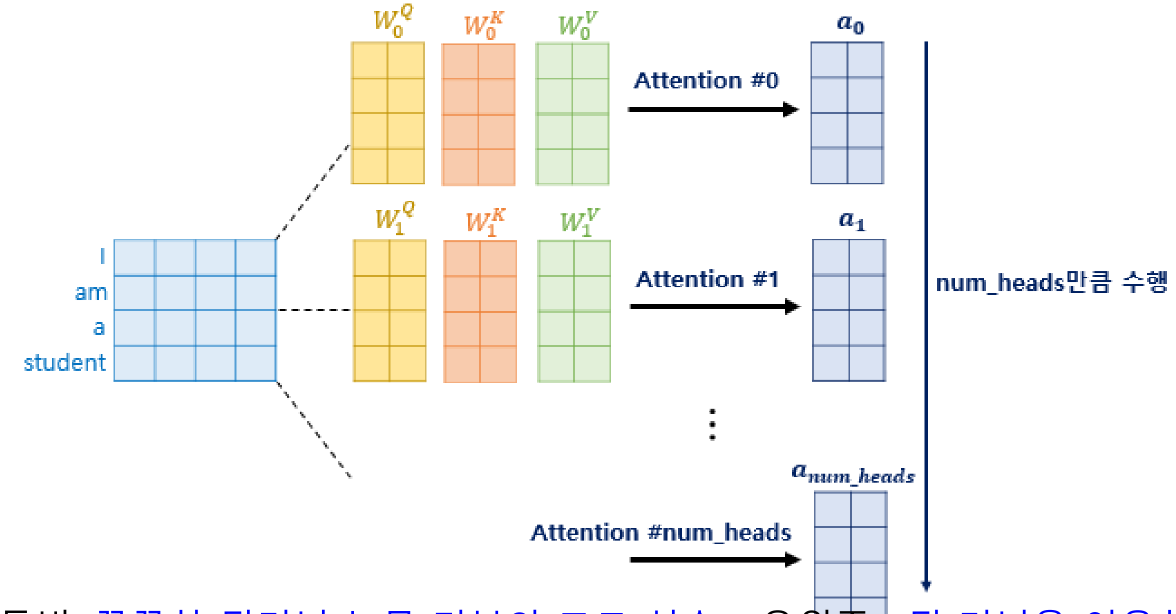

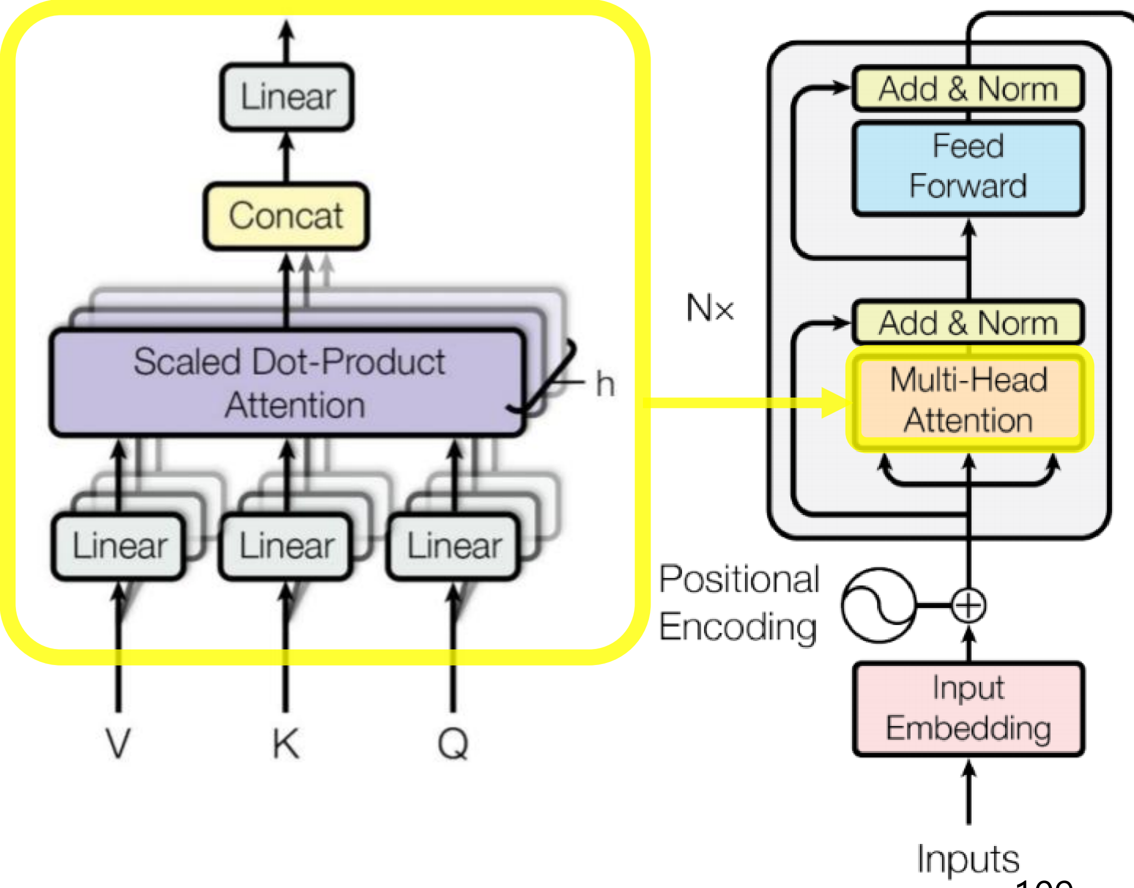

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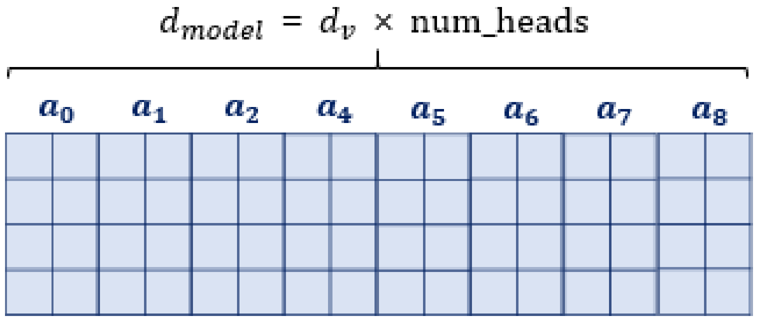
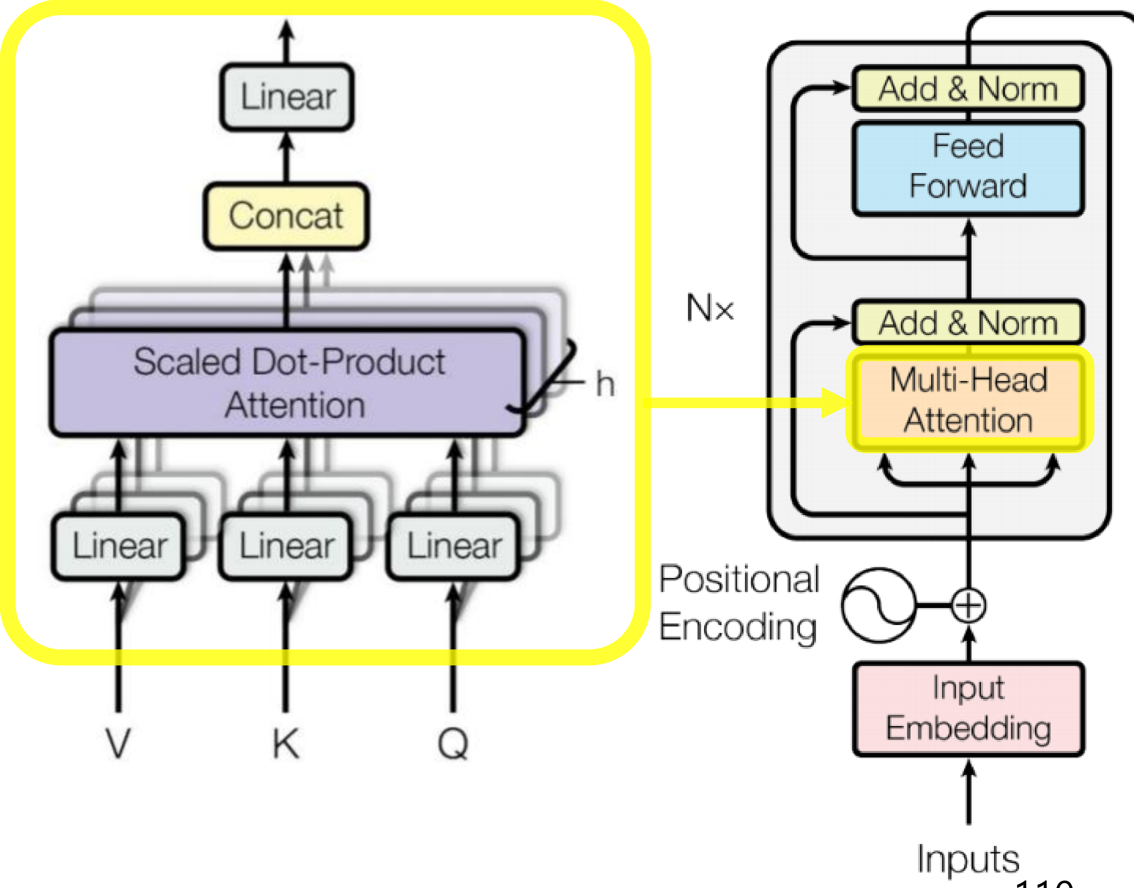


concatenate



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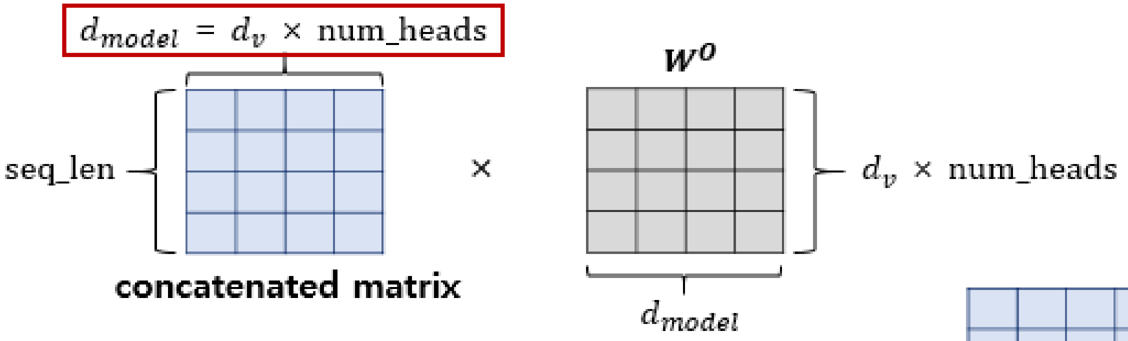
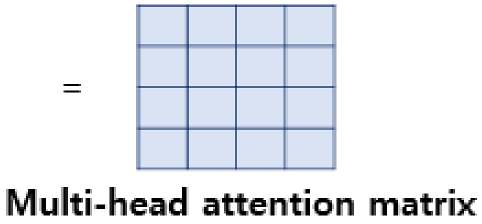
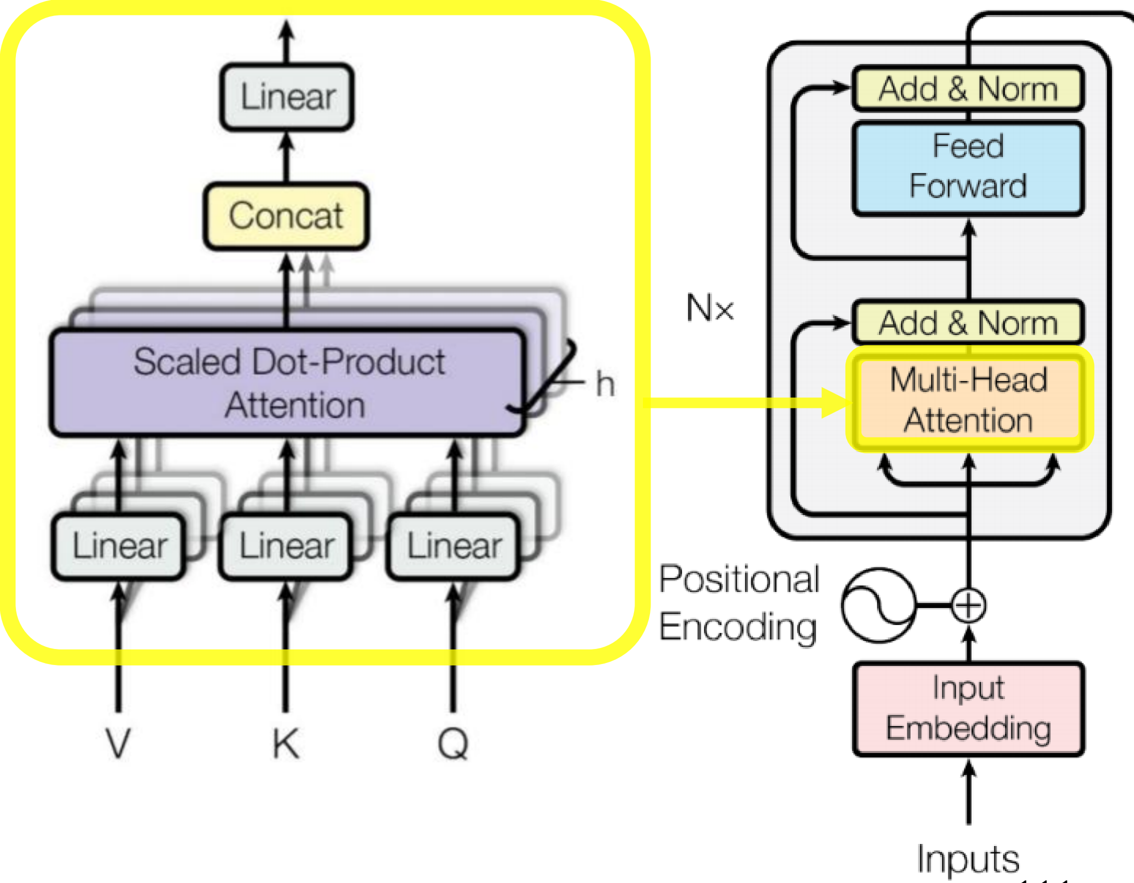




Q) Why does the output maintain the same dimensionality as the input?



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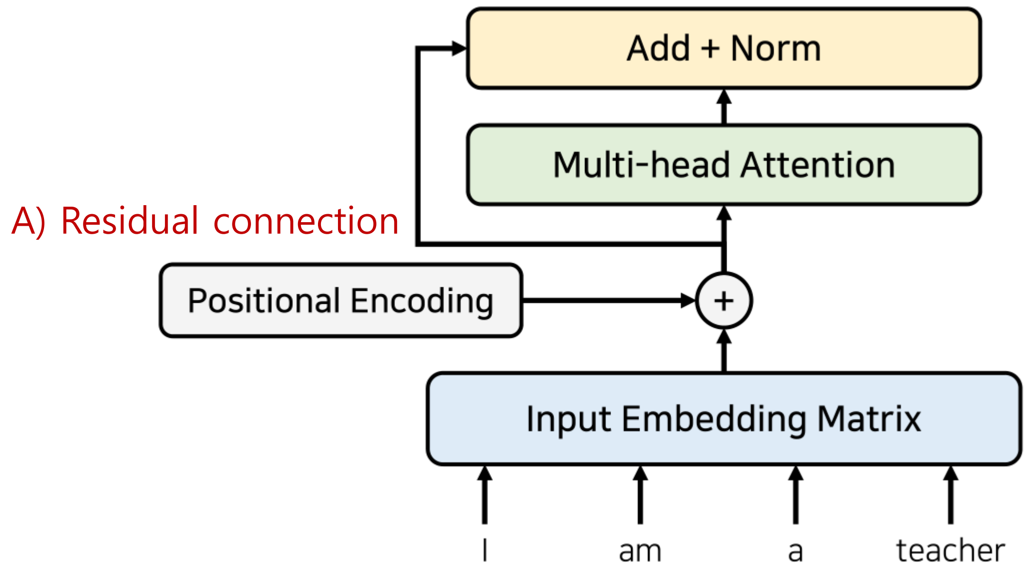
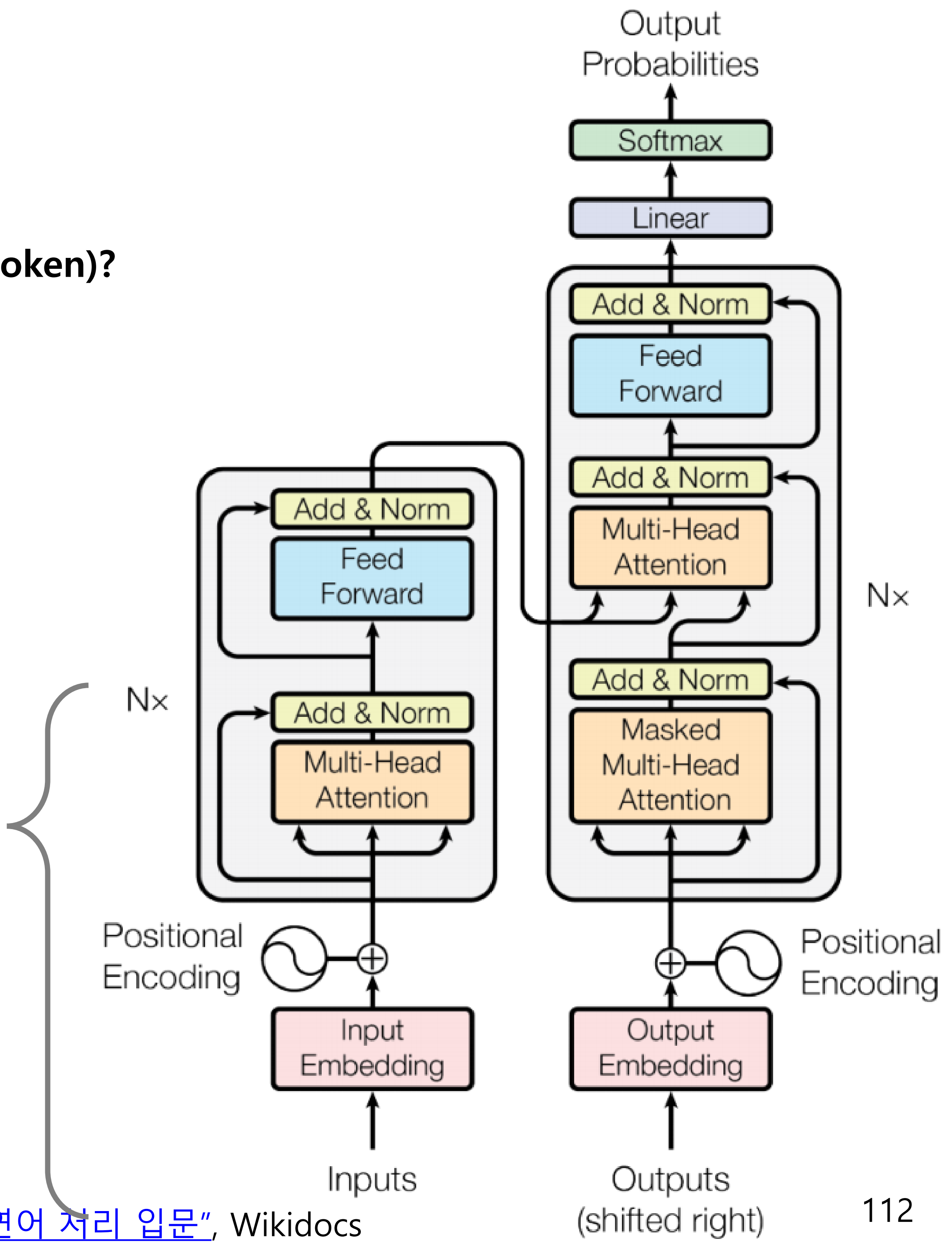
Transformer

- **Positional encoding**
  - **Q) How can the model capture the position of each word (token)?**
    - e.g. I am a student vs. am I a student
  - **A) Add positional information to the token embeddings!**





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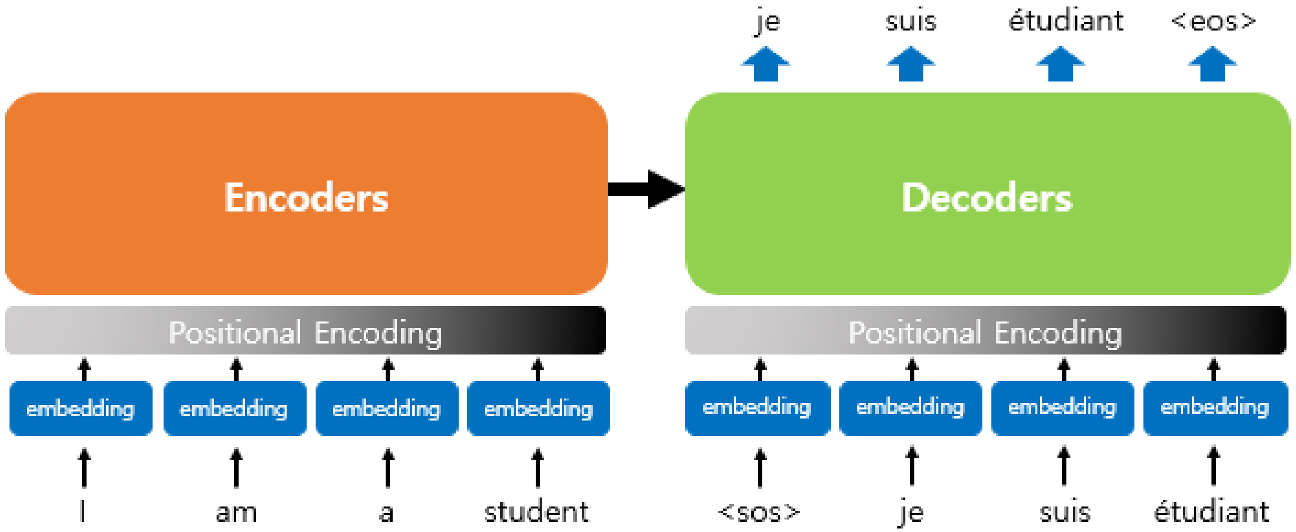

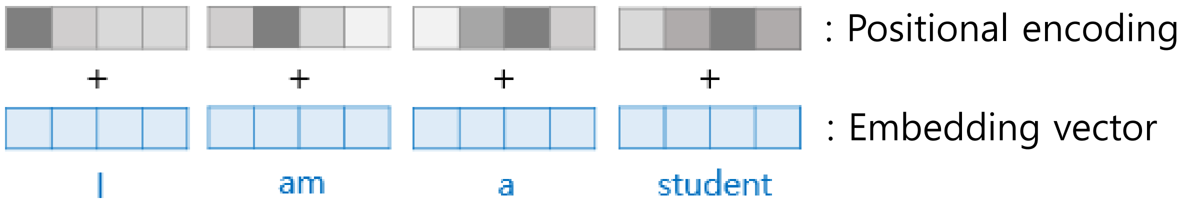

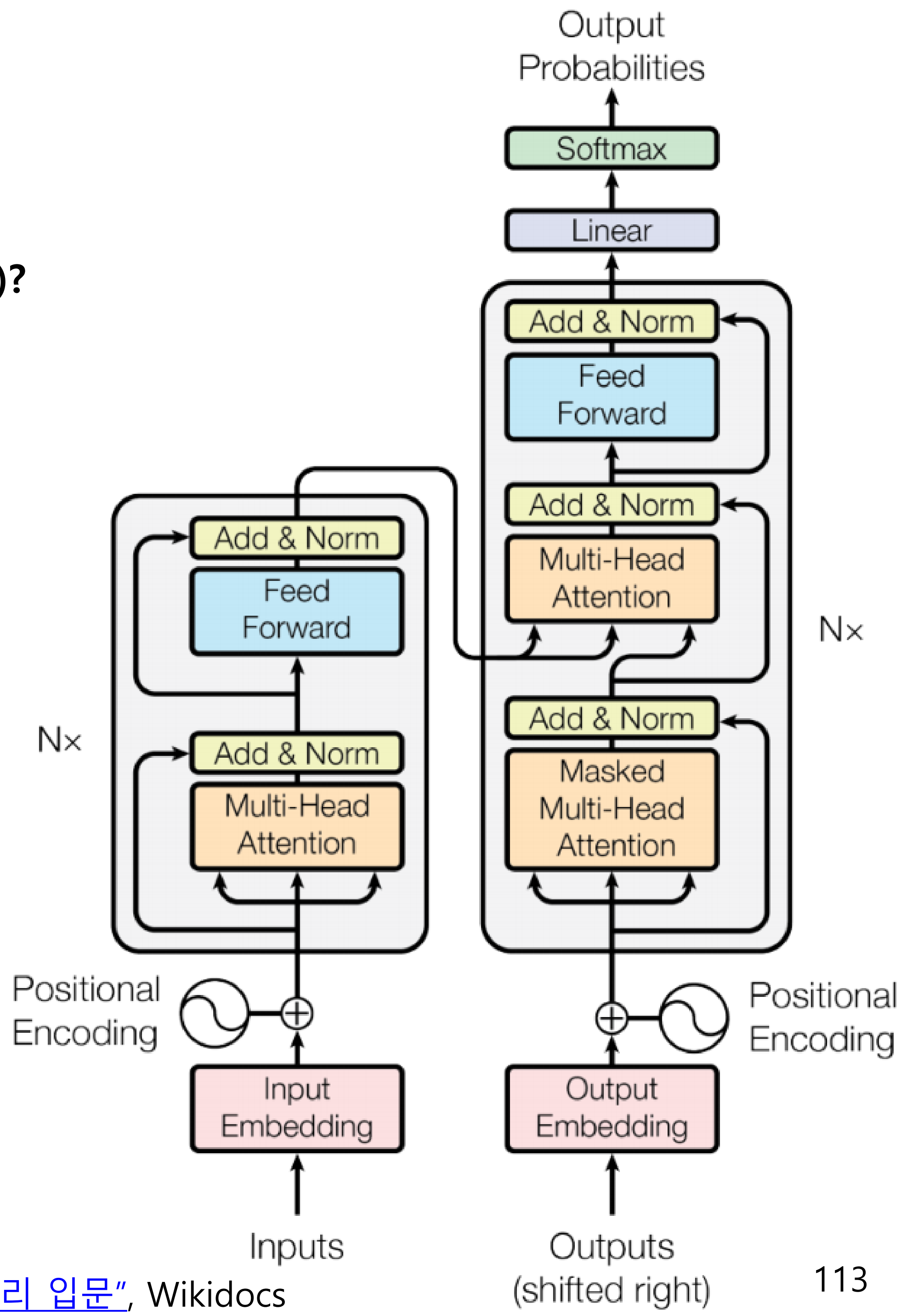

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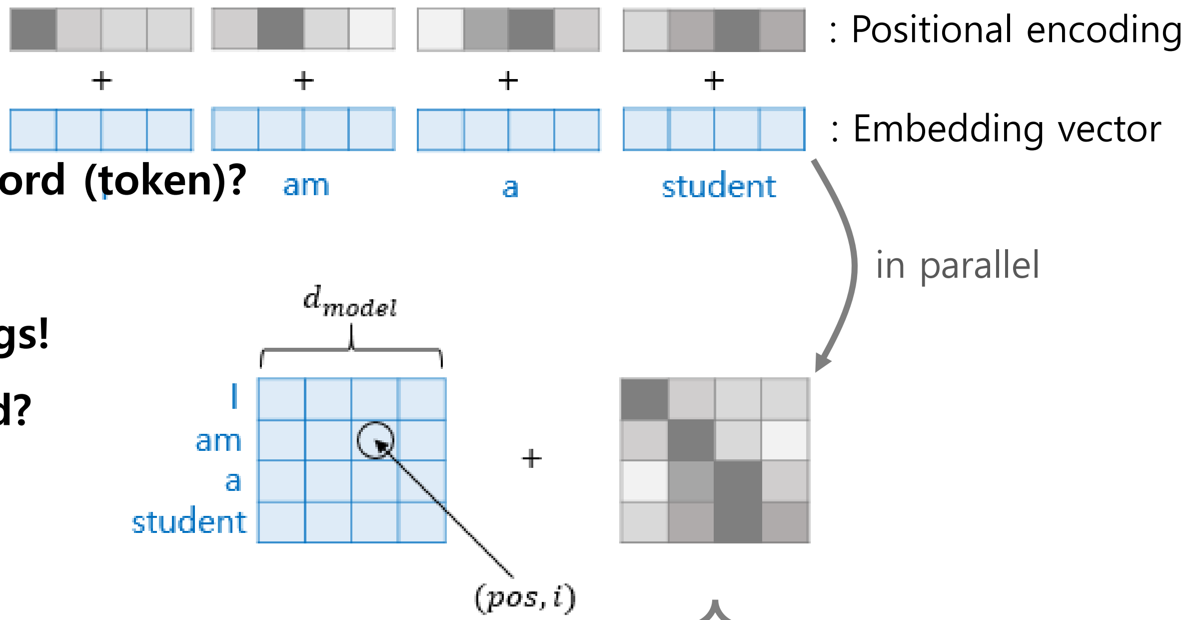
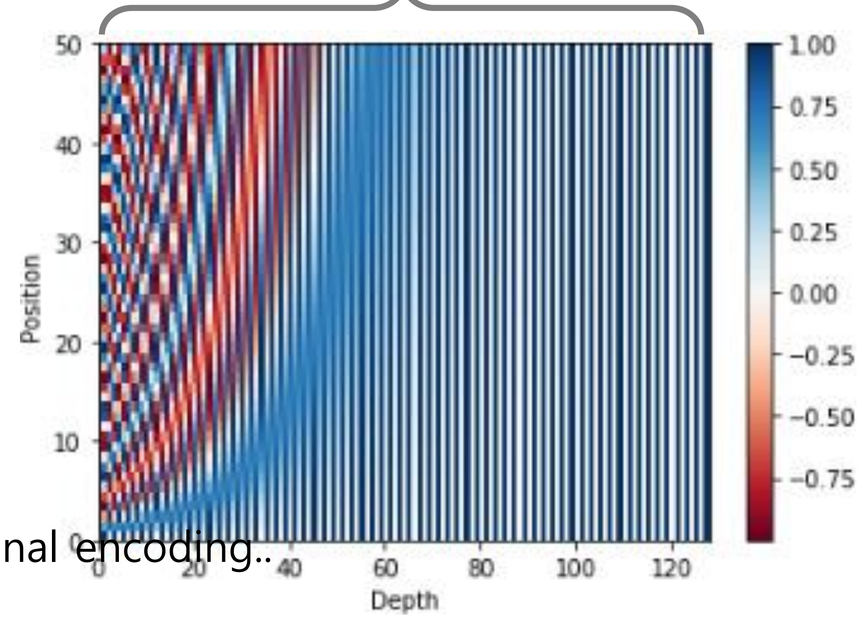
  - **A) Add positional information to the token embeddings!**
  - **Q) How are the positional encoding values determined?**
    - $\mathrm{PE}(pos, 2i) = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$
    - $\mathrm{PE}(pos, 2i+1) = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$
    - Example) d_{model}: 128, sequence length: 50 ([code available](#))
    - Note) Modern Transformer models often use learned positional embeddings, relative positional representations, or RoPE (Rotary Position Embedding) instead of fixed sinusoidal positional encoding..





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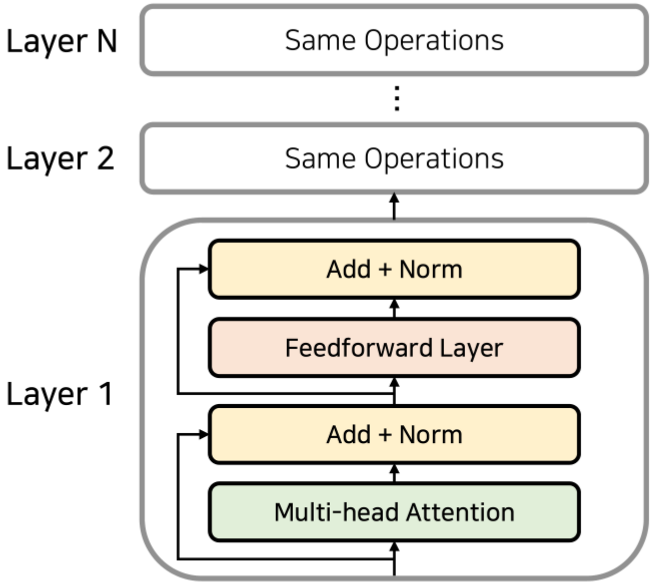
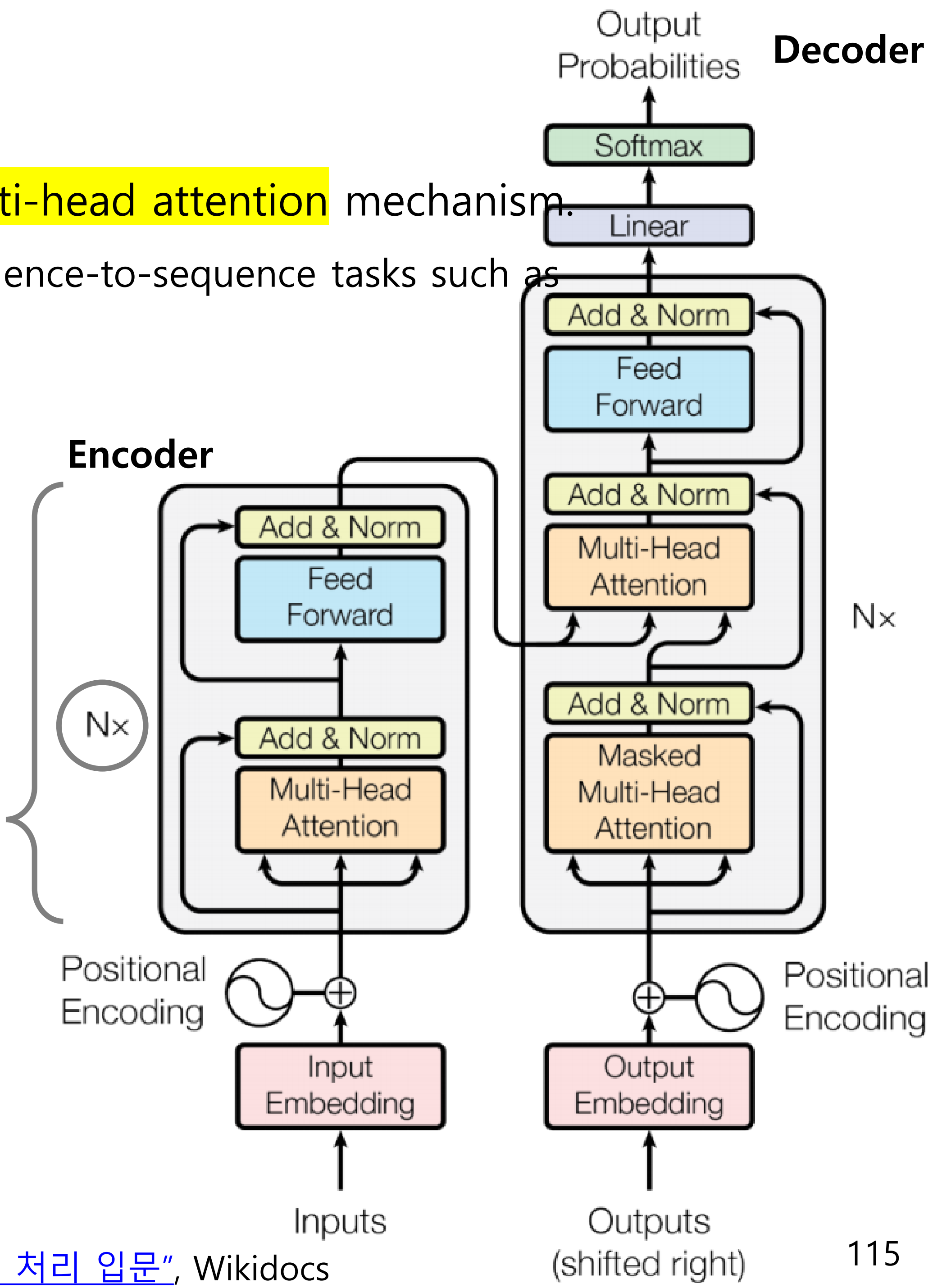
Transformer

- **Transformer** is a family of NN architectures based on the multi-head attention mechanism.
  - Originally proposed as an **encoder**-**decoder** architecture for sequence-to-sequence tasks such as machine translation.





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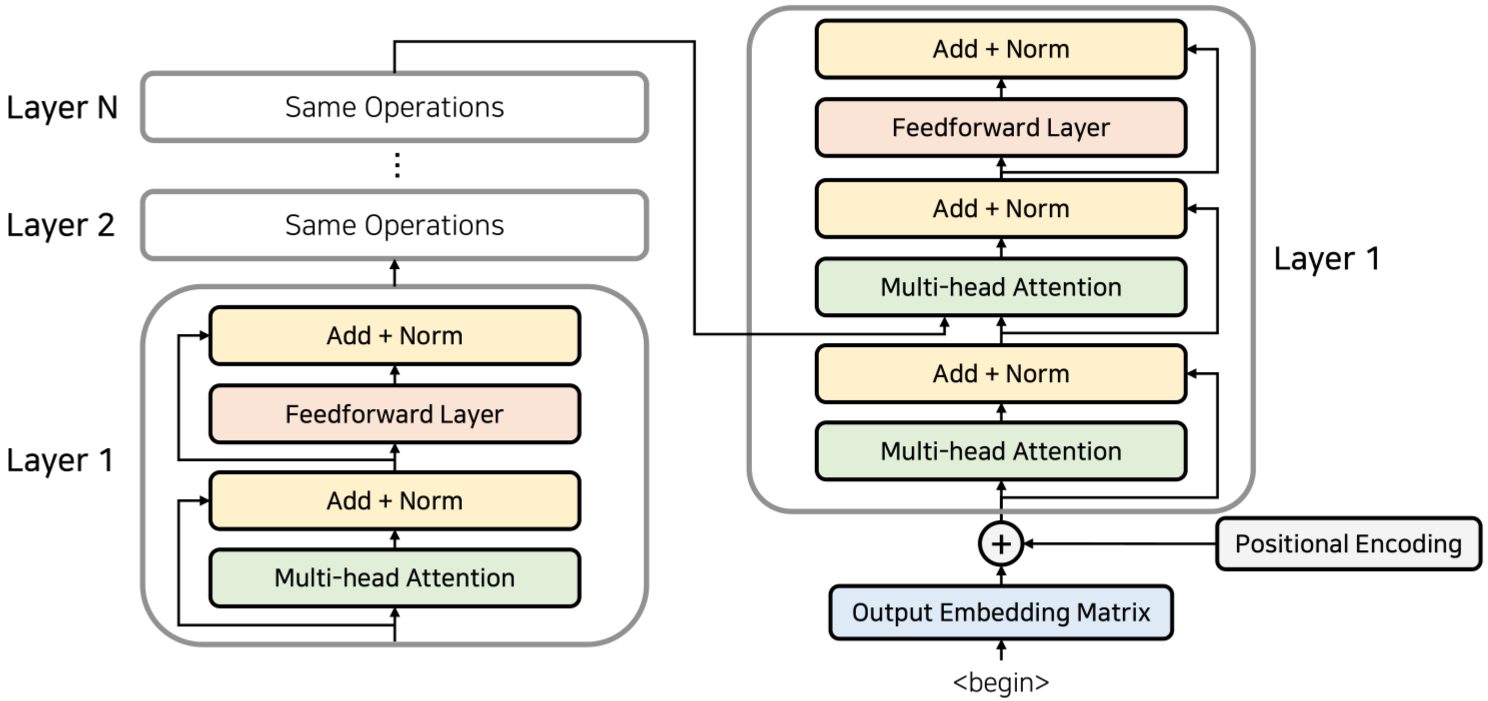

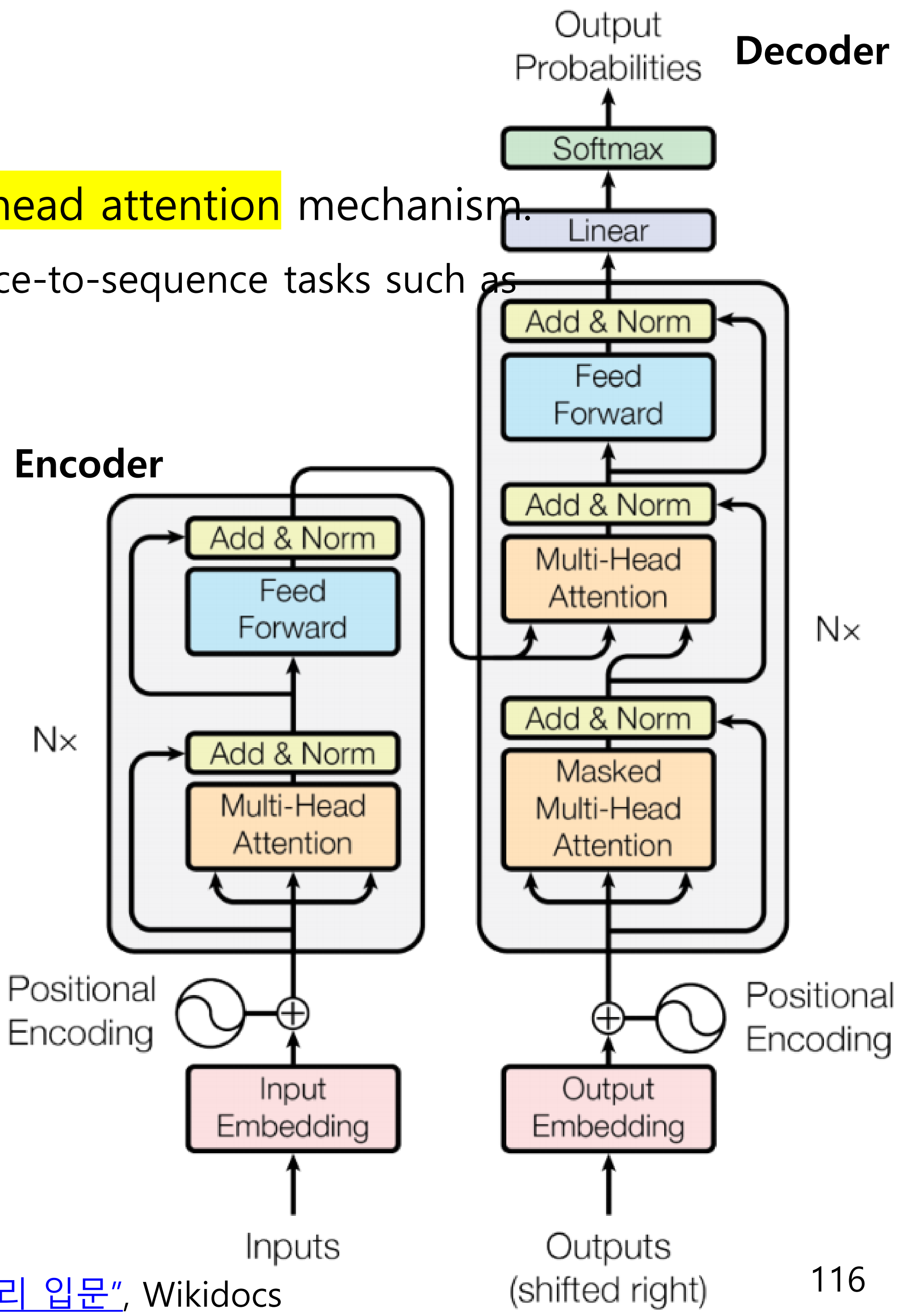

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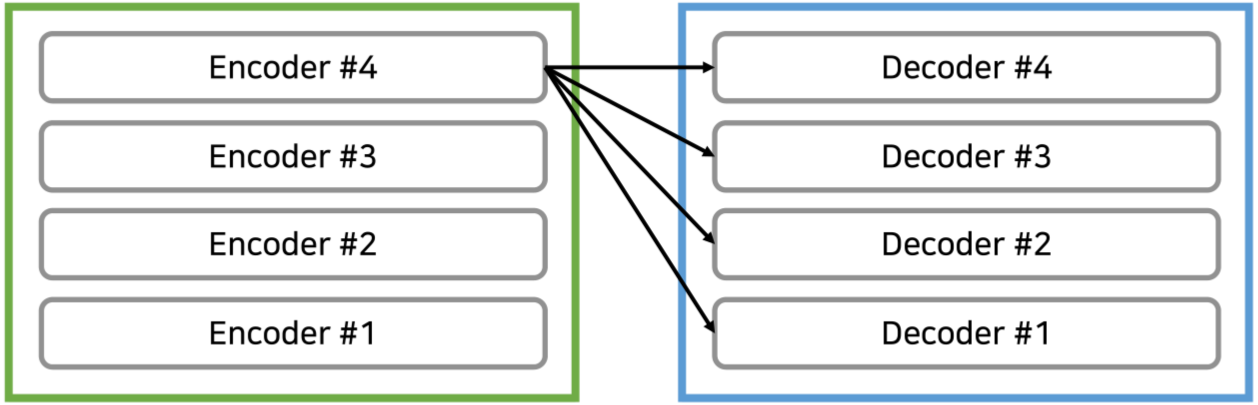
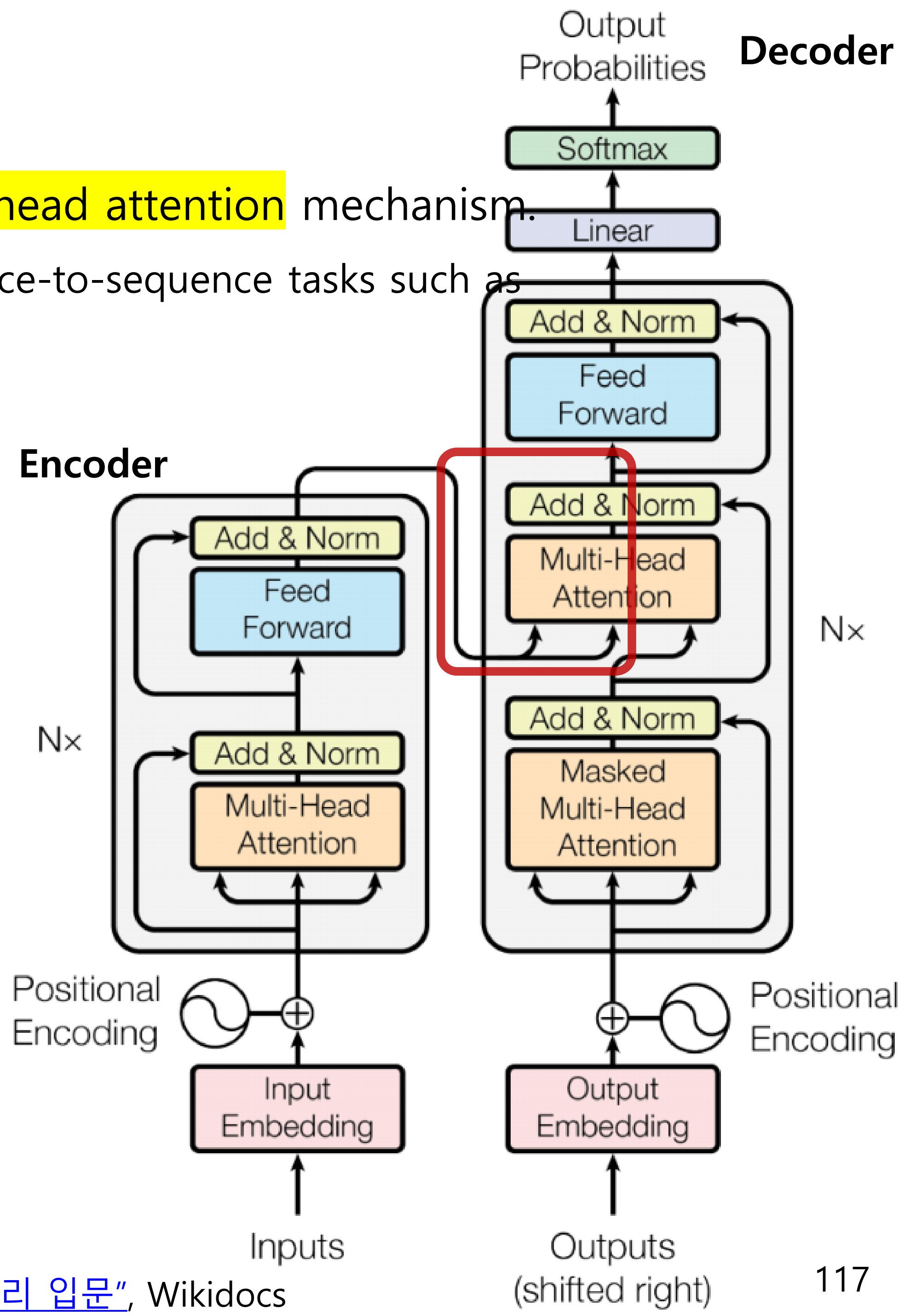
The output of the final encoder is provided as input to each decoder layer.





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    - **Encoder self-attention**: Q, K, V from encoders
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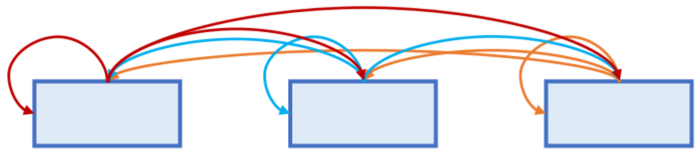
Encoder Self-Attention:



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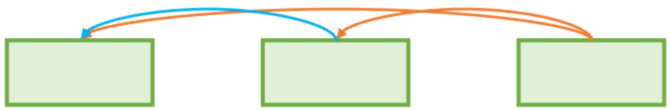
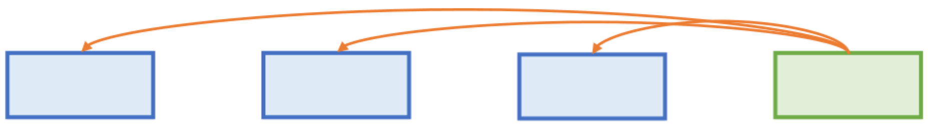
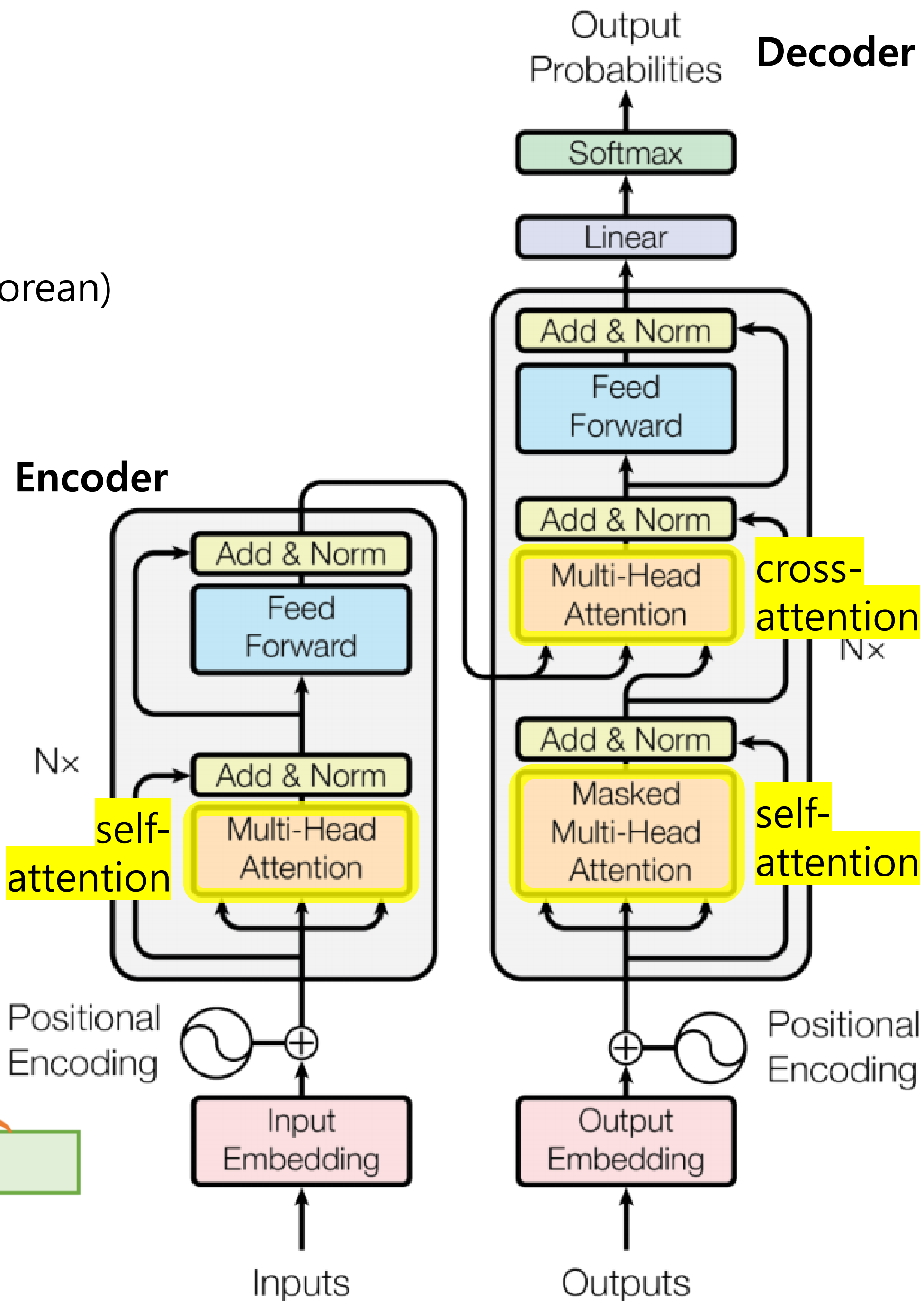


Encoder-Decoder Attention:





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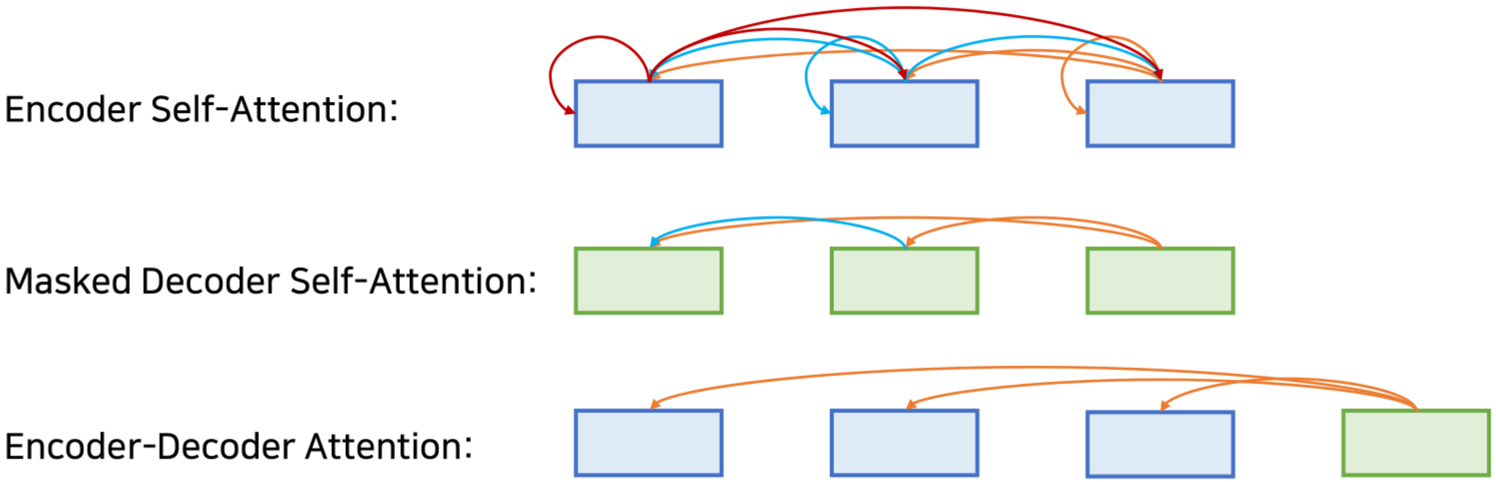
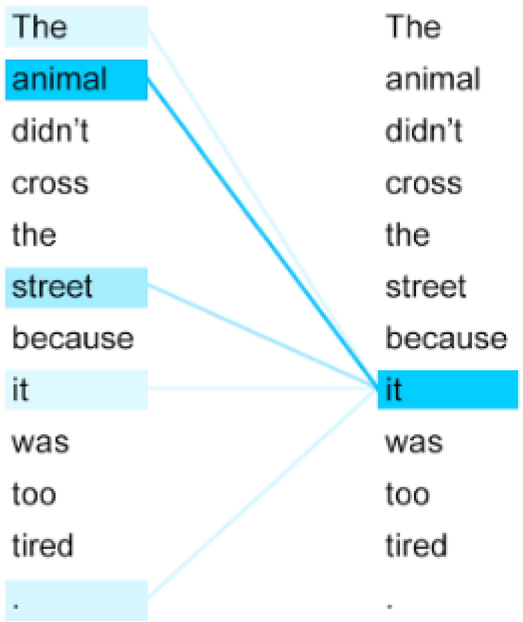


Self-attention?



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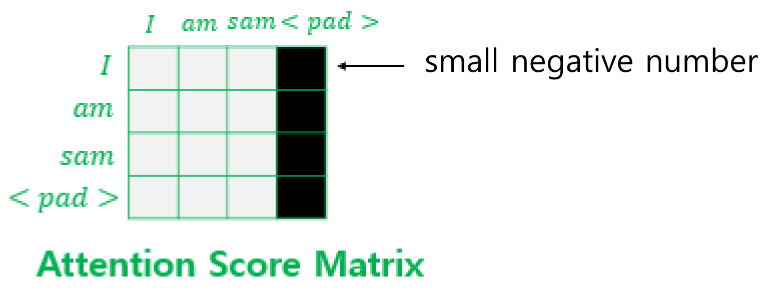
Transformer

Note) **Mask padding tokens** to prevent them from affecting the model's computation



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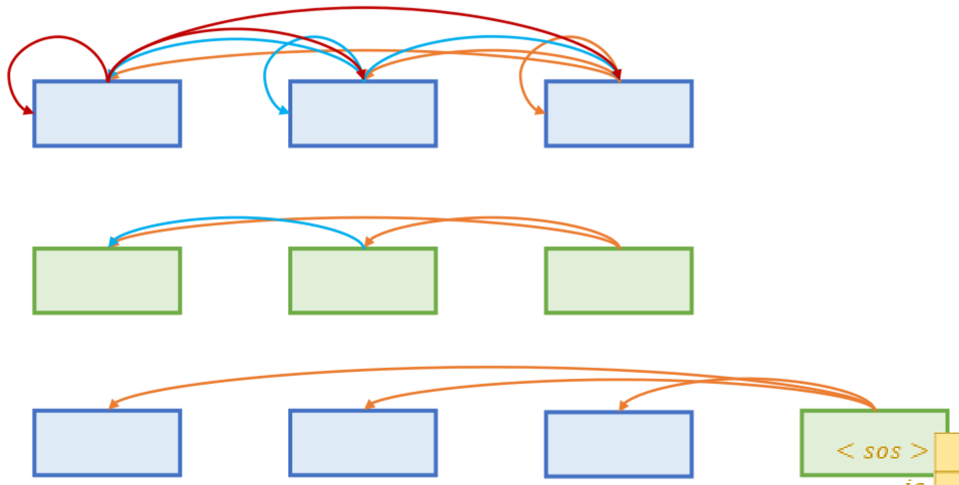

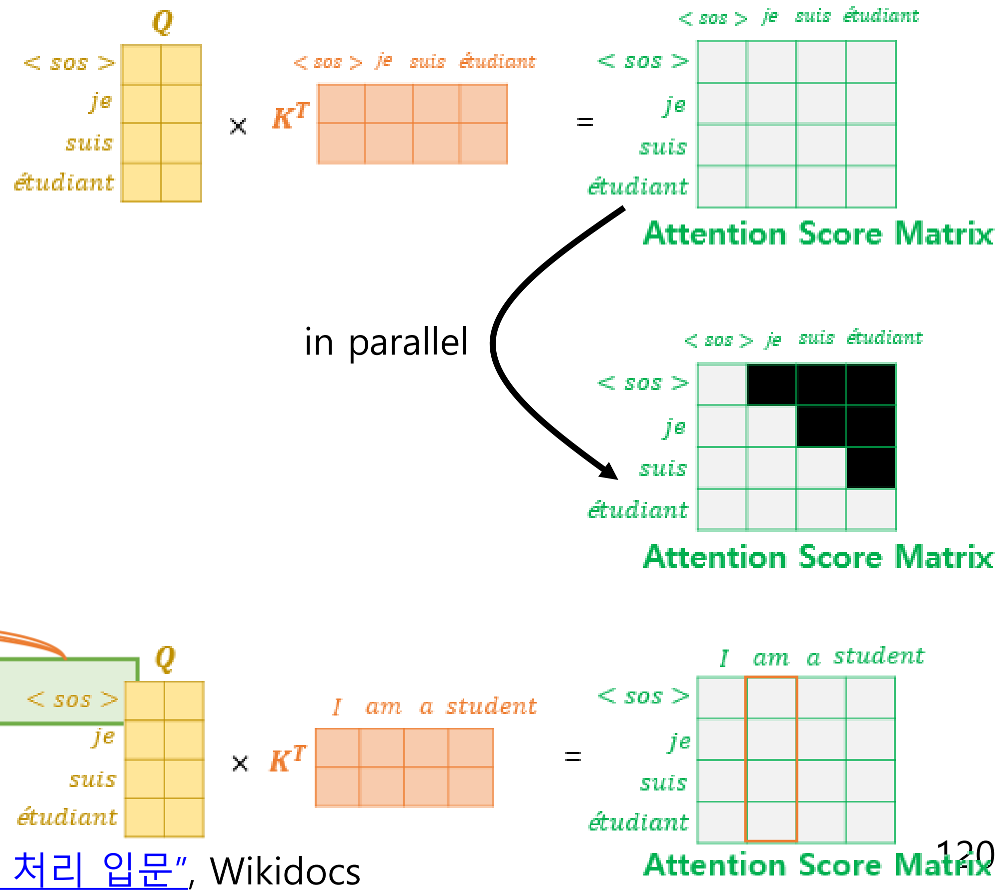

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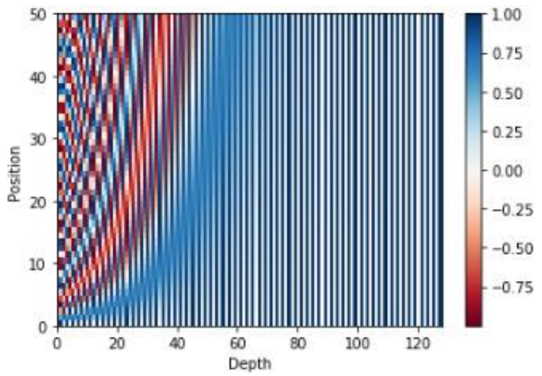
Example) Transformer from Scratch (by graykode)

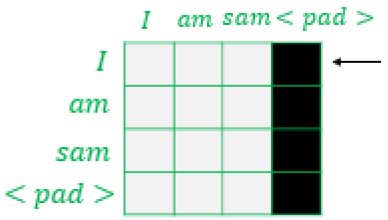
```python
if __name__ == '__main__':
    sentences = ['ich mochte ein bier P', 'S i want a beer', 'i want a beer E']

    # Transformer Parameters
    # Padding Should be Zero
    src_vocab = {'P': 0, 'ich': 1, 'mochte': 2, 'ein': 3, 'bier': 4}
    src_vocab_size = len(src_vocab)

    tgt_vocab = {'P': 0, 'i': 1, 'want': 2, 'a': 3, 'beer': 4, 'S': 5, 'E': 6}
    number_dict = {i: w for i, w in enumerate(tgt_vocab)}
    tgt_vocab_size = len(tgt_vocab)

    src_len = 5    # length of source
    tgt_len = 5    # length of target

    d_model = 512  # Embedding Size
    d_ff = 2048    # FeedForward dimension
    d_k = d_v = 64 # dimension of K(=Q), V
    n_layers = 6   # number of Encoder of Decoder Layer
    n_heads = 8    # number of heads in Multi-Head Attention

    model = Transformer()

    criterion = nn.CrossEntropyLoss()
    optimizer = optim.Adam(model.parameters(), lr=0.001)

    enc_inputs, dec_inputs, target_batch = make_batch(sentences)

    for epoch in range(20):
        optimizer.zero_grad()
        outputs, enc_self_attns, dec_self_attns, dec_enc_attns = model(enc_inputs, dec_inputs)
```

Example) Transformer from Scratch (by graykode)

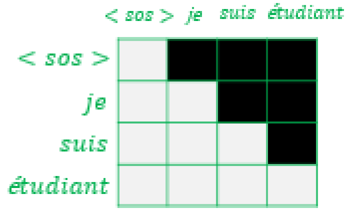
```
def make_batch(sentences):
    input_batch = [[src_vocab[n] for n in sentences[0].split()]]
    output_batch = [[tgt_vocab[n] for n in sentences[1].split()]]
    target_batch = [[tgt_vocab[n] for n in sentences[2].split()]]
    return torch.LongTensor(input_batch), torch.LongTensor(output_batch), torch.LongTensor(target_batch)

def get_sinusoid_encoding_table(n_position, d_model):
    def cal_angle(position, hid_idx):
        return position / np.power(10000, 2 * (hid_idx // 2) / d_model)
    def get_posi_angle_vec(position):
        return [cal_angle(position, hid_j) for hid_j in range(d_model)]

    sinusoid_table = np.array([get_posi_angle_vec(pos_i) for pos_i in range(n_position)])
    sinusoid_table[:, 0::2] = np.sin(sinusoid_table[:, 0::2])  # dim 2i
    sinusoid_table[:, 1::2] = np.cos(sinusoid_table[:, 1::2])  # dim 2i+1
    return torch.FloatTensor(sinusoid_table)
```




```
def get_attn_pad_mask(seq_q, seq_k):
    batch_size, len_q = seq_q.size()
    batch_size, len_k = seq_k.size()
    # eq(zero) marks PAD token as 1.
    pad_attn_mask = seq_k.data.eq(0).unsqueeze(1)          # batch_size x 1 x len_k(=len_q), one is masking
    return pad_attn_mask.expand(batch_size, len_q, len_k)  # batch_size x len_q x len_k
```

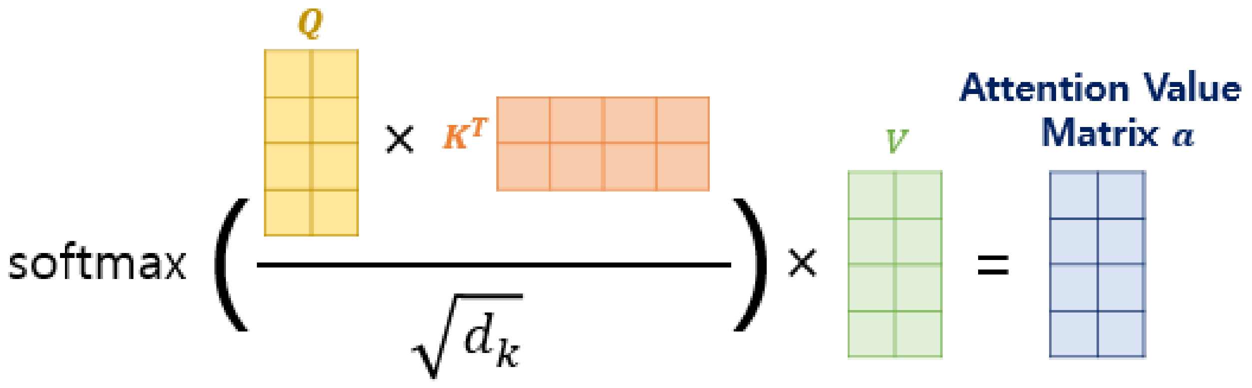



```
def get_attn_subsequent_mask(seq):
    attn_shape = [seq.size(0), seq.size(1), seq.size(1)]
    subsequent_mask = np.triu(np.ones(attn_shape), k=1)
    subsequent_mask = torch.from_numpy(subsequent_mask).byte()
    return subsequent_mask
```

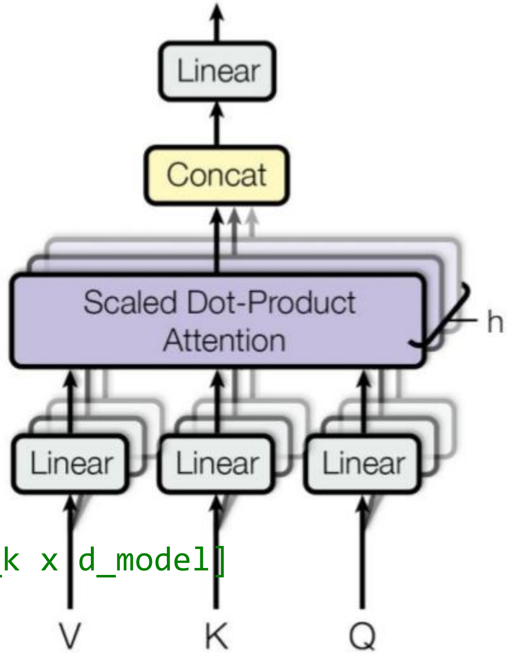


Example) Transformer from Scratch (by graykode)

```
class ScaledDotProductAttention(nn.Module):
    def __init__(self):
        super(ScaledDotProductAttention, self).__init__()

    def forward(self, Q, K, V, attn_mask):
        scores = torch.matmul(Q, K.transpose(-1, -2)) / np.sqrt(d_k) # [batch_size x n_heads x len_q(=len_k) x len_k(=len_q)]
        scores.masked_fill_(attn_mask, -1e9) # Fills elements of self tensor with value where mask is one.
        attn = nn.Softmax(dim=-1)(scores)
        context = torch.matmul(attn, V)
        return context, attn
```



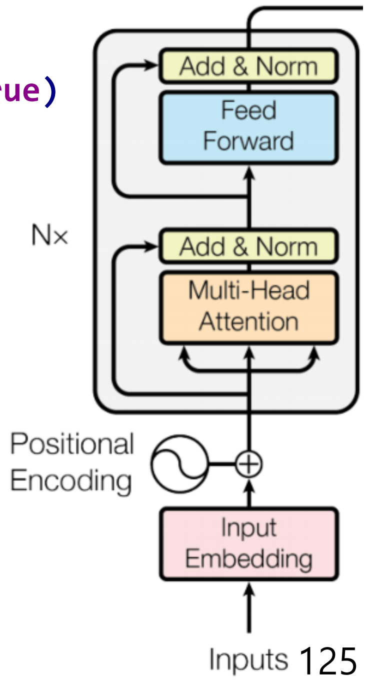
Example) Transformer from Scratch (by graykode)




```python
class MultiHeadAttention(nn.Module):
    def __init__(self):
        super(MultiHeadAttention, self).__init__()
        self.W_Q = nn.Linear(d_model, d_k * n_heads)
        self.W_K = nn.Linear(d_model, d_k * n_heads)
        self.W_V = nn.Linear(d_model, d_v * n_heads)
        self.linear = nn.Linear(n_heads * d_v, d_model)
        self.layer_norm = nn.LayerNorm(d_model)

    def forward(self, Q, K, V, attn_mask):
        # q: [batch_size x len_q x d_model], k: [batch_size x len_k x d_model], v: [batch_size x len_k x d_model]
        residual, batch_size = Q, Q.size(0)
        # (B, S, D) -proj-> (B, S, D) -split-> (B, S, H, W) -trans-> (B, H, S, W)
        q_s = self.W_Q(Q).view(batch_size, -1, n_heads, d_k).transpose(1,2)  # q_s: [batch_size x n_heads x len_q x d_k]
        k_s = self.W_K(K).view(batch_size, -1, n_heads, d_k).transpose(1,2)  # k_s: [batch_size x n_heads x len_k x d_k]
        v_s = self.W_V(V).view(batch_size, -1, n_heads, d_v).transpose(1,2)  # v_s: [batch_size x n_heads x len_k x d_v]

        attn_mask = attn_mask.unsqueeze(1).repeat(1, n_heads, 1, 1) # attn_mask : [batch_size x n_heads x len_q x len_k]

        # context: [batch_size x n_heads x len_q x d_v], attn: [batch_size x n_heads x len_q(=len_k) x len_k(=len_q)]
        context, attn = ScaledDotProductAttention()(q_s, k_s, v_s, attn_mask)
        context = context.transpose(1, 2).contiguous().view(batch_size, -1, n_heads * d_v) # context: [batch_size x len_q x n_heads
        output = self.linear(context)
        return self.layer_norm(output + residual), attn # output: [batch_size x len_q x d_model]
```

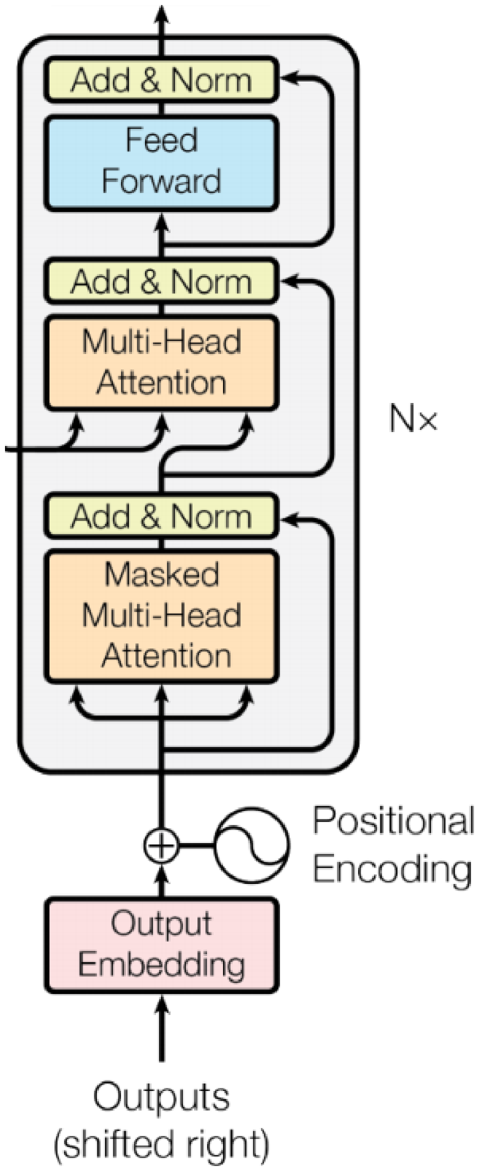
Example) Transformer from Scratch (by graykode)

```python
class EncoderLayer(nn.Module):
    def __init__(self):
        super(EncoderLayer, self).__init__()
        self.enc_self_attn = MultiHeadAttention()
        self.pos_ffn = PoswiseFeedForwardNet()

    def forward(self, enc_inputs, enc_self_attn_mask):
        enc_outputs, attn = self.enc_self_attn(enc_inputs, enc_inputs, enc_inputs, enc_self_attn_mask) # enc_inputs to same Q,K,V
        enc_outputs = self.pos_ffn(enc_outputs) # enc_outputs: [batch_size x len_q x d_model]
        return enc_outputs, attn

class Encoder(nn.Module):
    def __init__(self):
        super(Encoder, self).__init__()
        self.src_emb = nn.Embedding(src_vocab_size, d_model)
        self.pos_emb = nn.Embedding.from_pretrained(get_sinusoid_encoding_table(src_len+1, d_model),freeze=True)
        self.layers = nn.ModuleList([EncoderLayer() for _ in range(n_layers)])

    def forward(self, enc_inputs): # enc_inputs : [batch_size x source_len]
        enc_outputs = self.src_emb(enc_inputs) + self.pos_emb(torch.LongTensor([[1,2,3,4,0]]))
        enc_self_attn_mask = get_attn_pad_mask(enc_inputs, enc_inputs)
        enc_self_attns = []
        for layer in self.layers:
            enc_outputs, enc_self_attn = layer(enc_outputs, enc_self_attn_mask)
            enc_self_attns.append(enc_self_attn)
        return enc_outputs, enc_self_attns
```



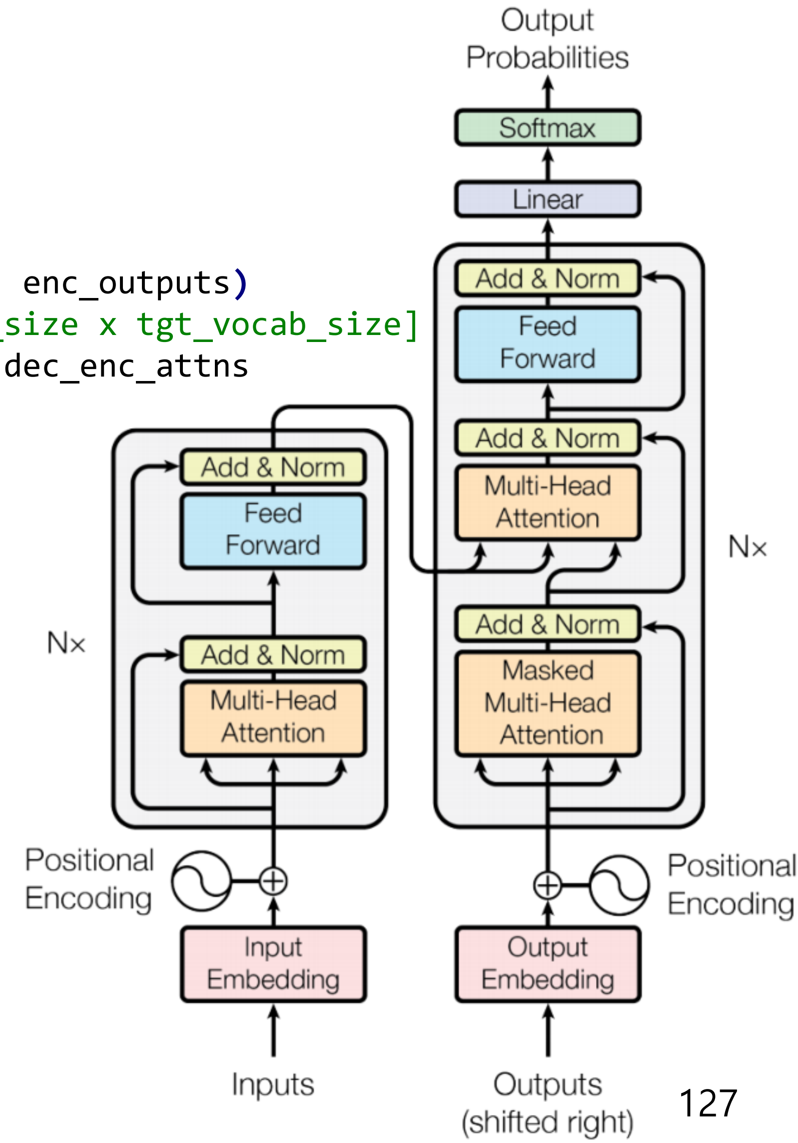
Example) Transformer from Scratch (by graykode)

```
class DecoderLayer(nn.Module):
    def __init__(self):
        super(DecoderLayer, self).__init__()
        self.dec_self_attn = MultiHeadAttention()
        self.dec_enc_attn = MultiHeadAttention()
        self.pos_ffn = PoswiseFeedForwardNet()

    def forward(self, dec_inputs, enc_outputs, dec_self_attn_mask, dec_enc_attn_mask):
        dec_outputs, dec_self_attn = self.dec_self_attn(dec_inputs, dec_inputs, dec_inputs, dec_self_attn_mask)
        dec_outputs, dec_enc_attn = self.dec_enc_attn(dec_outputs, enc_outputs, enc_outputs, dec_enc_attn_mask)
        dec_outputs = self.pos_ffn(dec_outputs)
        return dec_outputs, dec_self_attn, dec_enc_attn

class Decoder(nn.Module):
    def __init__(self):
        super(Decoder, self).__init__()
        self.tgt_emb = nn.Embedding(tgt_vocab_size, d_model)
        self.pos_emb = nn.Embedding.from_pretrained(get_sinusoid_encoding_table(tgt_len+1, d_model),freeze=True)
        self.layers = nn.ModuleList([DecoderLayer() for _ in range(n_layers)])

    def forward(self, dec_inputs, enc_inputs, enc_outputs): # dec_inputs : [batch_size x target_len]
        dec_outputs = self.tgt_emb(dec_inputs) + self.pos_emb(torch.LongTensor([[5,1,2,3,4]]))
        dec_self_attn_pad_mask = get_attn_pad_mask(dec_inputs, dec_inputs)
        dec_self_attn_subsequent_mask = get_attn_subsequent_mask(dec_inputs)
        dec_self_attn_mask = torch.gt((dec_self_attn_pad_mask + dec_self_attn_subsequent_mask), 0)

        dec_enc_attn_mask = get_attn_pad_mask(dec_inputs, enc_inputs)

        dec_self_attns, dec_enc_attns = [], []
        for layer in self.layers:
            dec_outputs, dec_self_attn, dec_enc_attn = layer(dec_outputs, enc_outputs, dec_self_attn_mask, dec_enc_attn_mask)
            dec_self_attns.append(dec_self_attn)
            dec_enc_attns.append(dec_enc_attn)
        return dec_outputs, dec_self_attns, dec_enc_attns
```



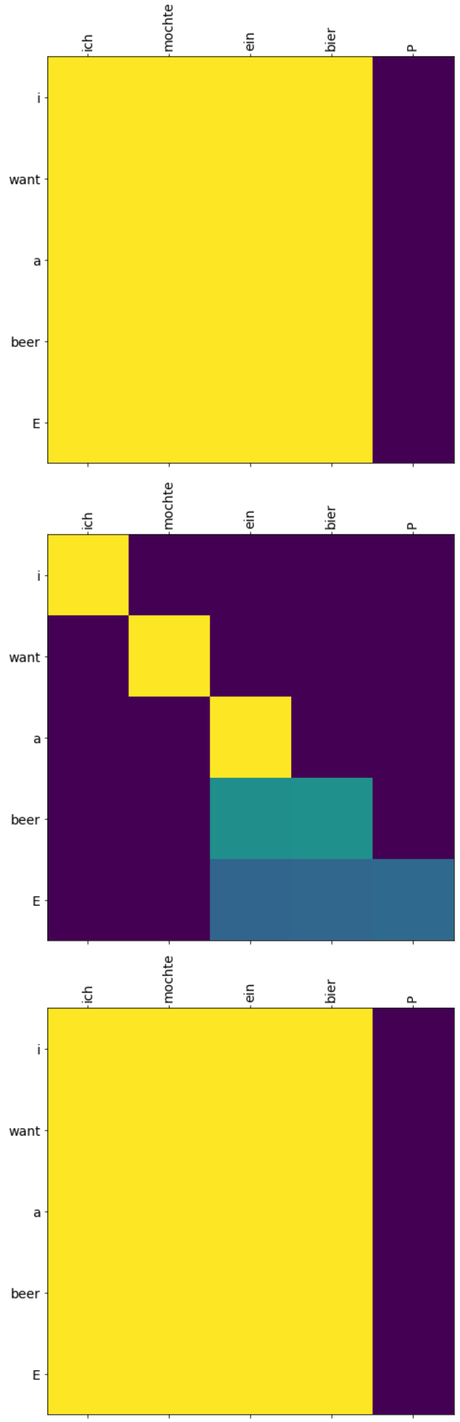
Example) Transformer from Scratch (by graykode)

```python
class Transformer(nn.Module):
    def __init__(self):
        super(Transformer, self).__init__()
        self.encoder = Encoder()
        self.decoder = Decoder()
        self.projection = nn.Linear(d_model, tgt_vocab_size, bias=False)

    def forward(self, enc_inputs, dec_inputs):
        enc_outputs, enc_self_attns = self.encoder(enc_inputs)
        dec_outputs, dec_self_attns, dec_enc_attns = self.decoder(dec_inputs, enc_inputs, enc_outputs)
        dec_logits = self.projection(dec_outputs) # dec_logits : [batch_size x src_vocab_size x tgt_vocab_size]
        return dec_logits.view(-1, dec_logits.size(-1)), enc_self_attns, dec_self_attns, dec_enc_attns
```



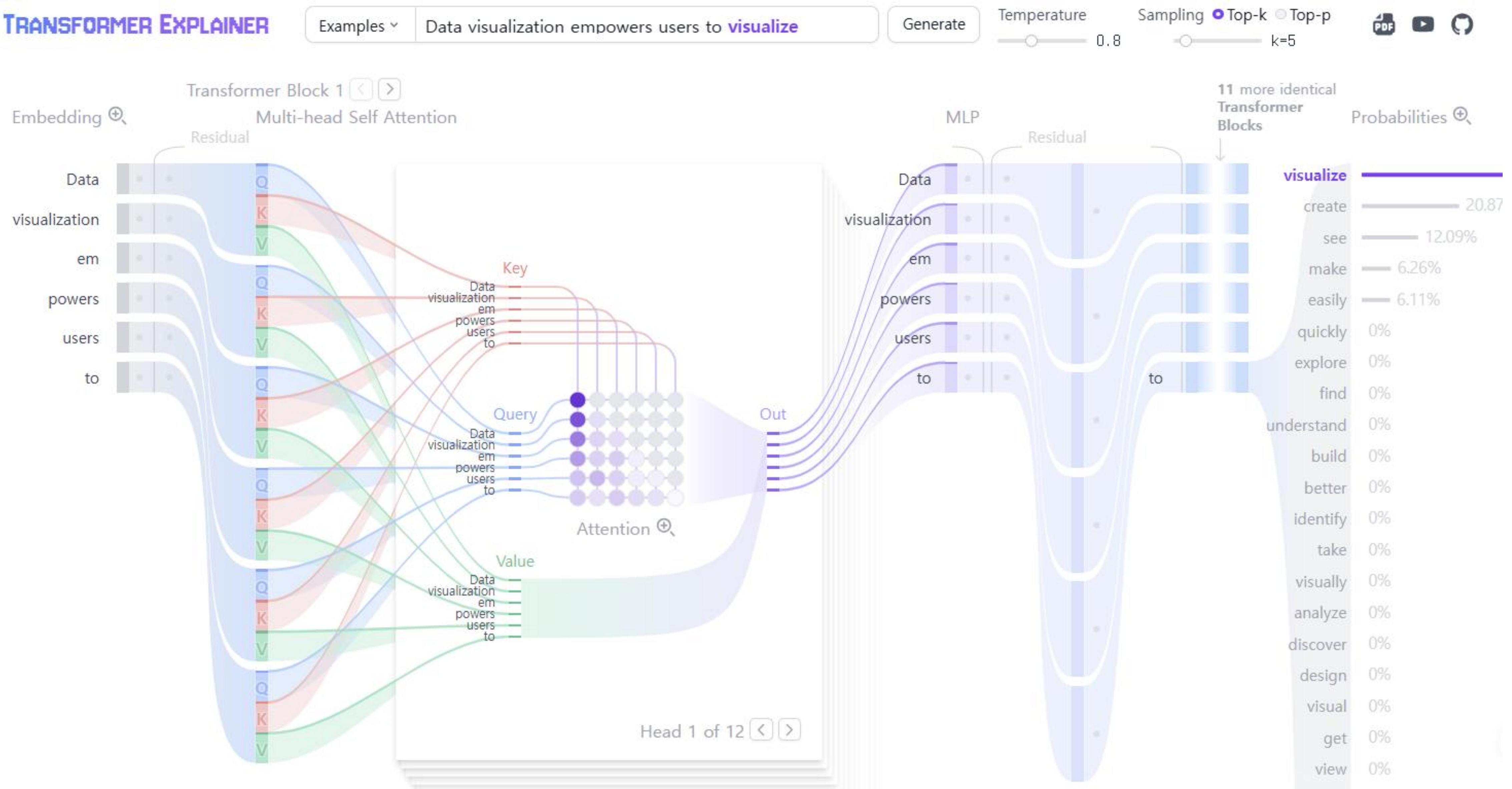
Example) Transformer from Scratch (by graykode)

```python
if __name__ == '__main__':
    sentences = ['ich mochte ein bier P', 'S i want a beer', 'i want a beer E’]

    ...

    for epoch in range(20):
        optimizer.zero_grad()
        outputs, enc_self_attns, dec_self_attns, dec_enc_attns = model(enc_inputs, dec_inputs)
        loss = criterion(outputs, target_batch.contiguous().view(-1))
        print('Epoch:', '%04d' % (epoch + 1), 'cost =', '{:.6f}'.format(loss))
        loss.backward()
        optimizer.step()

    # Test
    predict, _, _, _ = model(enc_inputs, dec_inputs)
    predict = predict.data.max(1, keepdim=True)[1]
    print(sentences[0], '->', [number_dict[n.item()] for n in predict.squeeze()])

    print('first head of last state enc_self_attns')
    showgraph(enc_self_attns)

    print('first head of last state dec_self_attns')
    showgraph(dec_self_attns)

    print('first head of last state dec_enc_attns')
    showgraph(dec_enc_attns)
```



Practice) GPT (Generative Pre-trained Transformer) Visualization

- Example) Transformer Explainer



Summary

- **DNN**: Deep neural network
- **CNN**: Feedforward with convolution (and pooling)
- **RNN**: NN with a loop → state/memory → sequential/temporal data
- **Transformer**: Multi-head attention mechanism, positional encoding → sequential/temporal data

- **Issue #1) Vanishing gradient problem**
  - Activation functions such as ReLU
  - Skip connection ~ LSTM and GRU
- **Issue #2) Overfitting problem**
  - More data by data collection or data augmentation or data synthesis
  - Data separation (train/validation/test data), cross-validation, and early stopping
  - More simplified models (e.g. CNN ← weight sharing and local connectivity; a.k.a. inductive bias)
  - Loss functions with regularization terms
- Other improvement
  - Dropout
  - Batch normalization
  - …

Further Information

- Natural Language Processing (NLP)
  - **Seq2seq** (2014), **attention mechanism** (2015)
  - **Transformer** (2017), **BERT** (Bidirectional Encoder Representations from Transformers), **GPT** (Generative Pretrained Transformer), …
- Computer Vision
  - CNN backbone networks: **AlexNet** (2012), **VGGNet** (2014), **Inception Net** (2014), **ResNet** (2015), …
  - CNN object detection networks
    - Two-stage detectors: **R-CNN** (2014), **Fast R-CNN** (2015), **Faster R-CNN** (2015), **Mask R-CNN** (2017), …
    - One-stage detectors: **YOLO** (You Only Look Once; 2015), **SSD** (Single Shot MultiBox Detector; 2015), …
  - Vision Transformers (ViT) and Vision Foundation Models (VFM)
    - **ViT** (Vision Transformer; 2020), **Swin Transformer** (2021), **DINO** (2021), …, **SAM** (Segment Anything; 2023), …
  - Generative models
    - **Autoencoder** (2006), …, **GAN** (Generative Adversarial Networks; 2014), **CycleGAN** (2017), …
    - **Diffusion model** (2020), **Stable Diffusion** (2022), …
  - Others
    - **NeRF** (Neural Radiance Field; 2020), **CLIP** (Contrastive Language-Image Pre-Training; 2021), …
- Others
  - Continual/incremental learning, …, transfer learning/domain adaptation, …, contrastive learning, …
  - Network compression/pruning, knowledge distillation (2015), …, ML model deployment, …

How To Understand New Methods (or Read Papers)

- **Model**
  - Network design
  - Input and output
    - e.g. Normalization, positional encoding
- **Training**
  - Dataset
    - e.g. Data augmentation, data synthesis
    - e.g. Data labeling, unsupervised or weakly-supervised approaches
  - Objective functions
    - e.g. Regularization terms, Huber loss (for robust regression), focal loss
  - Optimization
    - e.g. Optimization algorithms, hyperparameters